SIEMENS

AK 1703 ACP

Installation

DC2-009-2.00

Hint

Please observe Notes and Warnings for your own safety in the Preface.

Disclaimer of Liability
Although we have carefully checked the contents of this publication for conformity with the hardware and software described, we cannot guarantee complete conformity since errors cannot be excluded. The information provided in this manual is checked at regular intervals and any corrections that might become necessary are included in the next releases. Any suggestions for improvement are welcome.

Subject to change without prior notice
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Preface

This document is applicable to the following product(s):

- AK 1703 ACP

Purpose

This handbook describes how the hardware of an AK 1703 ACP system is to be installed and which preparations are to be made for the subsequent parameter settings.

Thereby the following points are described step-by-step:

- Conception of the architecture
- Assembly of the mounting rack
- Installation and removal of modules
- External communication
- Power Supply
- Parameter Setting Preparation

Target Group

This handbook is intended for persons who are entrusted with the design and installation of AK 1703 ACP systems.

Conventions used

Hint

is important information about the product, the handling of the product or the respective part of the documentation, to which special attention is to be given.

- Outputs visible on the screen are `described in this font`.
- Inputs via keyboard or mouse keys and visible on the screen are also `described in this font`
- An essential component of this guide are pictures, which document the activities described. If this activity is independent of the type of mounting rack, symbolically always only one type is represented in the pictures. The activity itself can in addition however also be performed analogously on another type of mounting rack.
  If however the type of mounting rack necessitates a different method of operation, then both are described in their own chapters.

Notes on Safety

This manual does not constitute a complete catalog of all safety measures required for operating the equipment (module, device) in question because special operating conditions might require additional measures. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger.

Danger

means that death, serious bodily injury or considerable property damage **will** occur, if the appropriate precautionary measures are not carried out.

Warning

means that death, serious bodily injury or considerable property damage **can** occur, if the appropriate precautionary measures are not carried out.

Caution

means that minor bodily injury or property damage could occur, if the appropriate precautionary measures are not carried out.

Hint

is important information about the product, the handling of the product or the respective part of the documentation, to which special attention is to be given.

Qualified Personnel

Commissioning and operation of the equipment (module, device) described in this manual must be performed by qualified personnel only. As used in the safety notes contained in this manual, qualified personnel are those persons who are authorized to commission, release, ground, and tag devices, systems, and electrical circuits in accordance with safety standards.

Use as Prescribed

The equipment (device, module) must not be used for any other purposes than those described in the Catalog and the Technical Description. If it is used together with third-party devices and components, these must be recommended or approved by Siemens.

Correct and safe operation of the product requires adequate transportation, storage, installation, and mounting as well as appropriate use and maintenance.

During operation of electrical equipment, it is unavoidable that certain parts of this equipment will carry dangerous voltages. Severe injury or damage to property can occur if the appropriate measures are not taken:

- Before making any connections at all, ground the equipment at the PE terminal.
- Hazardous voltages can be present on all switching components connected to the power supply.
- Even after the supply voltage has been disconnected, hazardous voltages can still be present in the equipment (capacitor storage).
- Equipment with current transformer circuits must not be operated while open.
- The limit values indicated in the manual or the operating instructions must not be exceeded; that also applies to testing and commissioning.

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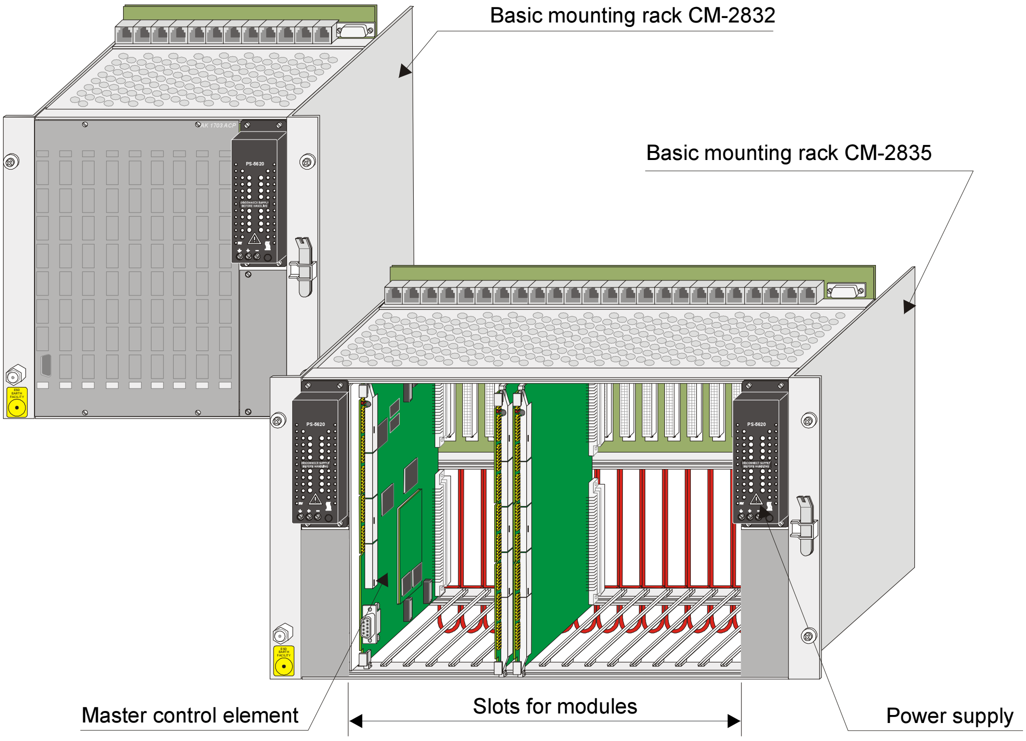
1. Architecture

Content

1.1. Mechanical Design

An AK 1703 ACP system consists of a mounting rack in double-Euroformat with slots for the seating of modules and power supplies.



There are 3 different sized mounting racks available.

The basic mounting rack CM-2835 has 17 slots for modules in double-Euroformat and 4 slots for power supplies in single-Euroformat. It can be used for rear panel assembly and 19" (swing-)frame installation.

The basic mounting rack CM-2832 has 9 slots for modules in double-Euroformat and 2 slots for power supplies in single-Euroformat. It can only be used for rear panel assembly.

The expansion mounting rack CM-2833 has 16 slots for modules in double-Euroformat and 4 slots for power supplies in single-Euroformat. It can be used for rear panel assembly and 19" (swing-)frame installation.

2. Components

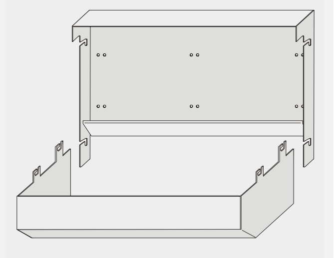
Content

2.1. Mounting Rack

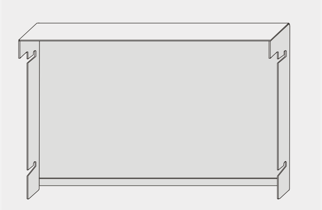
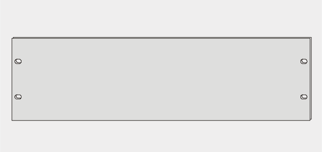
	Designation	Item Number / MLFB
	CM-2832 AK 1703 ACP Board R. 9 Board SI (Mounting rack with wall bracket and interface cover)	GC2-832 6MF11130CJ320AA0
	CM-2835 AK 1703 ACP Board R. 17 Board SI (Mounting rack without wall bracket and without interface cover)	GC2-835 6MF11130CJ350AA0
	AK 1703 ACP Board Rack Extension (Mounting rack without wall bracket and without interface cover)	GC2-833 6MF11130CJ330AA0

2.1.1. Accessories for CM-2835 and CM-2833

Rear Panel Installation

	Designation	Item Number/ MLFB
	Wall fastening kit for CM-2835 AK 1703 ACP	TC2-008 6MF13130CA080AA0

(Swing) Frame Installation

	Designation	Item Number/ MLFB
	IP20-Backside Cover kit for CM-2835 (for 19" assembly)	TC2-050 6MF13130CA500AA0
	FRBL19"-3HE Front Panel -RAL7032 (for 19" assembly)	TF3-018 6MF13140DA180AA0

2.2. Master Control Element

	Designation	Item Number/ MLFB
	CP-2010 Master Control Element	BC2-010 6MF10130CA100AA0

2.3. Processing and Communication Element

	Designation	Item-Number/ MLFB
	CP-2012 Processing and Communication Element	BC2-012 6MF10130CA120AA0
	CP-2017 Processing and Communication Element	BC2-017 6MF10130CA170AA0

2.4. Connection Boards

	Designation	Item-Number/ MLFB
	CM-2837 Connect. Comm./Sys-I/O (CP-2010)	BC2-837 6MF10130CJ370AA0
	CM-2838 Connection Communication (CP-2012)	BC2-838 6MF10130CJ380AA0

2.5. Serial Interface Modules

2.5.1. Serial Interface Processor (SIP)

	Designation	Item-Number/ MLFB
	SM-2551 Serial Interface Prozessor 2 SS	BC2-551 6MF10130CF510AA0
	SM-0551 Ser. Interface Prozessor 2 SS	BC0-551 6MF10130AF510AA0

2.5.2. Network Interface Processor (NIP)

Designation	Item-Number/ MLFB
SM-2556 Ser.Interface+Ethernet 10/100FX	BC2-556 6MF10130CF560AA0

2.5.3. Field Bus Interface Processor (FIP)

Designation	Item-Number/ MLFB
SM-2545 Profibus Interface (Master)	BA2-545 6MF10110CF450AA0

2.6. Bus Interface Module

	Designation	Item-Number / MLFB
	Ax 1703 bus interface electrical	GA0-843 6MF11110AJ430AA0
	CM-0842 Ax 1703-Bus-Interface 4x fiber optical	GA0-842 6MF11110AJ420AA0

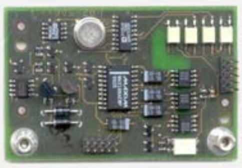
2.7. Peripheral Elements

	Designation	Item-Number / MLFB
	AI-2300 Ana. Inp.16x ±20mA + 4x opt.IOM	BA2-300 6MF10110CD000AA0
	DI-2100 Digital Input 8x8, 24-60VDC	BA2-100 6MF10110CB000AA0
	DI-2110 Digital Input 8x8, 24-60VDC,1ms	BA2-110 6MF10110CB100AA0
	DI-2111 Digital Input 8x8, 110/220VDC, 1ms	BA2-111 6MF10110CB110AA0
	DO-2201 Digital Output 40x1, 24-60VDC	BA2-201 6MF10110CC010AA0

	Designation	Item-Number / MLFB
	DO-2210 Command Output 24-60VDC	BA2-210 6MF10110CC100AA0
	MX-2400 In-/Output 24-60VDC, 1x opt. IOM	BA2-400 6MF10110CE000AA0

2.7.1. Zubehör

	Designation	Item-Number / MLFB
	SM-0570 Analog input extension (2x ±20mA) (for AI-2300)	BA0-570 6MF10110AF700AA0
	SM-0571 Analog input extension (2x Pt100) (for AI-2300)	BA0-571 6MF10110AF710AA0

	Designation	Item-Number / MLFB
	SM-0572 Analog output extension (2x ±20mA/±1V/±10V) (for AI-2300)	BA0-572 6MF10110AF720AA0
	SM-0574 Count pulse input extension (2x 24-60VDC) (for AI-2300)	BA0-574 6MF10110AF740AA0
	SM-2506 Measuring circuit module for command output (for DO-2210)	BA2-506 6MF10110CF060AA0

3. Mounting Rack Installation

Content

3.1. Necessary Material

Due to the different circumstances with the local installation of the AK 1703 ACP systems, not all details such as e.g. screw sizes, cable lengths and the like can be dealt with in this document. Therefore only recommendations and minimum requirements are specified concerning consumables.

The fitter himself must ensure that suitable materials and tools necessary for the installation are available.

3.2. How and where can Installation take place?

3.2.1. Installation Location

An AK 1703 ACP can be used for the rear panel installation in a cabinet. Additionally, systems that use a CM-2835 mounting rack, can be installed in a 19" (swing-)frame.

3.2.2. Environmental Conditions

The permissible temperature range for the operation of the system is about 0°....+55°C.

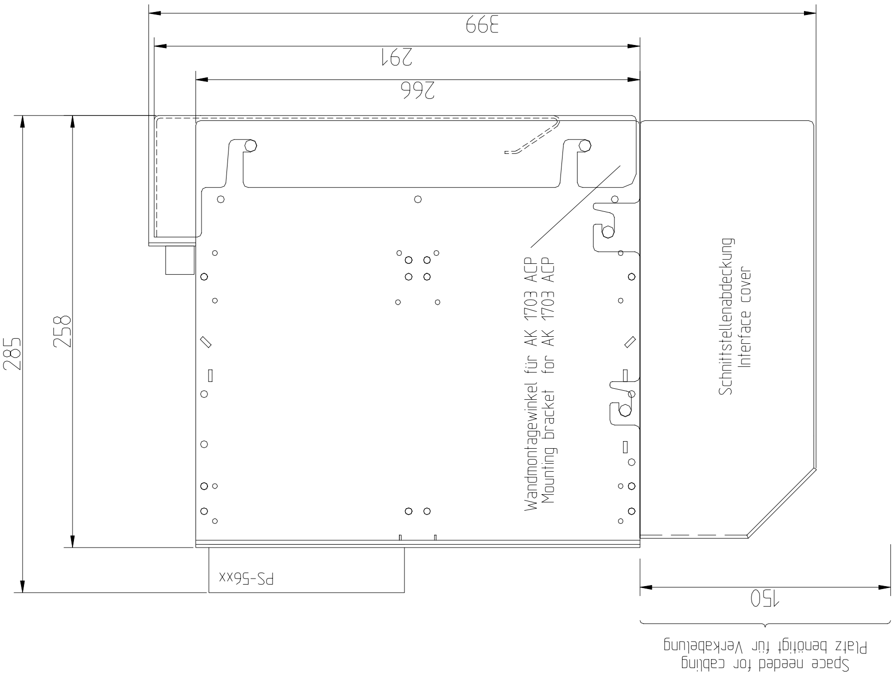
3.2.3. Installation Position

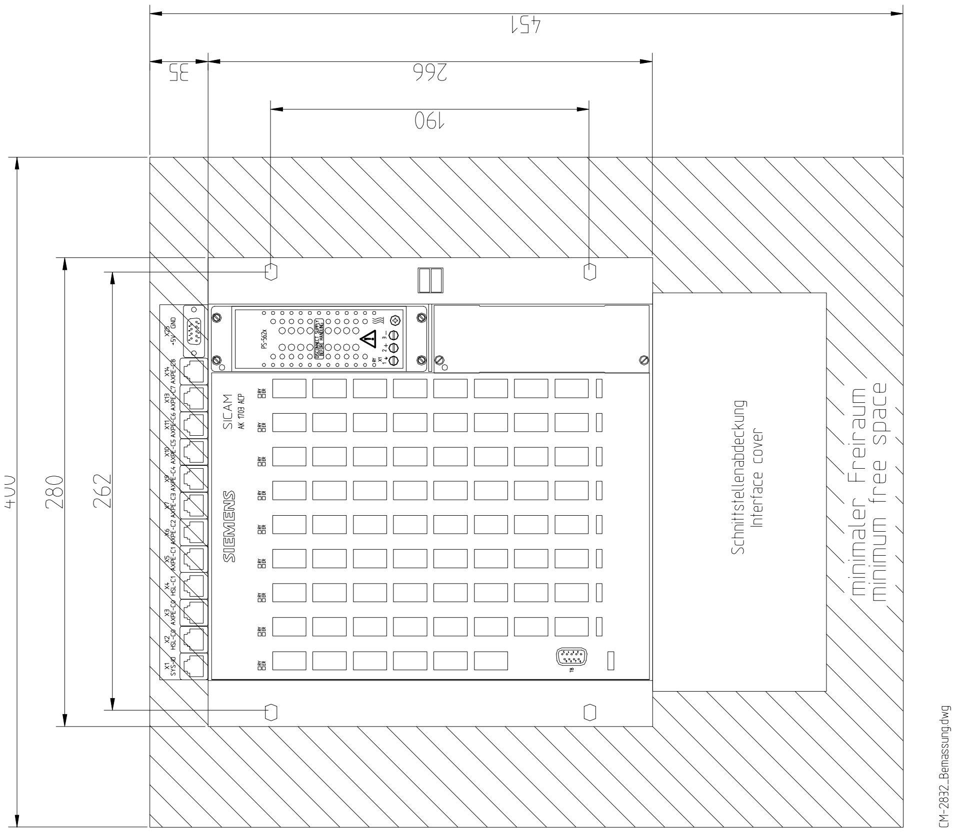
AK 1703 ACP systems may only be installed horizontally.

3.2.4. Space Requirement

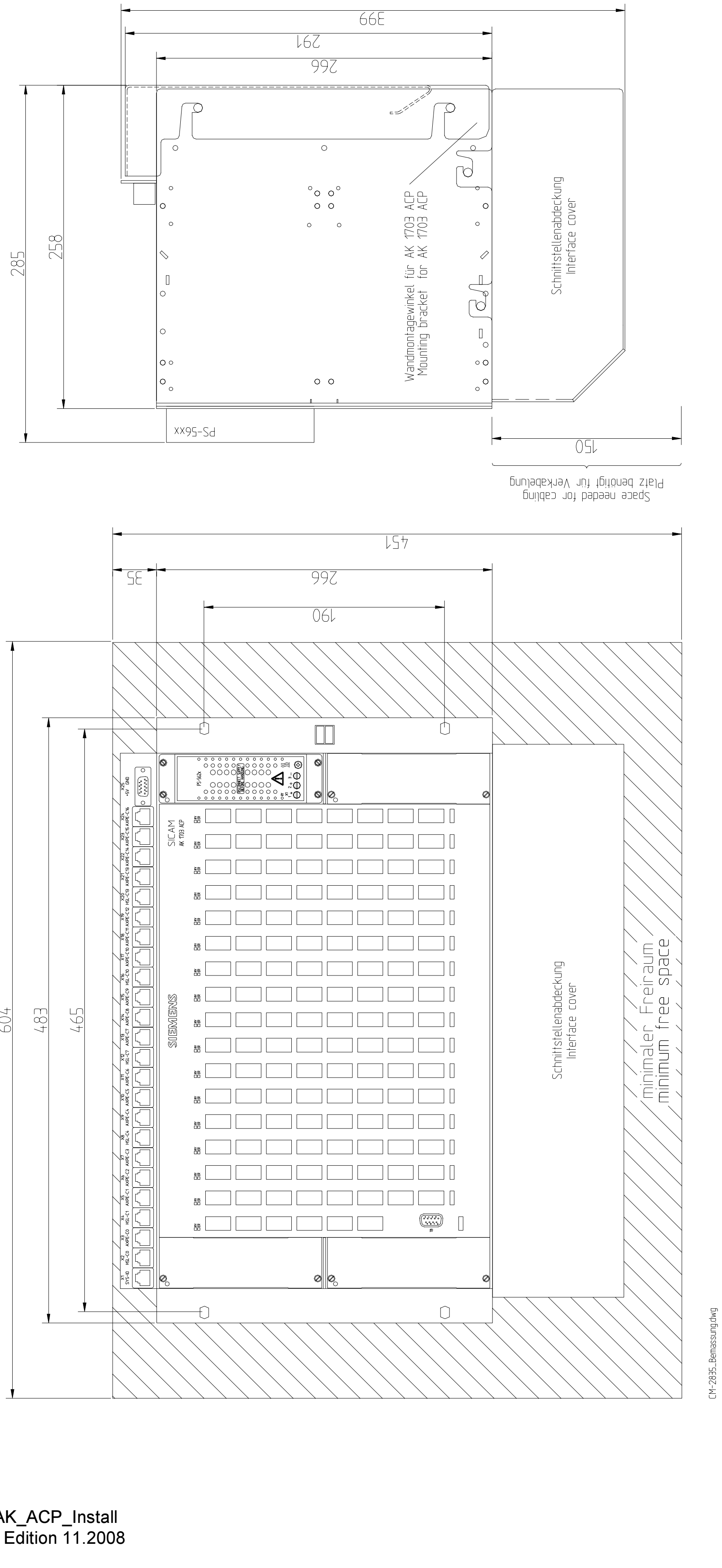
Certain minimum distances must be observed between mounting rack and adjacent equipment. These minimum distances serve the installation and removal of components, the cabling and the ventilation of the device in operation.

3.2.4.1. CM-2832 - Rear Panel Installation

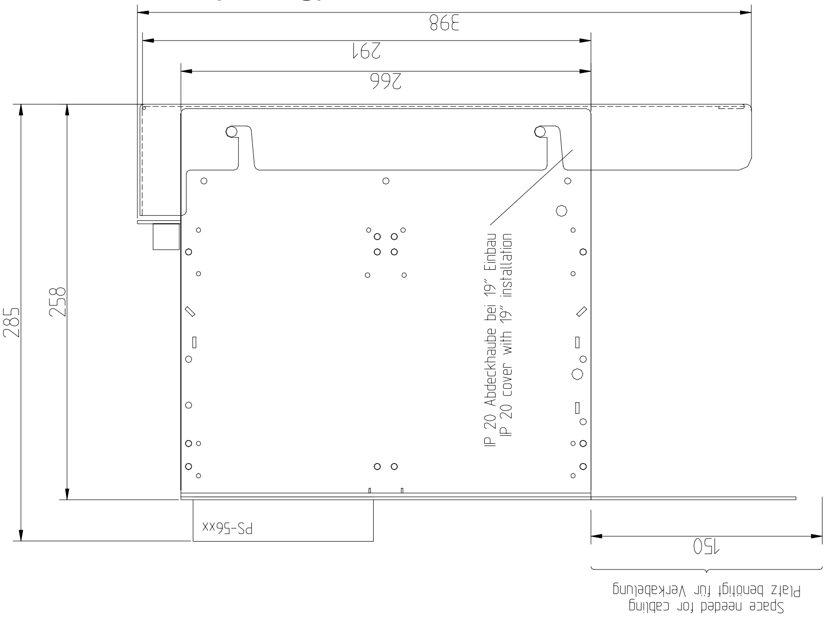


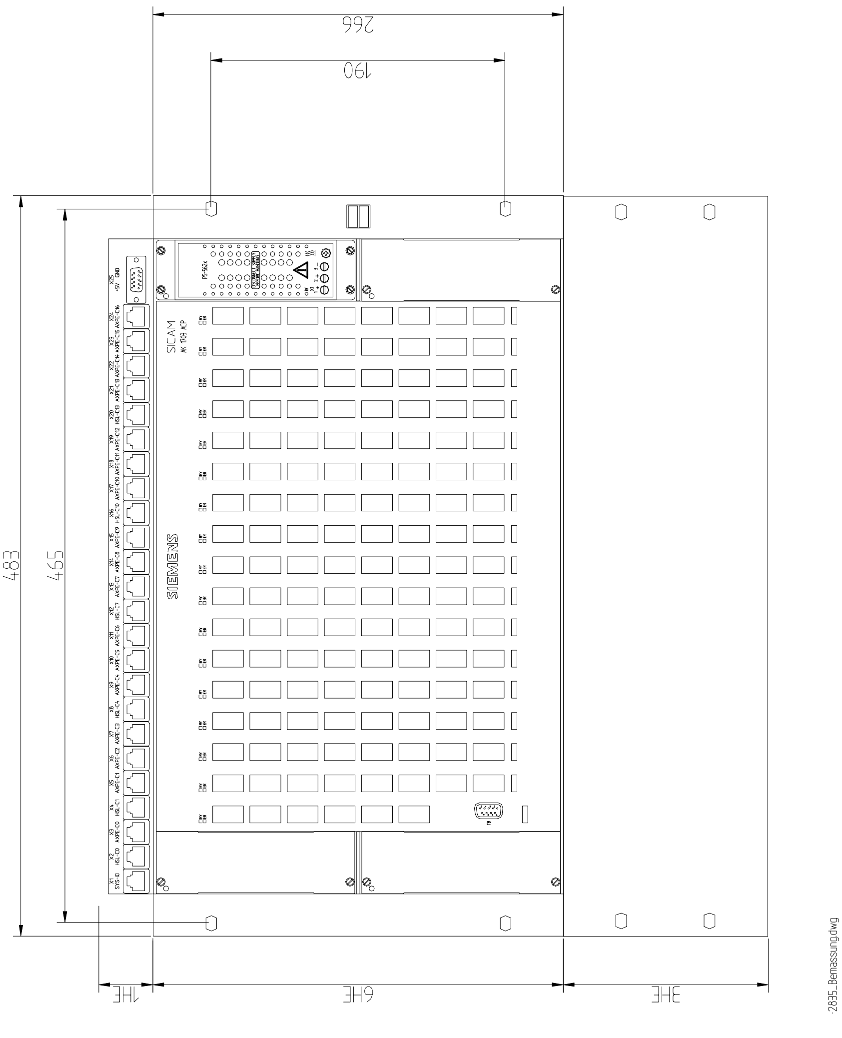


3.2.4.2. Rear Panel Installation

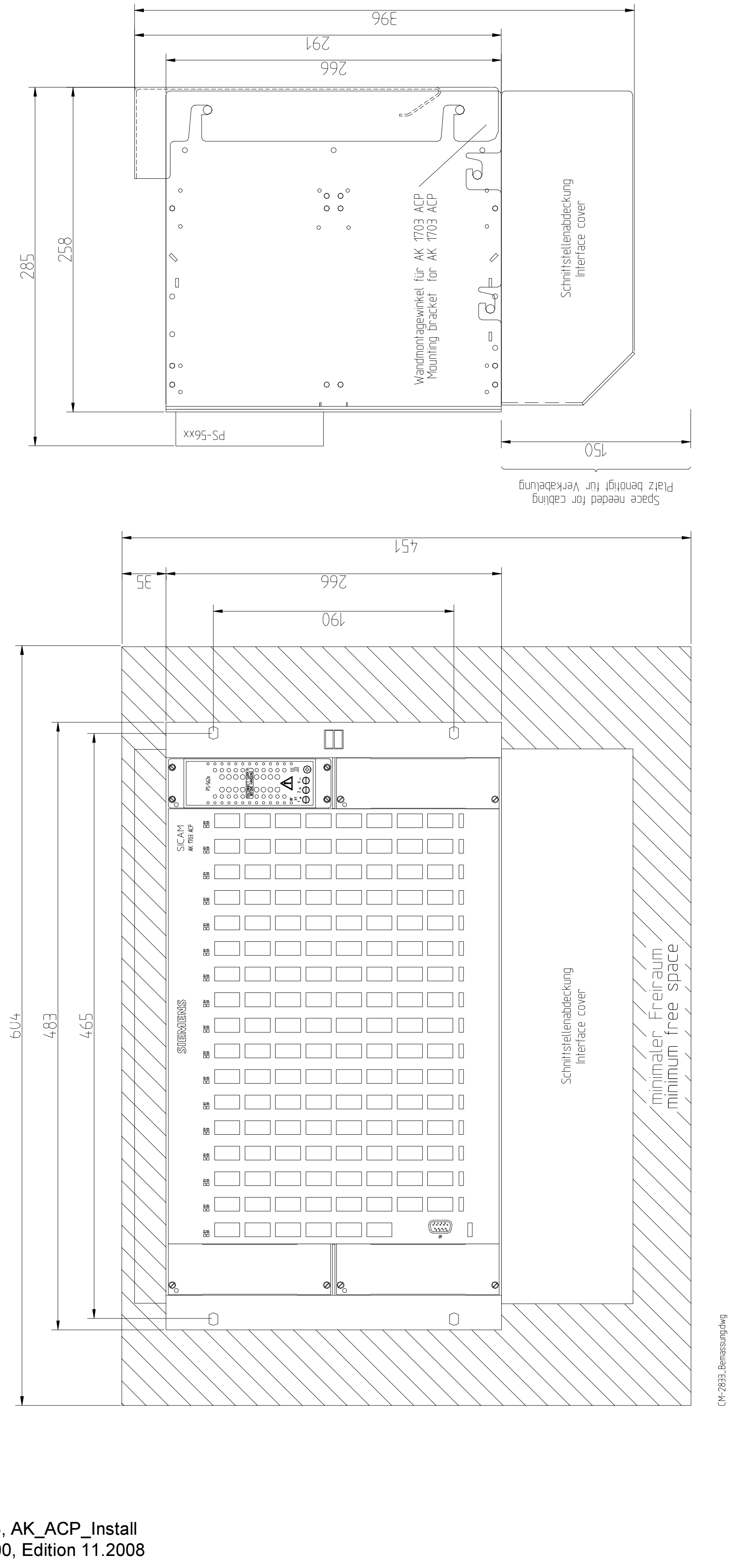


3.2.4.3. CM-2835 - 19" (Swing) Frame Installation

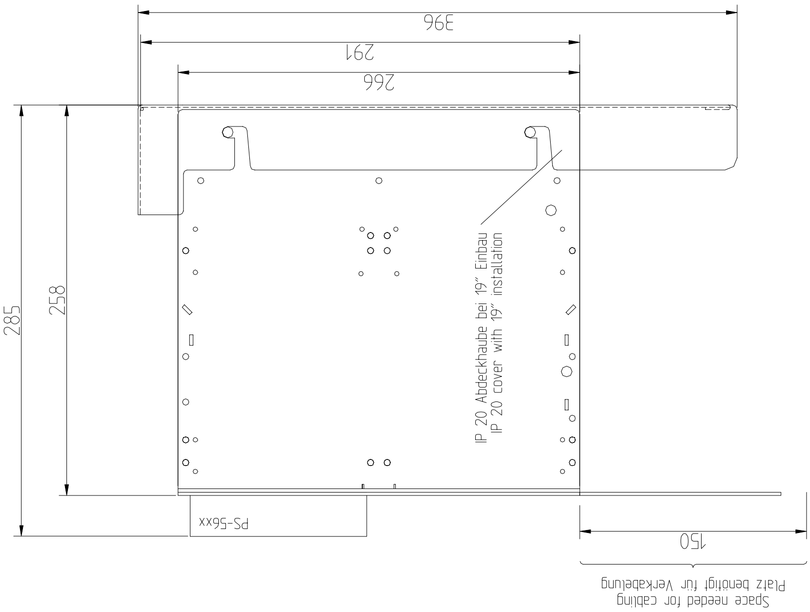


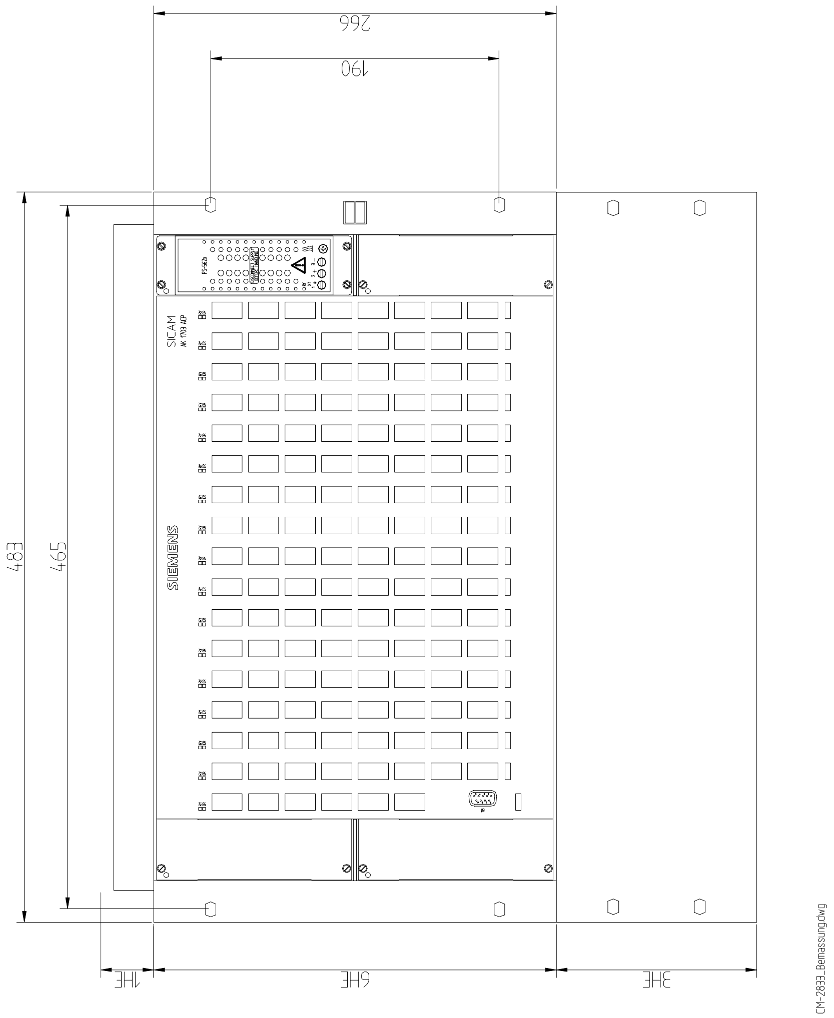


3.2.4.4. CM-2833 - Rear Panel Installation

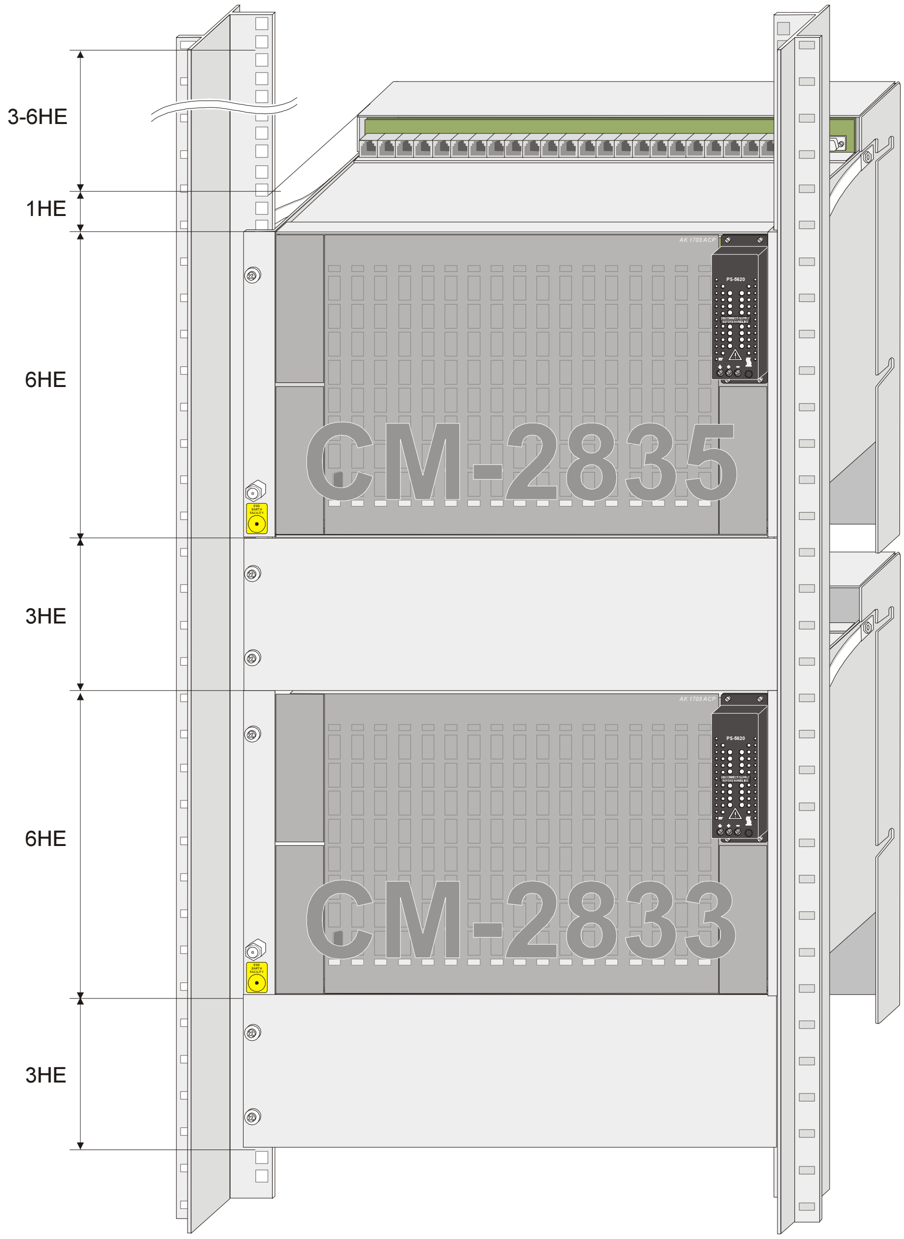


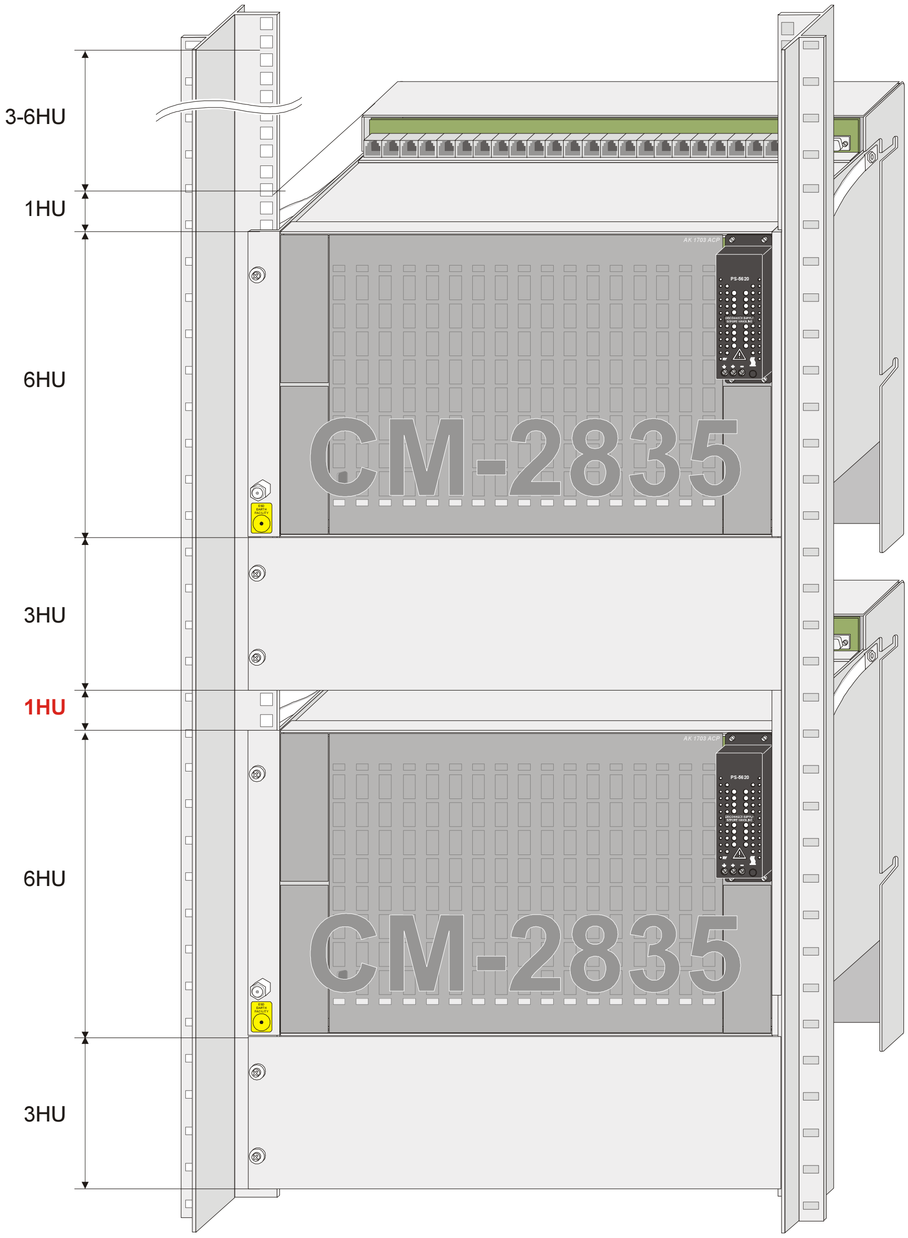
3.2.4.5. CM-2833 - 19" (Swing) Frame Installation





3.2.4.6. Examples for 19" (swing) frame installation of 2 Mounting Racks



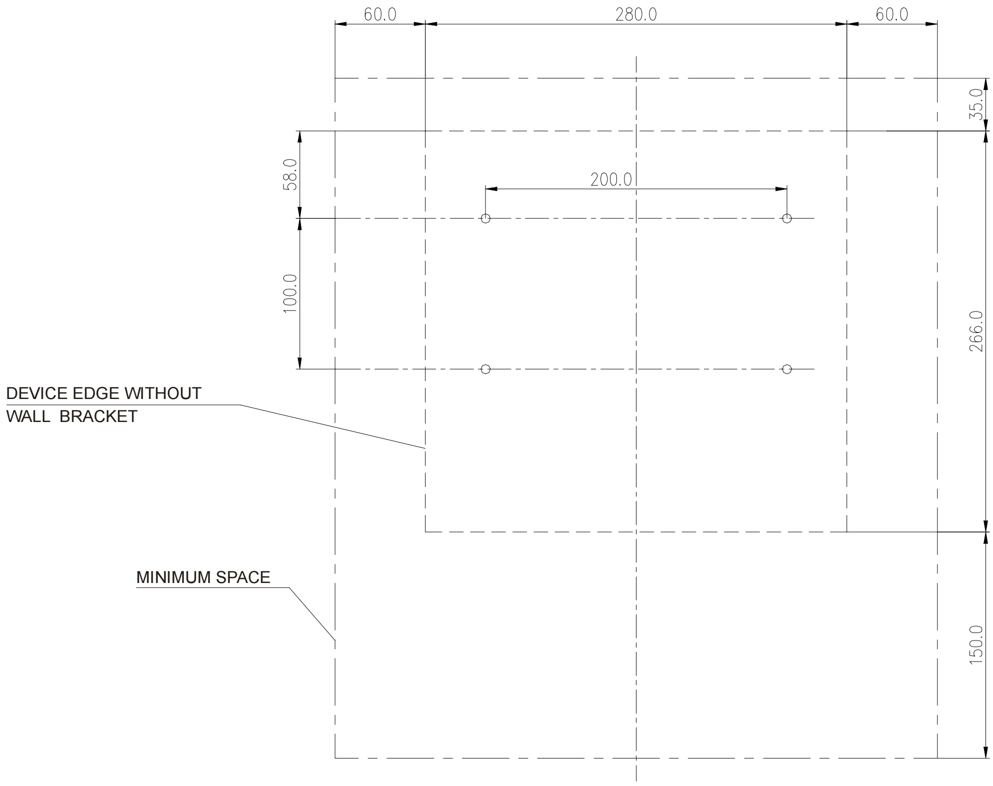

3-6HU
1HU
6HU
3HU
1HU
6HU
3HU
CM-2835
CM-2835

3.3. Mounting Rack CM-2832

The mounting rack CM-2832 is supplied as set with wall bracket and interface cover.

Proceed as follows:

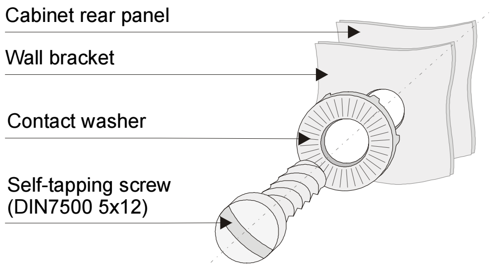
- Remove interface cover and wall bracket from the mounting rack.
- Mark out the drill holes for the wall bracket on the rear panel of the cabinet according to the enclosed drilling diagram (DC2-000-1).



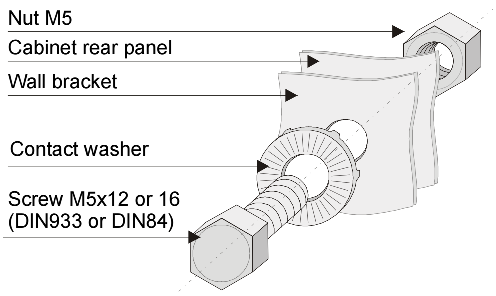
- Drill the holes according to the specifications on the drilling diagram.

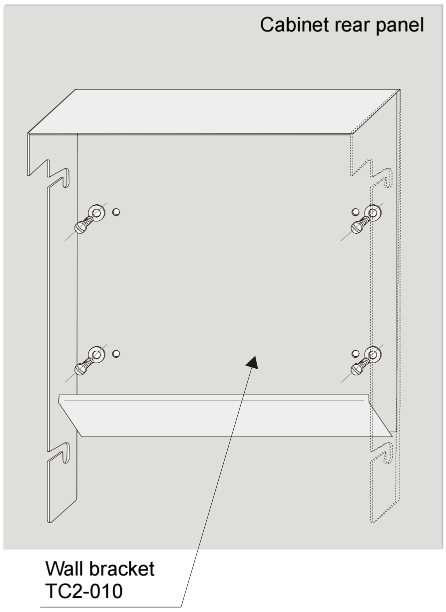
- Mount the wall bracket

Fitting with self-tapping screws

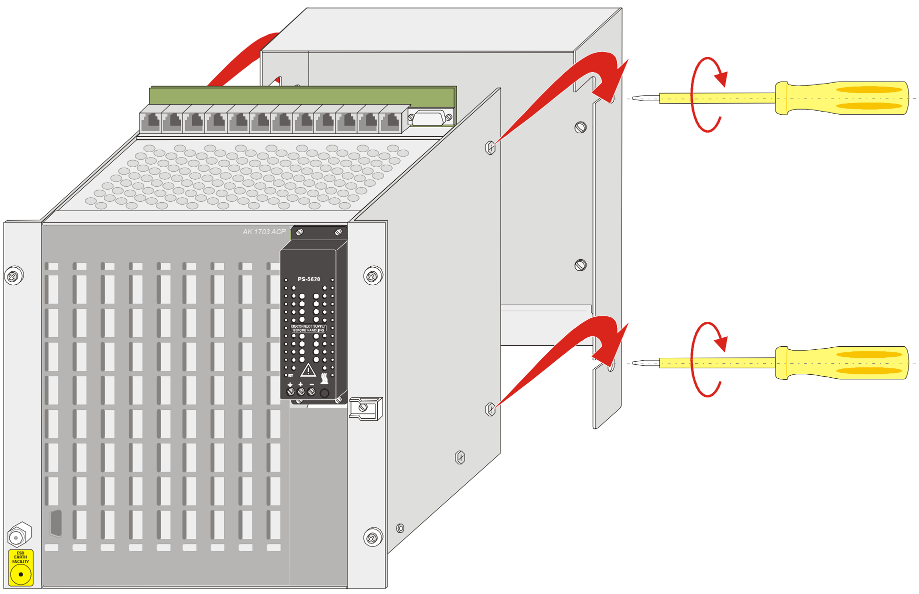


Fitting with conventional screws





- Hang the mounting rack in the wall bracket and tighten the fastening screws M5x6 (DIN88933)



3.4. Mounting Rack CM-2835 and CM-2833

The mounting racks CM-2835 and CM-2833 can be used for rear panel -and 19" (swing-)frame installation. Therefore the basic set does not include any installation-specific components. If required, these must be ordered separately.

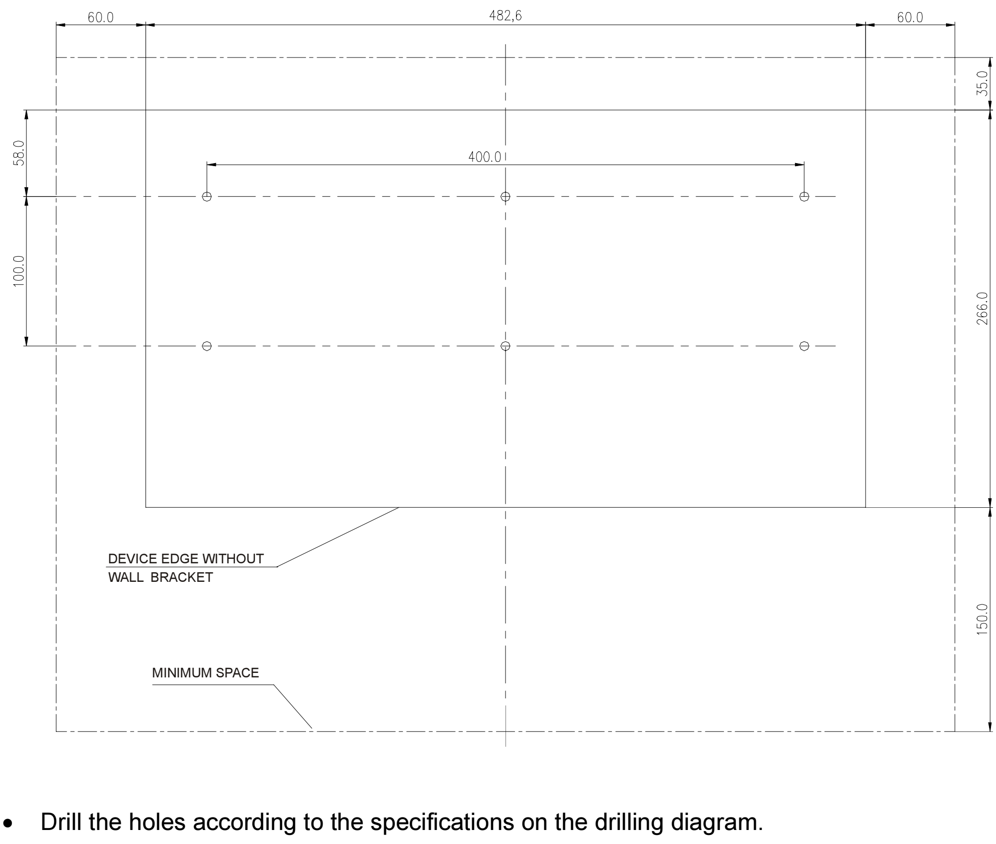
These are:

- **Wall-Mounting Set** (TC2-008)
  for systems that are mounted on the rear panel (mounting plate).
- **19" Cover Set** (TC2-050)
  for systems that are installed in a 19" cabinet/rack.

3.4.1. Rear Panel Assembly

Proceed as follows:

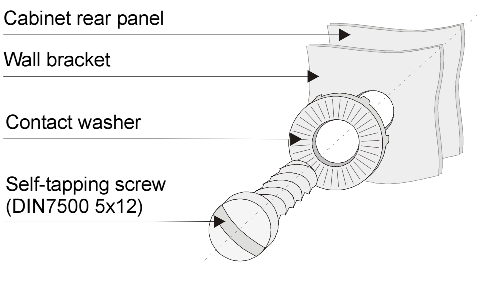
- Mark out the drill holes for the wall bracket on the rear panel of the cabinet according to the enclosed drilling diagram (DC2-001-1).



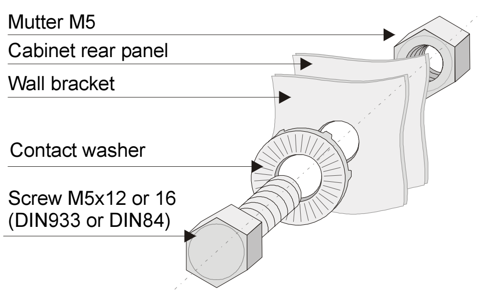
- Drill the holes according to the specifications on the drilling diagram.

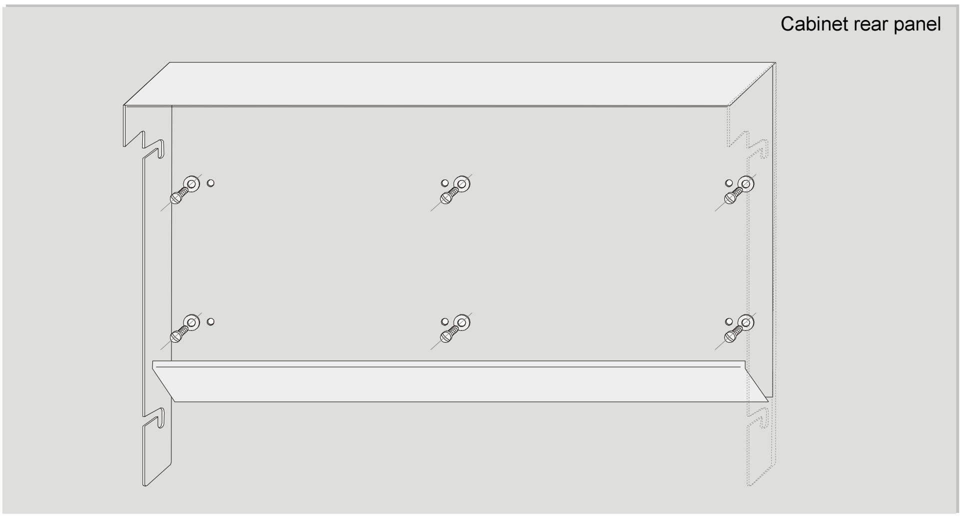
- Mount the wall bracket

Mounting with self-tapping screws



Mounting with conventional screws

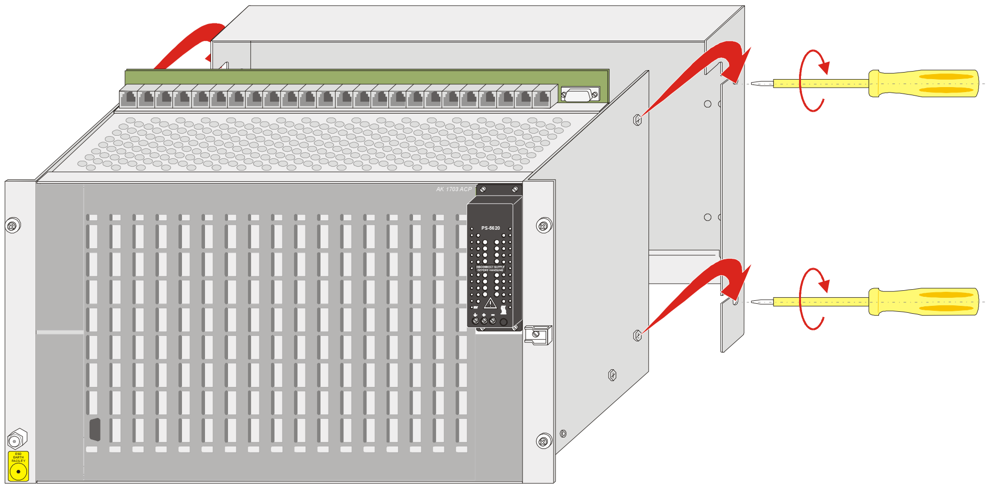




Attention

The mounting rack must be fastened with at least 6 screws.

- Hang the mounting rack in the wall bracket and tighten the fastening screws M5x6 (DIN88933)

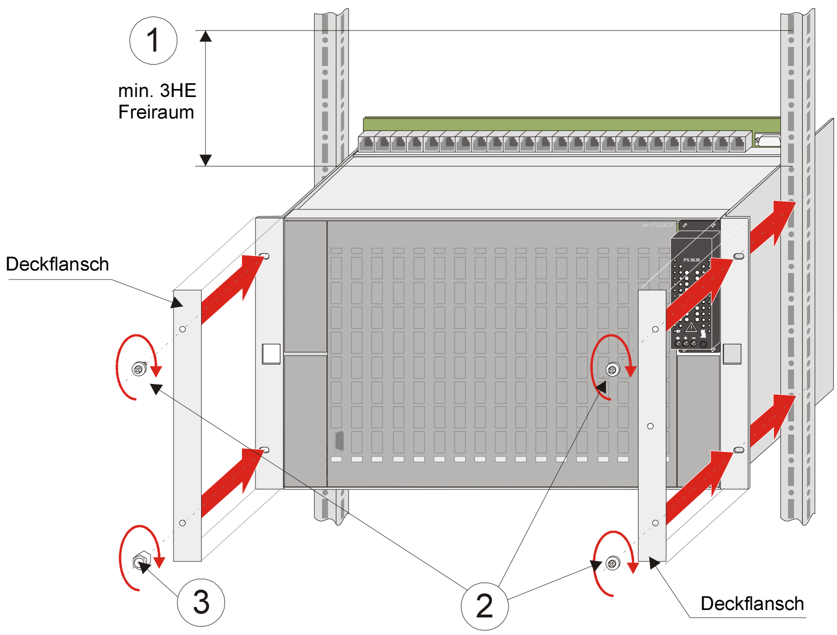


3.4.2. 19" (Swing) Frame Installation

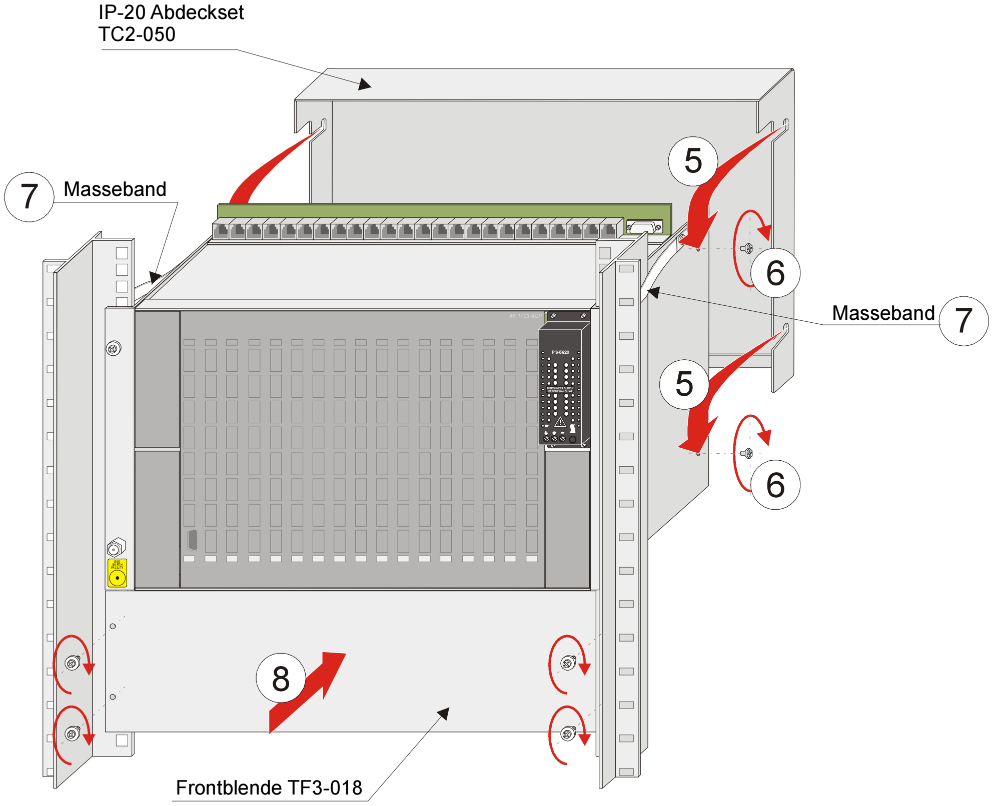
Proceed as follows:

- Determine the position at which the mounting rack is to be installed. Please take note, that a space of 3HU must be maintained above the mounting rack ①.
- Screw the mounting rack and the cover flange onto the 19" frame.

  To do this, use the screws M6x16 (DIN85) and the plastic washers 6.4-KST (DIN125) at position ②, and the ESD-screw M6x14 at position ③.
- Put the unlocking tool into the bracket on the right cover flange ④.



- Hang the IP-20 cover set in the mounting rack ⑤ and tighten the fastening screws M5x6 (DIN88933) ⑥.
- Attach the grounding straps to the cabinet/frame ⑦
- Screw on the front panel ⑧.



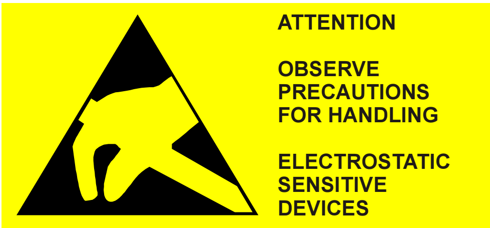
4. Installation and Removal of Modules

Content

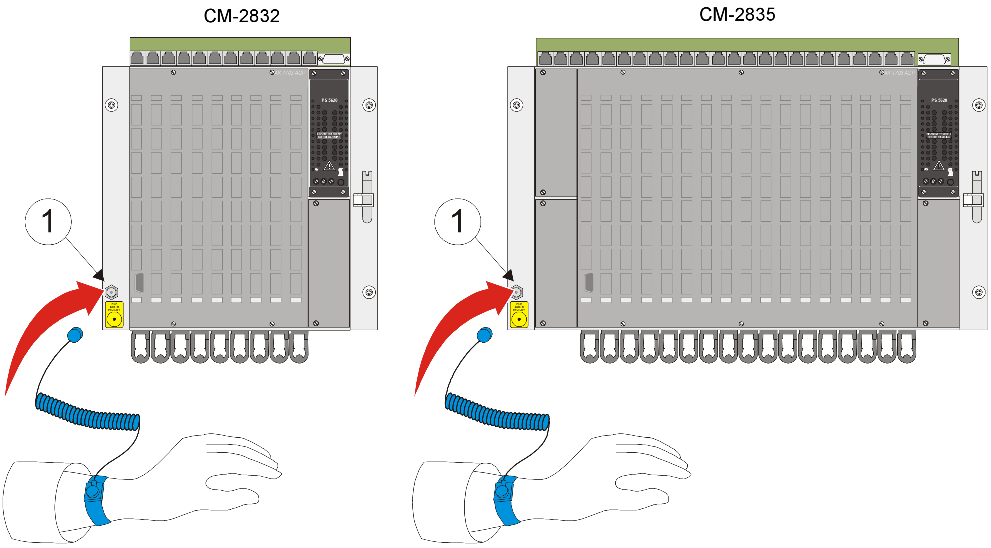
Warning
The activities described in this chapter presume that the grounding connection between AK 1703 ACP and cabinet or rack has been properly carried out and that these are earthed.

4.1. Preparations

4.1.1. Connect and put on Grounding Strap



The grounding strap can be connected directly to the mounting rack. The connection point is located on the left side panel of the mounting rack ①.

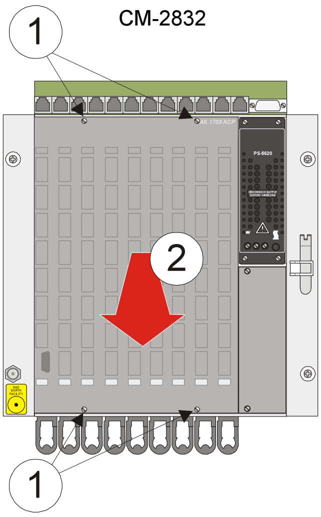


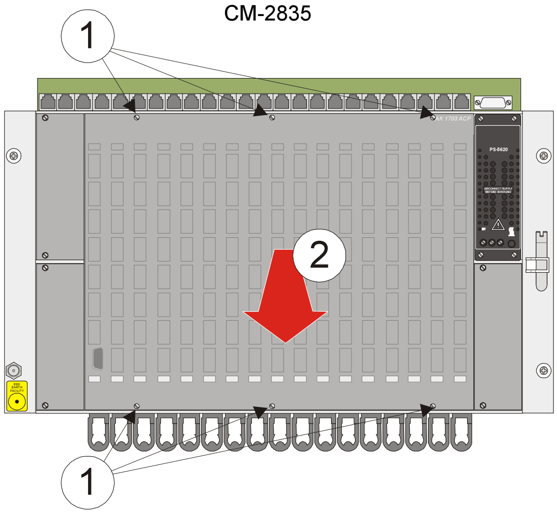
Designation	Item Number/ MLFB
Grounding Strap	TF3-133 6MF13140DB330AA0

4.1.2. Removing Front Panel

Proceed as follows:

- Open the knurled screws* ① which connect the front panel with the mounting rack.
  (*with CM-2832 there are 4 knurled screw to open, with CM-2835 6)
- Take hold of the front panel at the knurled screws and remove it ②.





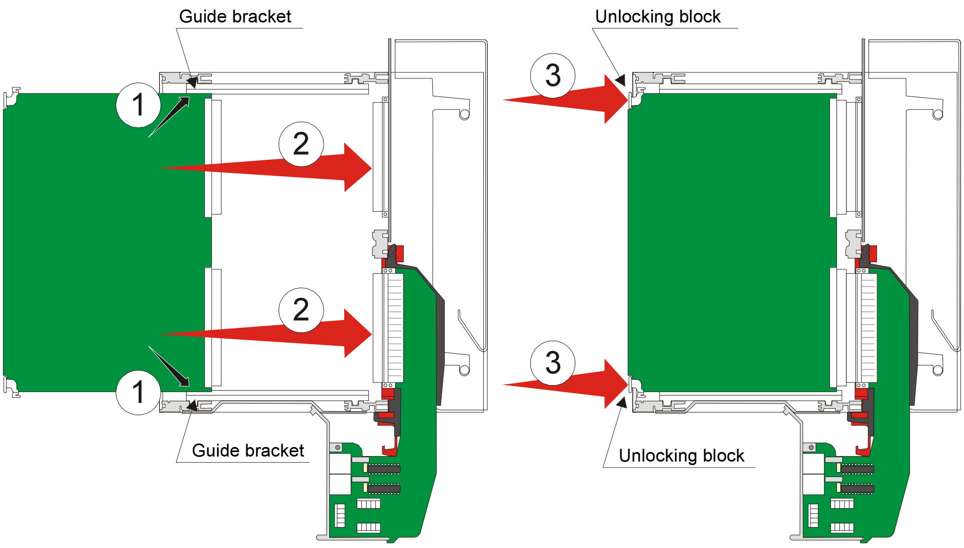
4.2. Installation

Attention

Before the installation of a module the associated connection board (CM-2837, CM-2838) or the peripheral cable (CM-2890) must already be fitted.

The installation of modules is carried by hand without any tools. Proceed as follows:

- Connect/put on grounding strap and remove front panel.
- Place the module in the guide bracket of the desired slot ① and push it carefully up to the connectors located at the bottom ②.
- Then apply the same pressure at both unlocking blocks ③ in order to connect the module with the backplane and connection board or peripheral cable.



4.3. Removal

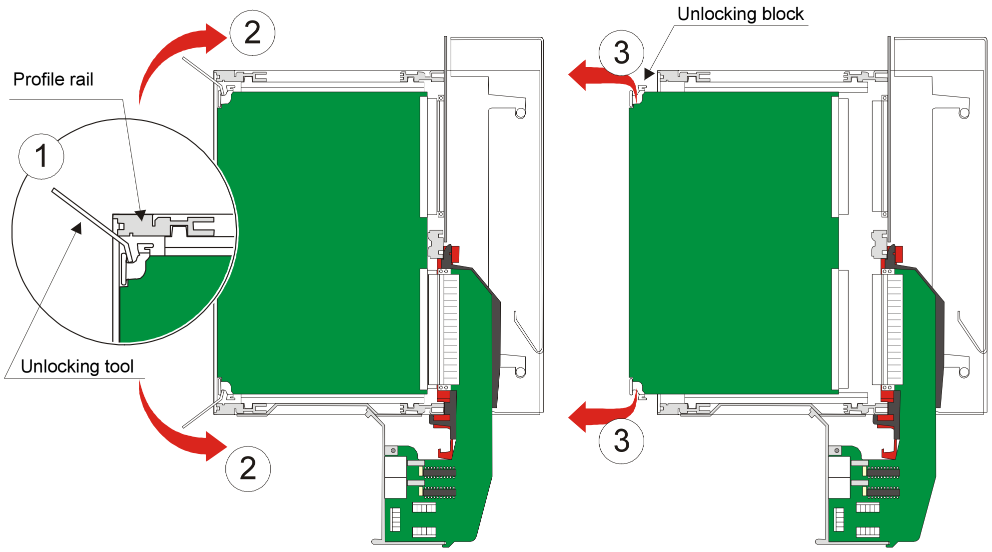
Attention

The master control element CP-2010 may only be removed in de-energized state. This does not apply to the other modules (processing and communication element, peripheral element).

For the removal of a module, you will need the unlocking tool TA2-105. This is included in the scope of supply of every mounting rack and must be inserted in the bracket provided on the mounting rack during installation.

Proceed as follows:

- Connect/put on grounding strap and remove front panel.
- Position the unlocking tool between module and profile rail ①.
- Loosen the plug-in connector by pushing the unlocking clip upwards ②, or downwards ②.
- Grasp the module at the unlocking blocks and withdraw it from the mounting rack ③.



Designation	Item Number/ MLFB
Extracting Tool for AK-Boards	TA2-105 6MF13110CB050AA0

4.4. Labeling Strip

After the installation/removal of a module, the associated labeling strip is to be inserted in the front panel or removed.

The openings for inserting the labeling strip are located on the reverse side of the front panels.

4.5. Mounting Front Panel

Proceed as follows:

- Position the front panel
- Tighten the knurled screws *
  (*with CM-2832 there are 4 knurled screws to tighten, with CM-2835 6)

5. Setup of external Communication Connections

Content

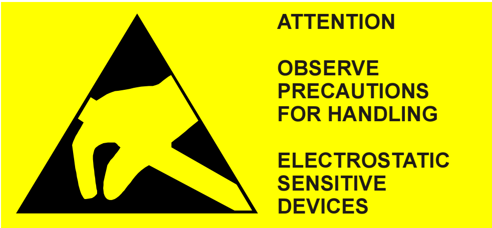
5.1. General

By means of appropriate expansions, AK 1703 ACP can be expanded with the following methods of communication:

- Serial communication (V.28, V.11)
  - Point-to-point traffic according to IEC 60870-5-101
  - Multi-point traffic according to IEC 60870-5-101
  - Dial-up traffic
- LAN Communication (Ethernet TCP/IP)
- Field Bus Communication

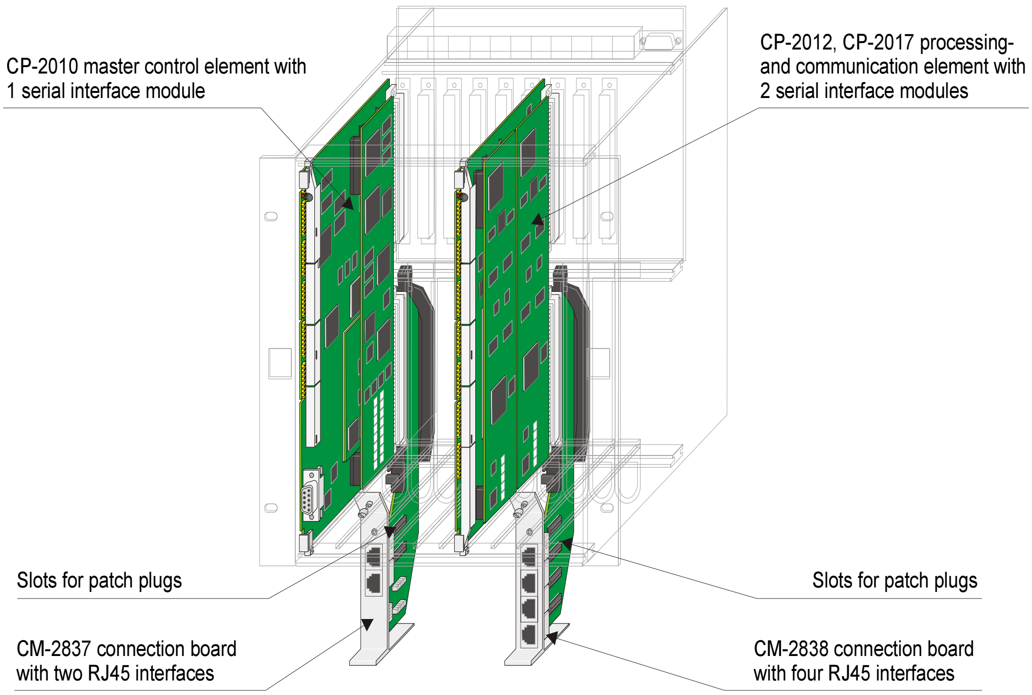
For this it is necessary to expand the system internally with additional modules (refer to section 5.2; Installation of SIM's and communication connections) and then to connect with the external communication device (Modem) (refer to section 5.3; Communication Cabling).

5.2. Installation of SIM's and communication connections



Regardless of the method of communication the following tasks are to be carried out:

- Installation of a Serial Interface Module (SIM) on master control element or on processing- and communication element
- Fitting of the connection boards with Patch Plugs
- Installation of the connection boards on master control element or on processing- and communication element



5.2.1. Installation of the Serial Interface Module

The serial interface module can be installed on master control element or processing- and communication element.

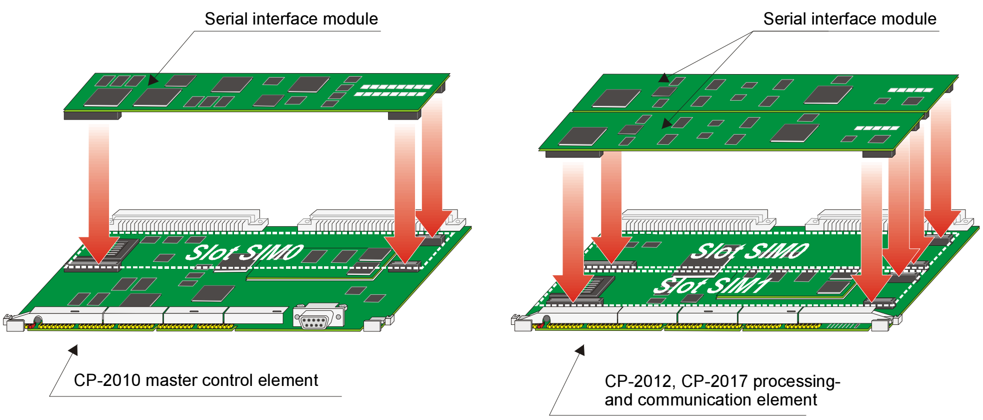
The master control element CP-2010 can carry one SIM, the processing- and communication element CP-2012 or CP-2017 two.

The installation can only take place with the module removed.

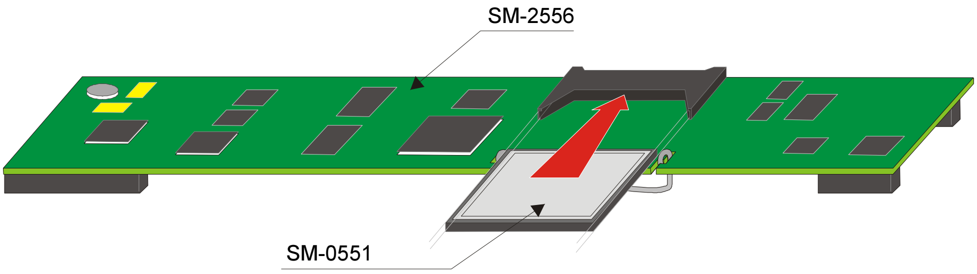
The following table shows which SIM may be installed on which module at which slot.

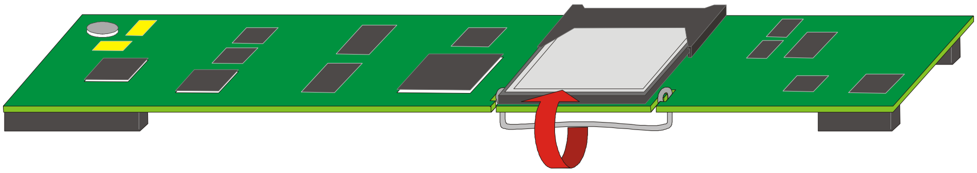
	CP-2010	CP-2012/CP-2017	
SIM	**SIM0**	**SIM0**	**SIM1**
SM-2551 Universal Serial Interface Processor (SIP)	✓	✓	✓
SM-2556 Network-Interf.Ethernet 10/100TX (NIP)	✓	✓	✓
SM-2545 Profibus Interface Processor (FIP)	✓	–	✓
SM-0551 Ser. Interface Prozessor 1 SS (SIP) [1]	✓ [1]	✓ [1]	✓ [1]

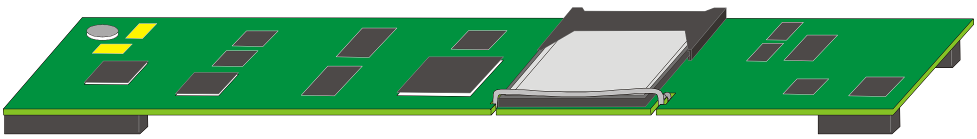
[1] SM-0551 must be mounted on SM-2556



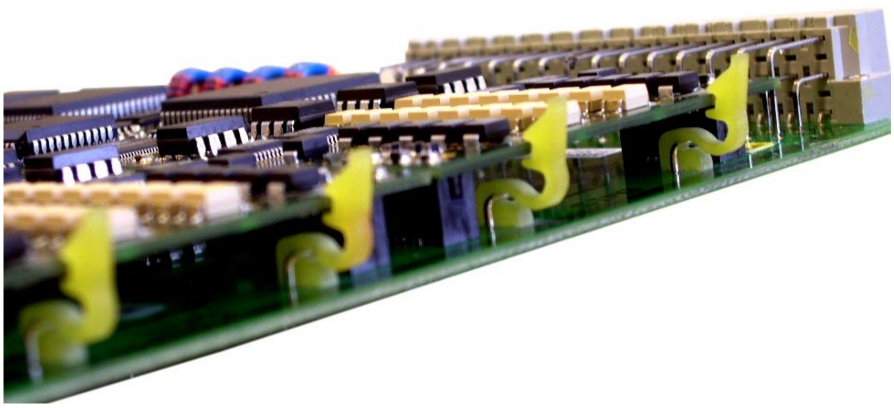
An exception is SM-0551. It is not mounted directly on a basic system element like the other SIM, it must be mounted on a SM-2556 before.





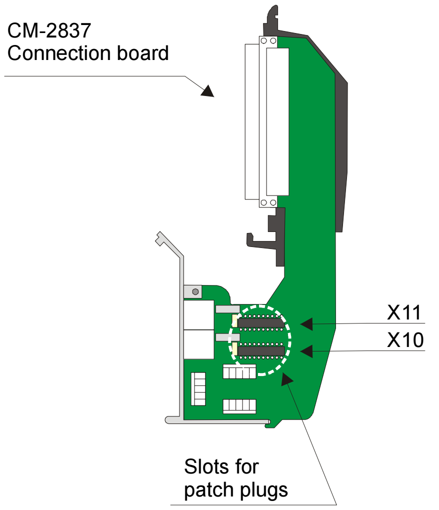


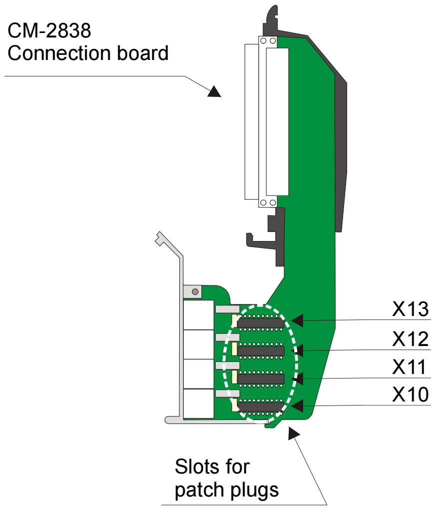
SIM and module are connected by means of 3 connectors and 2 retainer clips. The connectors serve the mechanical and electrical connection of both components. The clips provide an additional mechanical hold.



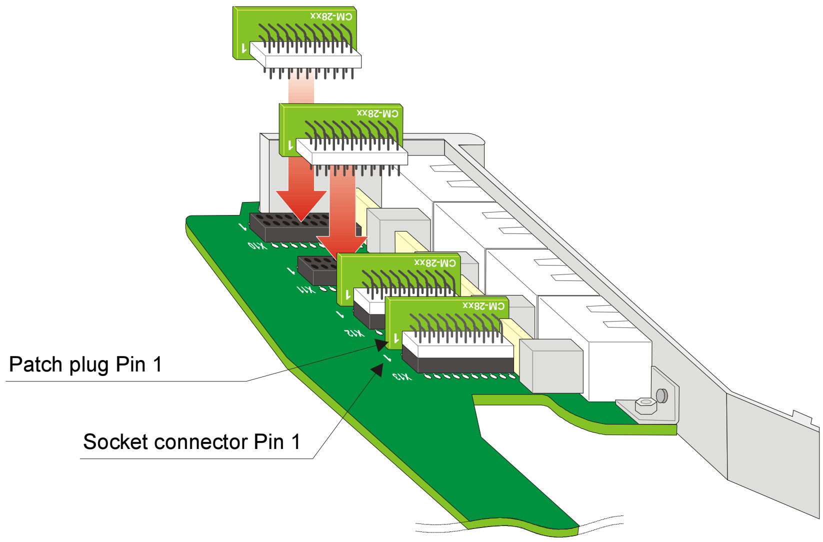
5.2.2. Fitting Connection Board with Patch Plugs

On the connection boards CM-2837 and CM-2838 there is a socket connector available for the Patch Plug for each of the interfaces provided. These Patch Plugs must be fitted before the connection boards are installed on the mounting rack.





It is to be observed thereby, that Pin 1 of the Patch Plug is plugged into the opening marked with 1 of the respective socket connector. In addition, components are positioned around the socket connector that prevent a distorted fitting of the patch plugs.



Attention

The connection of the signal lines for Watchdog, Error and Time Signal must also be carried out before the installation of the associated connection board (CM-2837). See section 7, *Wiring Watchdog and Error*, and section 6; *Wiring for the Reception of Time Signals*.

5.2.3. Installing Connection Board

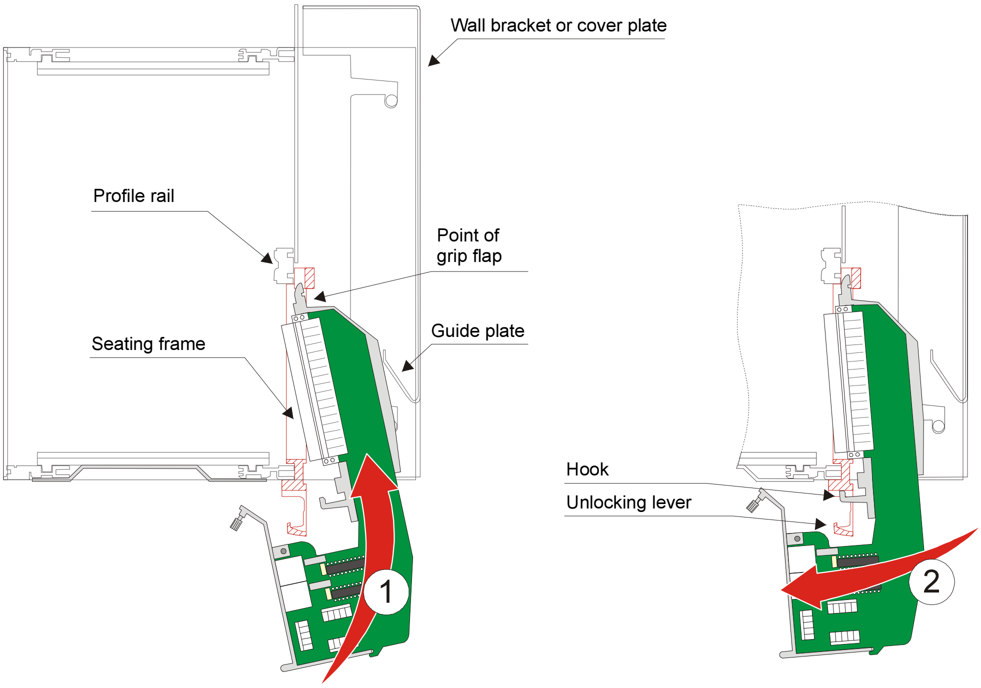
Attention

Master control elements and processing- and communication elements must **NOT** be installed in the mounting rack during the fitting of the associated connection boards..

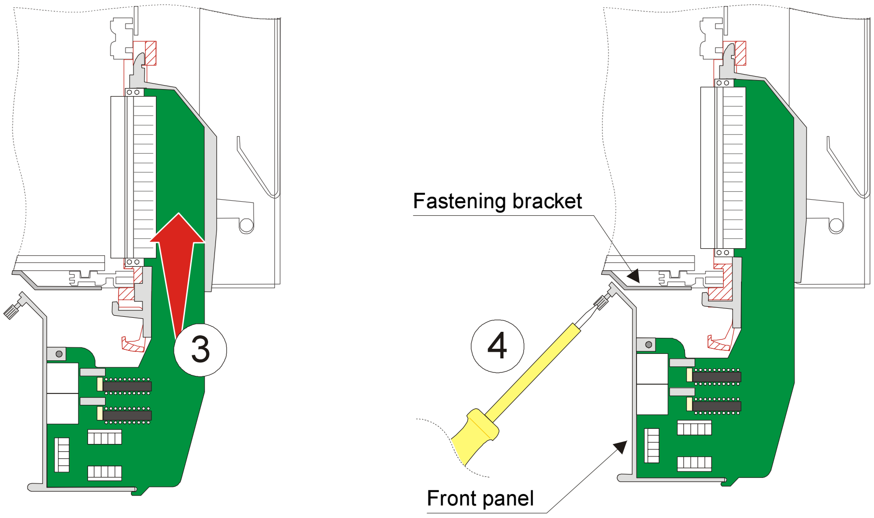
The fitting is carried out by engaging the connection board in the red seating frame and subsequent screwing with the fastening bracket on the underside of the mounting rack.

Proceed as follows:

- Insert the connection board into the mounting rack from below, until the point of the grip flap lies between profile rail and seating frame ①. (Wall bracket and cover plate support this activity with the help of an integrated guide plate.)
- Then swing the lower part of the connection board forwards ②. Thereby the hook of the grip flap must be guided through the unlocking lever.



- Now push the connection board upwards ③. Thereby grip flap and seating frame are finally fixed together.
- Subsequently the front panel of the connection board is screwed with the fastening bracket on the underside of the mounting rack ④.

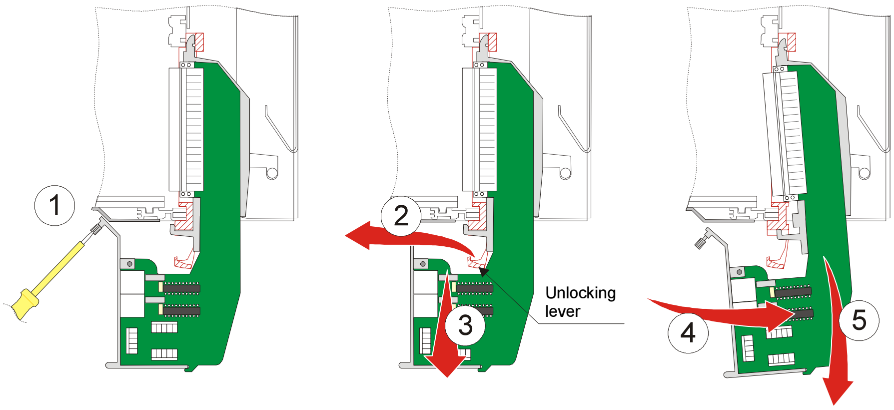


5.2.4. Removing Connection Board

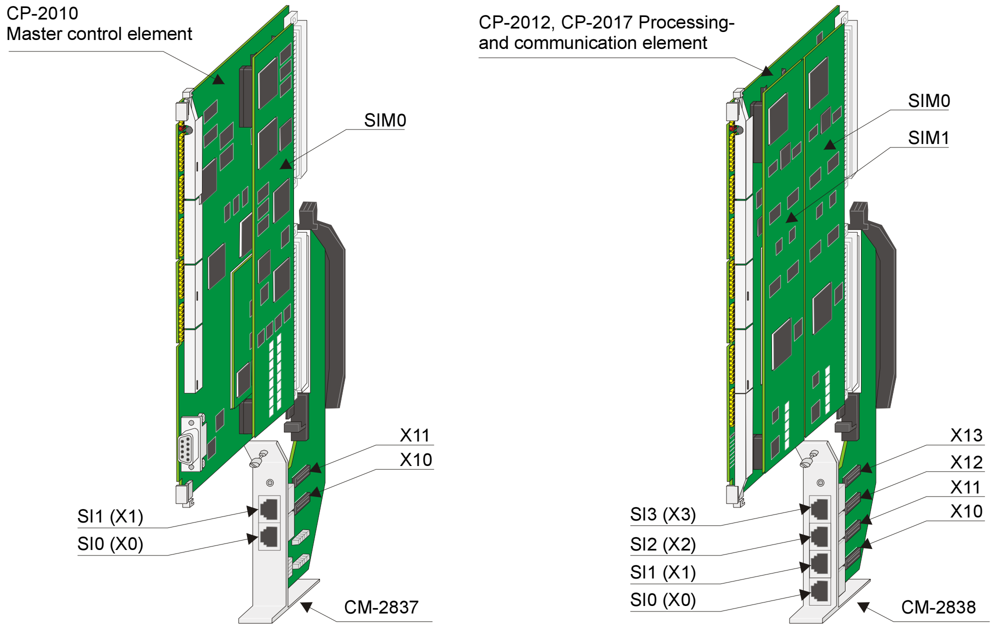
Attention

Master control elements and processing- and communication elements must **NOT** be installed in the mounting rack during the fitting of the associated connection boards.

- Open the screw connection between front panel and fastening bracket ①.
- Push the unlocking lever of the seating frame slightly forwards and hold it in this position ②.
- Pull the connection board vertically 4-5mm downwards ③.
- Swing the connection board to the back ④ (the hook of the grip flap is thereby guided through the unlocking lever) and pull it downwards out of the mounting rack ⑤.



5.2.5. Configuration Notes



Depending on the type of SIM used and the slot in which this is installed, there are interfaces available on the associated connection board. Which of these interfaces is assigned to which SIM and in which slot the associated patch plug must be plugged in is shown in the following tables:

Master control element CP-2010 with Connection Board CM-2837

SIM-Type	Slot	Interfaces on CM-2837				Slots for Patch Plugs on CM-2837			
SM-2551 (SIP)	SIM0	SI0	SI1	–	–	X10	X11	–	–
SM-2556 (NIP)	SIM0	–	SI1	–	–	–	X11	–	–
SM-2545 (FIP)	SIM0	SI0	–	–	–	X10	–	–	–
SM-0551 (SIP) 1)	SIM0 1)	SI0	–	–	–	X10	–	–	–

1) SM-0551 must be mounted on SM-2556

Processing- and communication element CP-2012 or CP-2017 with Connection Board CM-2838

SIM-Type	Slot	Interfaces on CM-2838				Slots for Patch Plugs on CM-2838			
SM-2551 (SIP)	SIM0	SI0	SI1	–	–	X10	X11	–	–
SM-2556 (NIP)	SIM0	–	SI1	–	–	–	X11	–	–
SM-0551 (SIP) 1)	SIM0 1)	SI0	–	–	–	X10	–	–	–
SM-2551 (SIP)	SIM1	–	–	SI2	SI3	–	–	X12	X13
SM-2556 (NIP)	SIM1	–	–	–	SI3	–	–	–	X13
SM-2545 (FIP)	SIM1	–	–	SI2	–	–	–	X12	–
SM-0551 (SIP) 1)	SIM1 1)	–	–	SI2	–	–	–	X12	–

1) SM-0551 must be mounted on SM-2556

5.3. Communication Cabling

This chapter describes how the various methods of communication (serial, ethernet, field bus) can be realized by means of standard modems and cables.

Following kinds of communication are shown:

- Serial Communication
  - Point-to-Point Traffic / Multi-Point Traffic (CM-070x)
  - Multi-Point Traffic via Glass Fibre Optic and Star Connection (CM-0827)
  - Analog Dial-up Traffic
  - ISDN Dial-up Traffic
  - GSM Dial-up Traffic
  - Serial Communication with DMS
- LAN Communication
- Field Bus Communication

The mounting racks and their configuration shown in the pictures are examples. They are used in order to show which connection options are available with corresponding configuration.

Hint

Communication cables are, if possible, to be installed separately from the supply and peripheral cables.

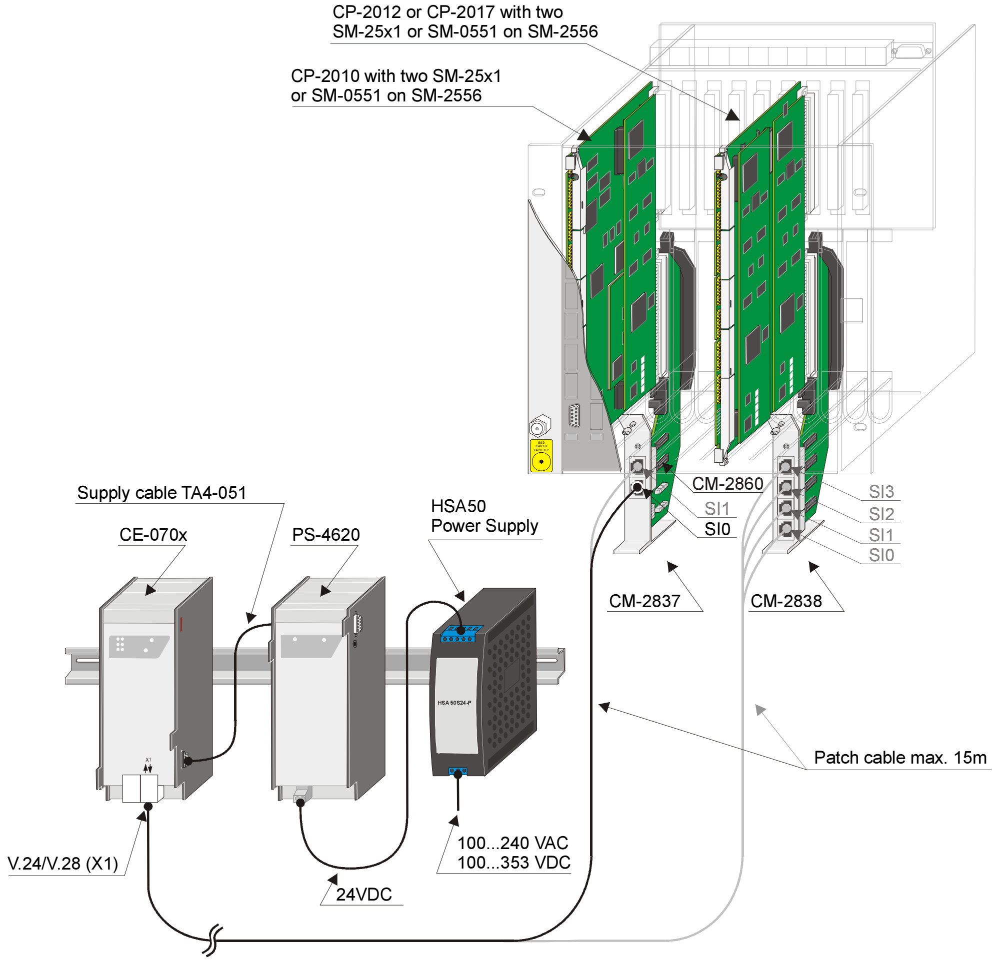
5.3.1. Serial Communication

5.3.1.1. Point-to-Point Traffic / Multi-Point Traffic

Standard Modem and Cable for Mulit-Point- and Point-to-Point Traffic

	Designation	Item Number/ MLFB
	CE-0700 V.23 dedicated line modem f. 1703	G21-200 6MF11020BC000AA0
	CE-0701 WT channel modem f. DIN-rail	G22-081 6MF11020CA810AA0
	Patch Cable CAT5 (4x2) AWG26/7	T41-255 (1m) 6MF13040BC550AA0 T41-251 (2m) 6MF13040BC510AA0 T41-252 (3m) 6MF13040BC520AA0 T41-253 (5m) 6MF13040BC530AA0 T41-254 (10m) 6MF13040BC540AA0

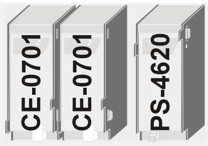
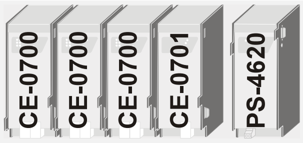
5.3.1.1.1. Cabling with external power supply of the modems



This kind of power supply offers the possibility to supply several modems (CE-0700 and/or CE-0701).

Which variants are possible can be seen in table *Configuration Variants*.

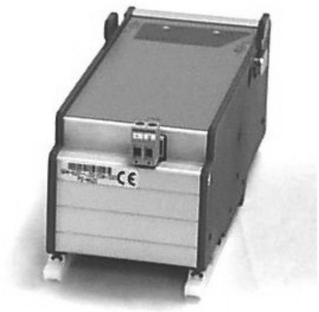
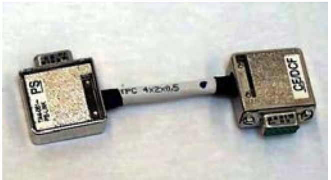
Configuration Variants

	Comment	<-- Configuration succession
A	Maximal number of CE-0700 which can be supplied	
B	Maximal number of CE-0701 which can be supplied	
C	One CE-0701 and maximal number of CE-0700	
D	One DCF77 and maximal number of CE-0700 which can be supplied	
E	One DCF77 and maximal number of CE-0701 which can be supplied	

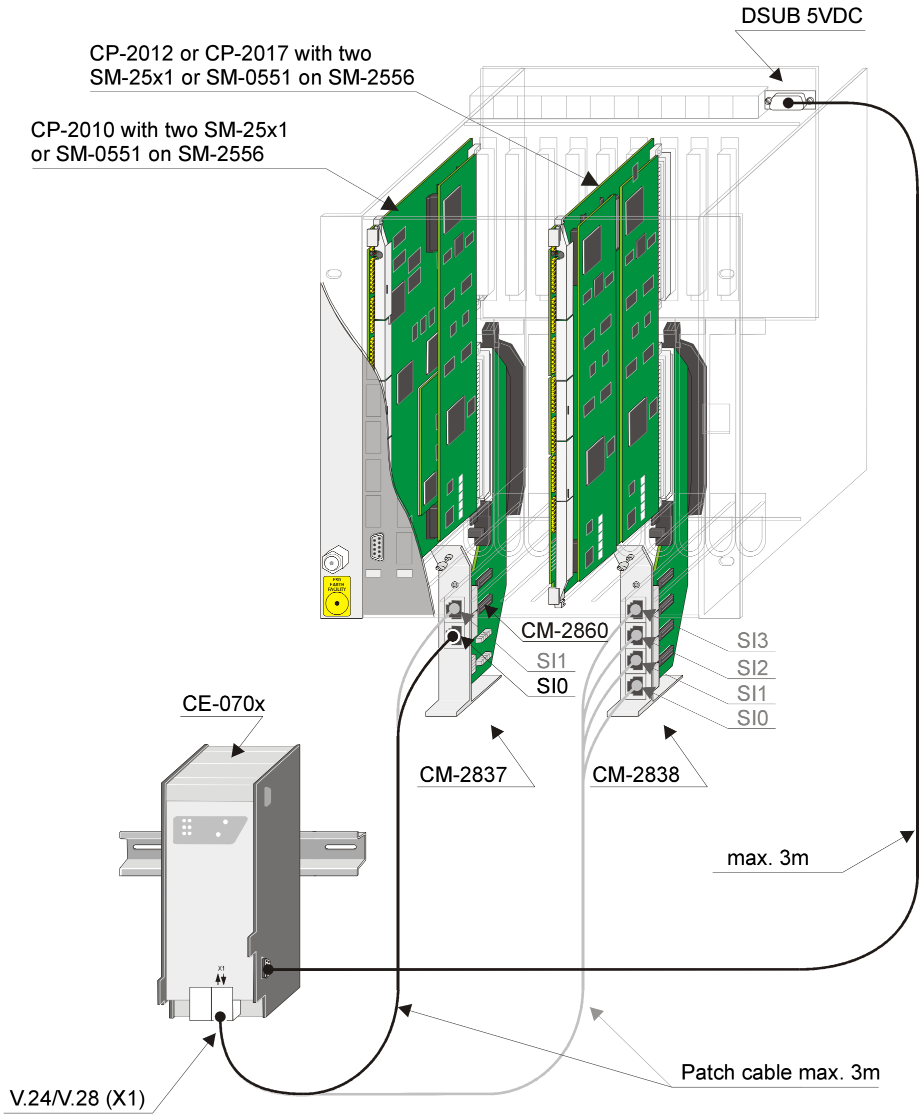
Attention

The succession of the modules, from right to left, must be absolutely followed during configuration.

Accessoires for external Power Supply of Modems

	Designation	Item Number/ MLFB
	PS-4620 Additional Power Supply 24-60VDC	GA4-620 6MF11110EG200AA0
	Power supply cable for modem and DCF77 receiver	TA4-051 6MF13110EA510AA0
	Power S. 115/230VAC,100-375VDC-24VDC-50W	www.meinberg.de

5.3.1.1.2. Cabling with internal power supply of the modems



With this kind of power supply either one CE-0701 or two CE-0700 can be supplied.

Accessoires for internal Power Supply of Modems

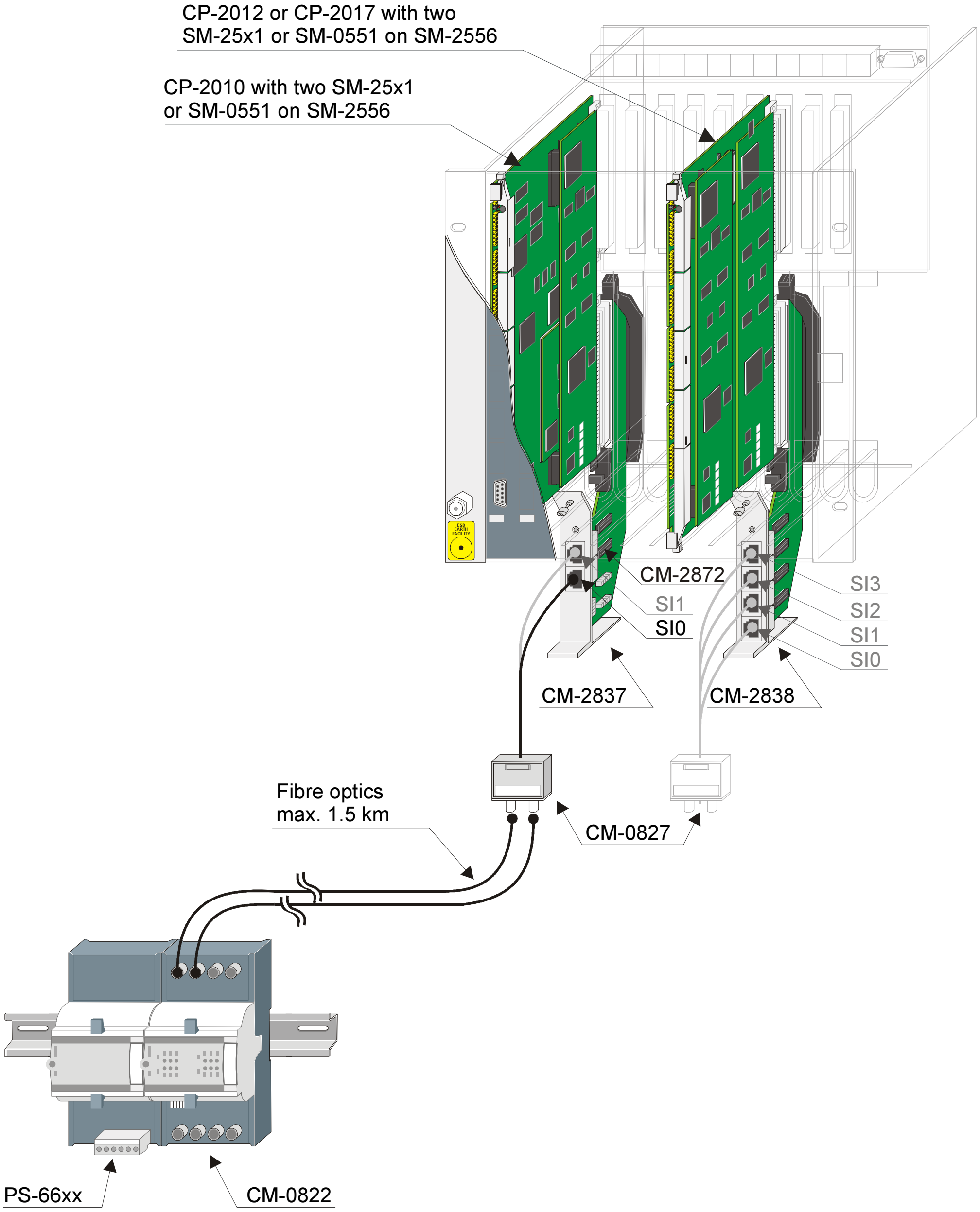
Designation	Item Number/ MLFB
Cable, 3m, 2xDSUB-9pol, 1:1 (Attention, modification required)	T41-289 6MF13042BC800AA0
D-SUB Socket Connector , 9-pole, solder version	SAT105104
D-SUB Housing 9-pole	TF0-004 6MF13140AA040AA0

Attention

The cable used for the internal power supply (T41-289) must be modified for this use. The female D-SUB plug must be pinched off and be repaced by a new, which is wired according the following cable circuitry.

Cable Circuitry of Power Supply Cable: AK 1703 ACP – Dedicated Line Modem/Channel Modem

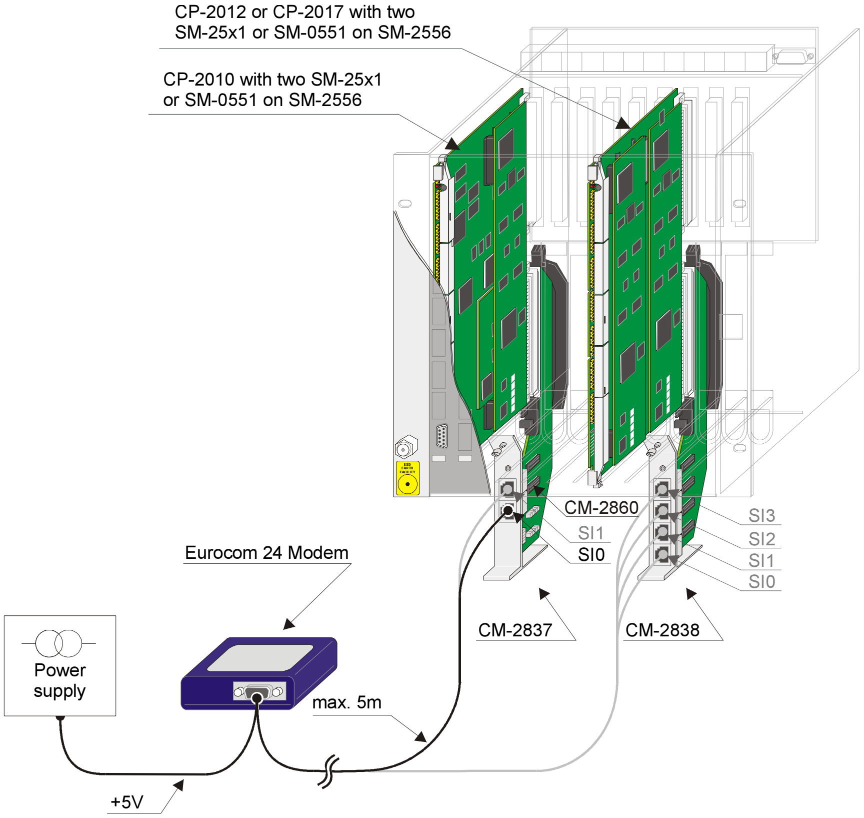
5.3.1.2. Multi-Point Traffic via Glass Fibre Optic and Star Connection



	Designation	Item Number/ MLFB
	CM-0827 - Fiberoptik-Interface (el.-LWL)	GA0-827 6MF11110AJ270AA0
Fibre optics length: max. 0,5km	FO-**INDOOR**CABLE-50-DUP-LSOH FO-**OUTDOOR**CABLE-50-2FIB-ARM FO-CONNECTOR-ST-50/62	TF7-027 6MF13140HA270AA0 TF7-028 6MF13140HA280AA0 TF7-016 6MF13140HA160AA0
Fibre optics length: max. 1,5km	FO-**INDOOR**CABLE-62-DUP-LSOH FO-**OUTDOOR**CABLE-62-2FIB-ARM FO-CONNECTOR-ST-50/62	TF7-006 6MF13140HA060AA0 TF7-007 6MF13140HA070AA0 TF7-016 6MF13140HA160AA0

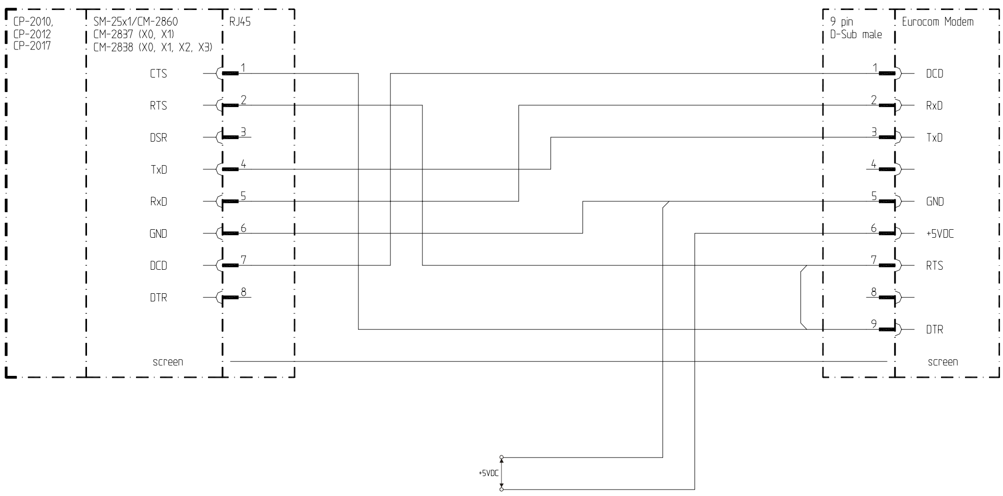
5.3.1.3. Analog Dial-up Traffic

5.3.1.3.1. Eurocom 24 Modem with external Power Supply

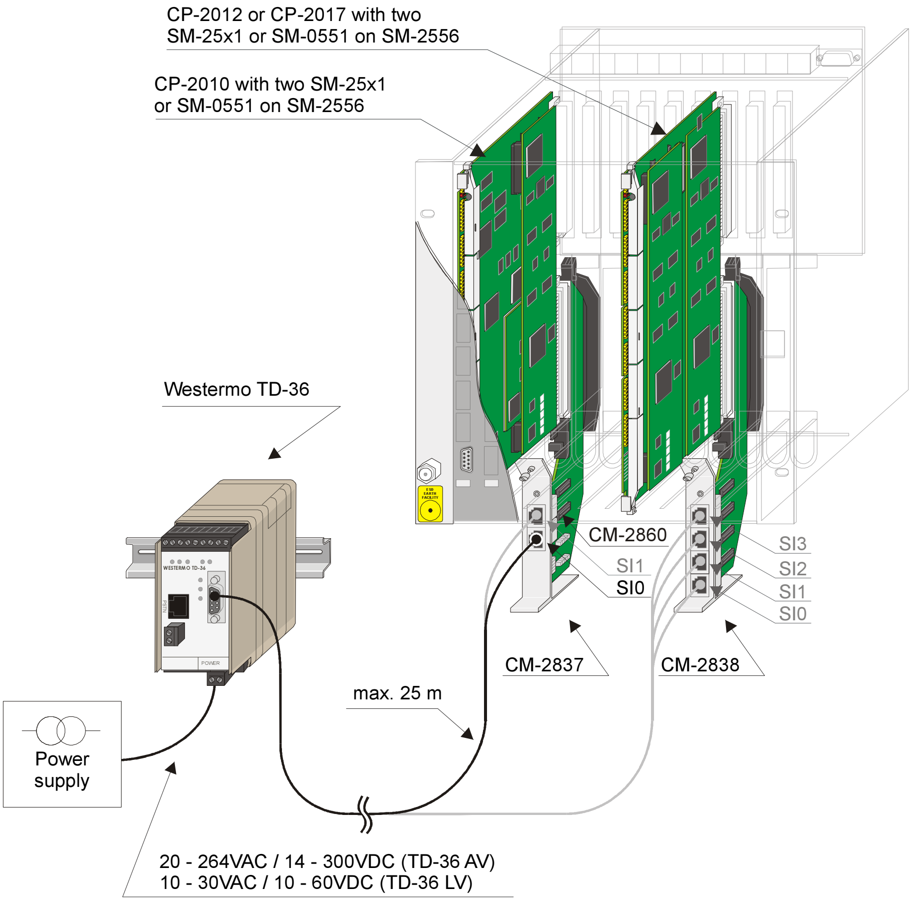


Accessories see Appendix A, Recommended and tested purchased products

Cable Circuitry: Connection Board – Eurocom 24 Modem

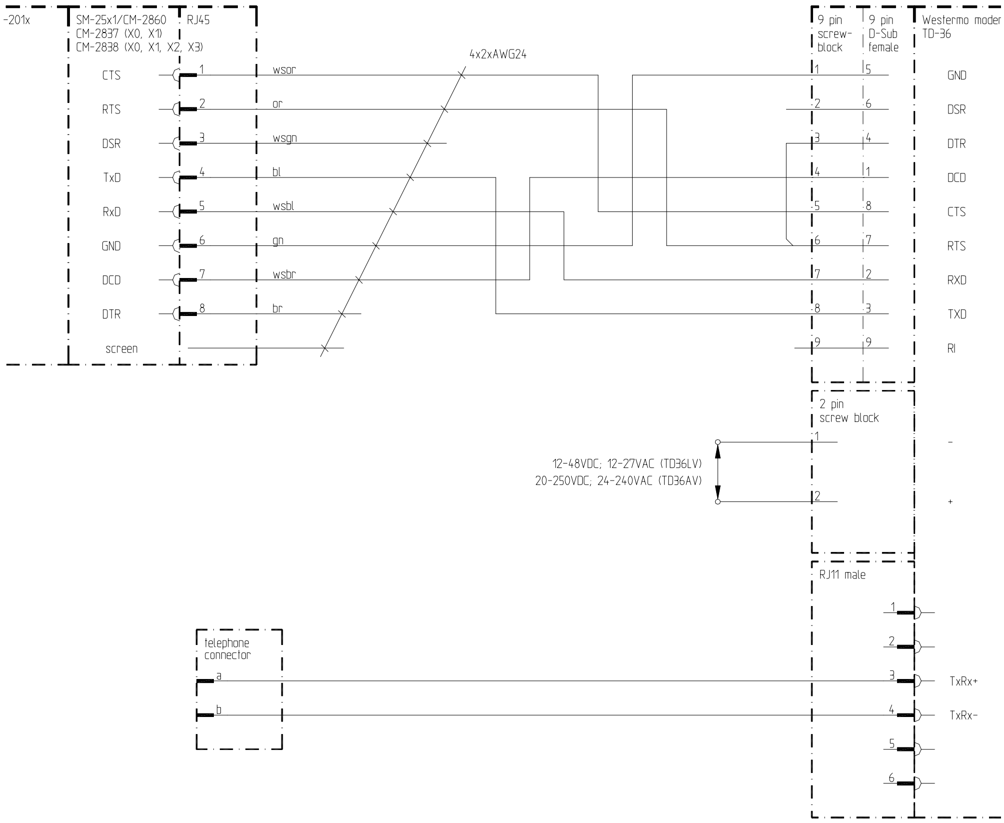


5.3.1.3.2. Westermo TD36 Modem with external Power Supply

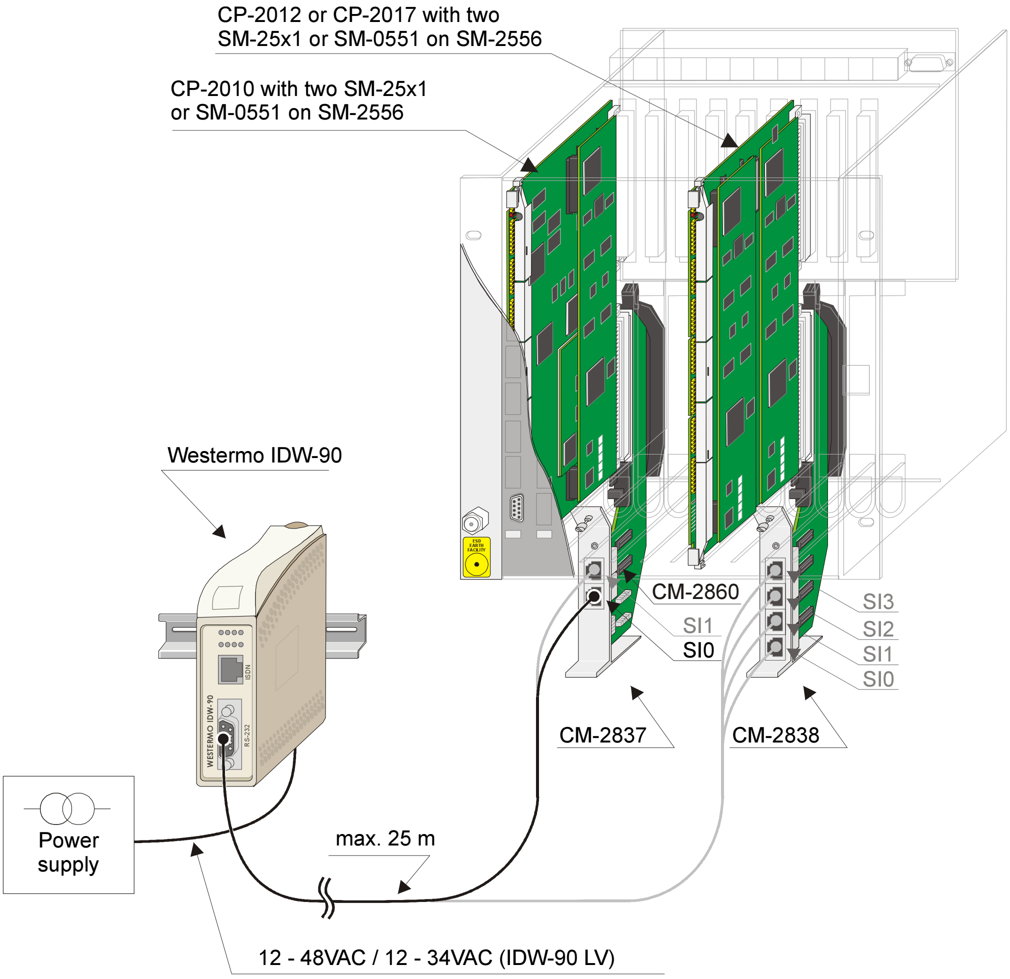


Accessories see Appendix A, Recommended and tested purchased products

Cable Circuitry: Connection Board – Westermo TD-36

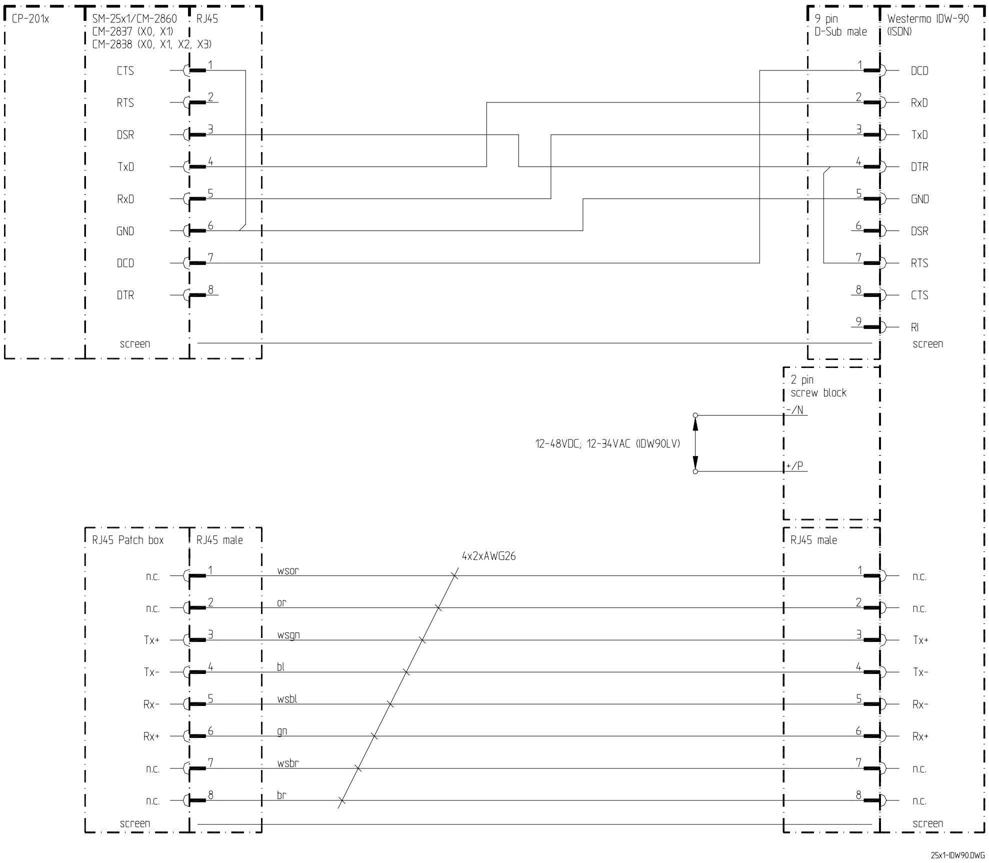


5.3.1.4. Dial-up Traffic Analog/GSM/ISDN

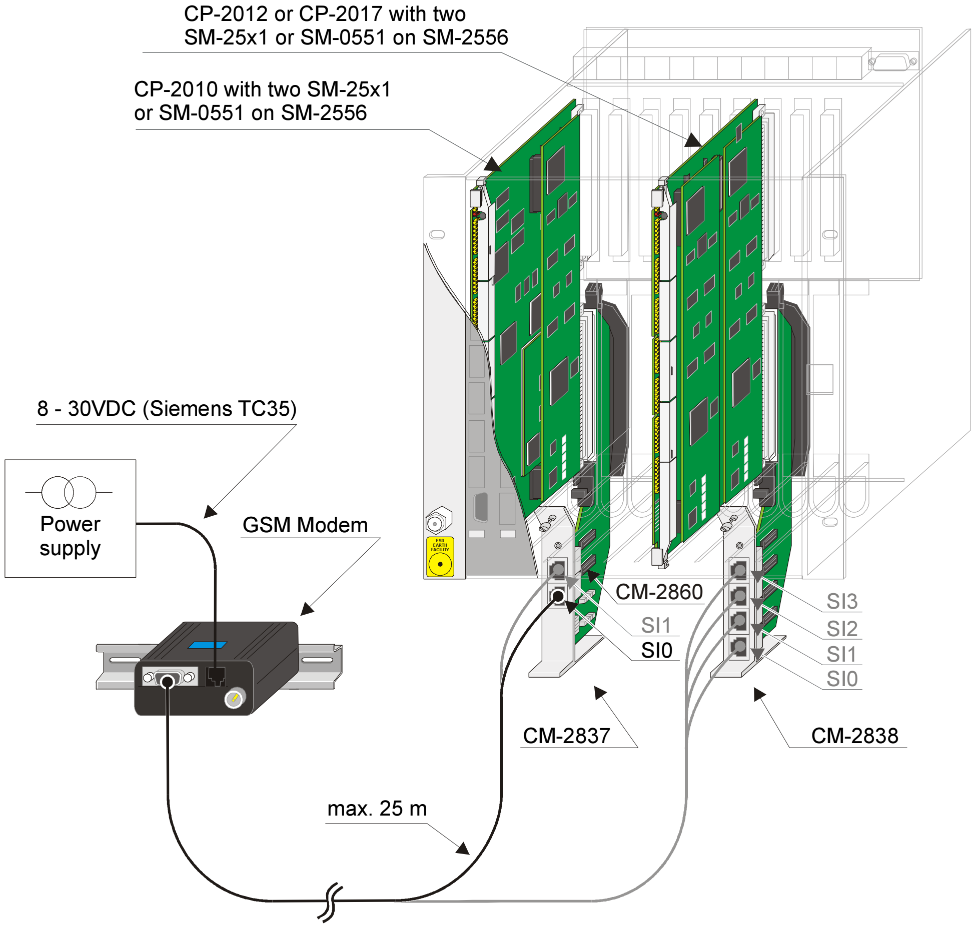


Accessories see Appendix A, Recommended and tested purchased products

Cable Circuitry Connection Board – ISDN Modem



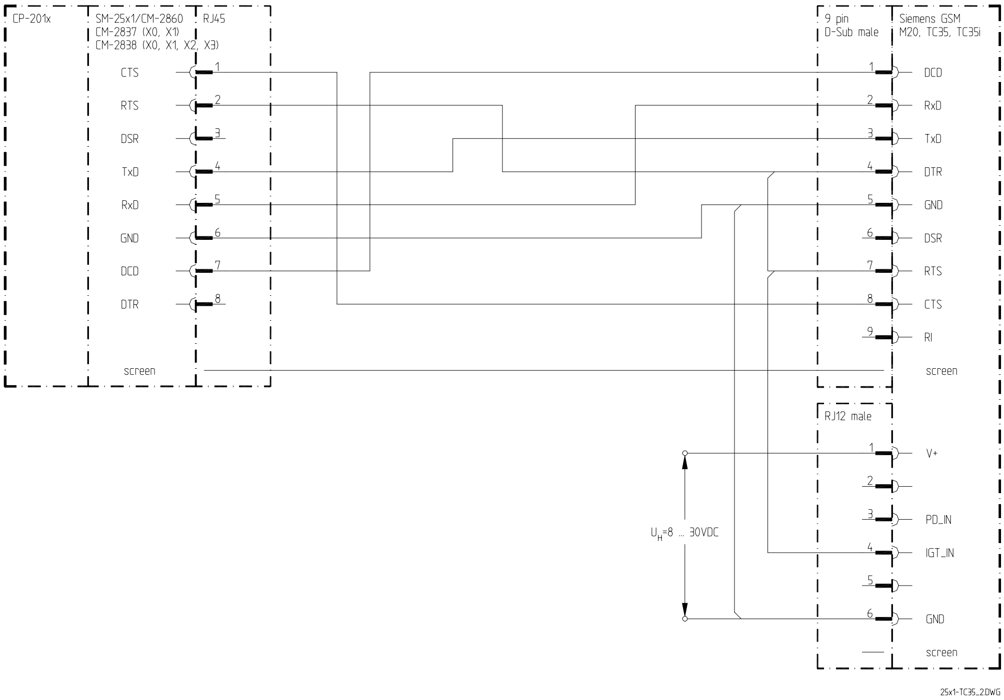
5.3.1.5. GSM Dial-up Traffic



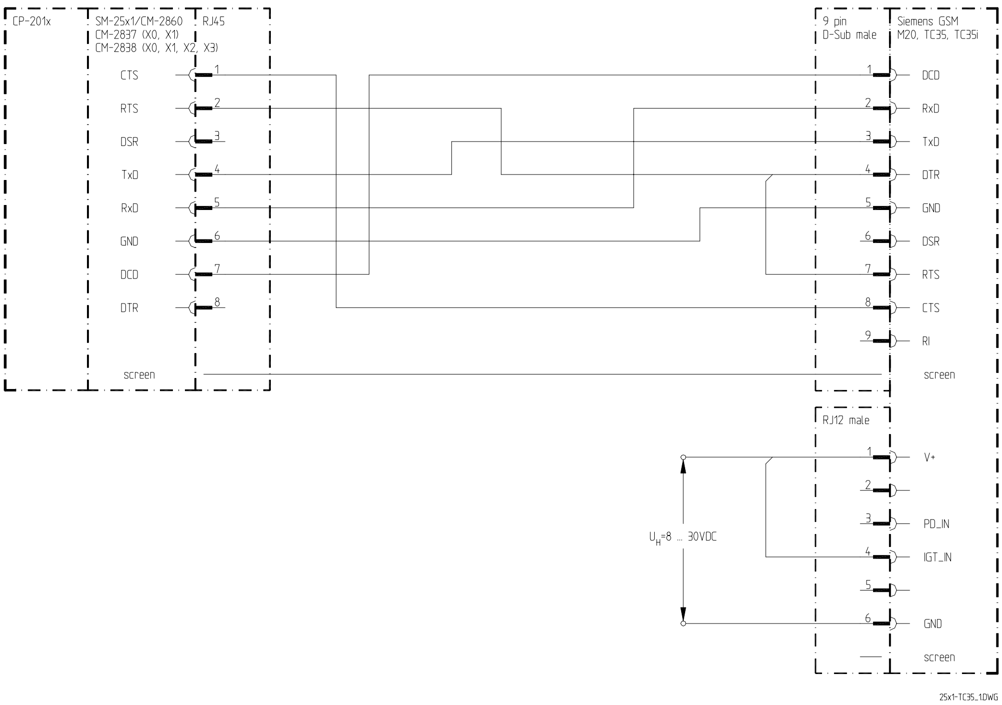
Accessories see Appendix A, Recommended and tested purchased products

Cable Circuitry Connection Board – GSM Modem

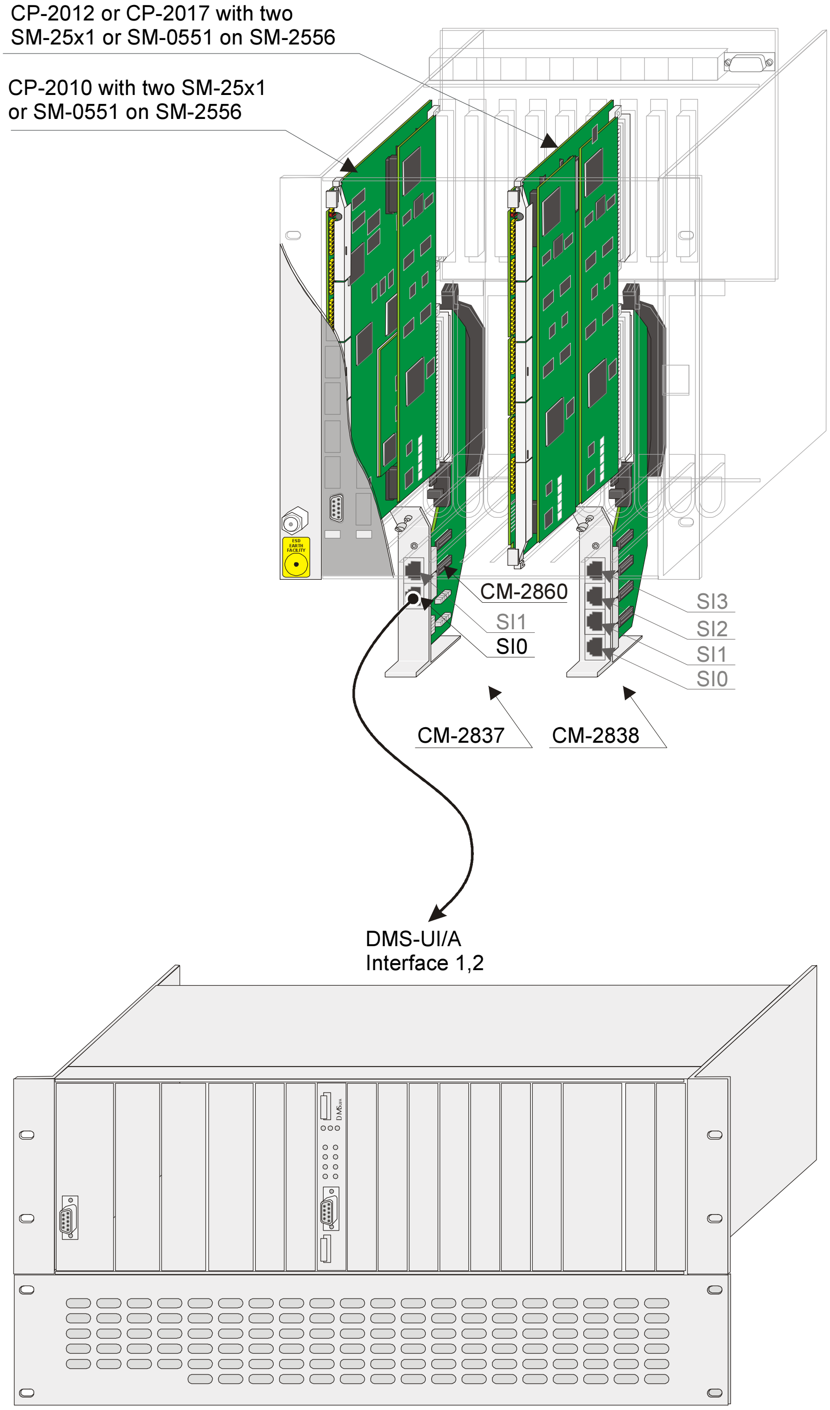
Variant for the deactivation function of the TC35 GSM modem controlled by ACP 1703.



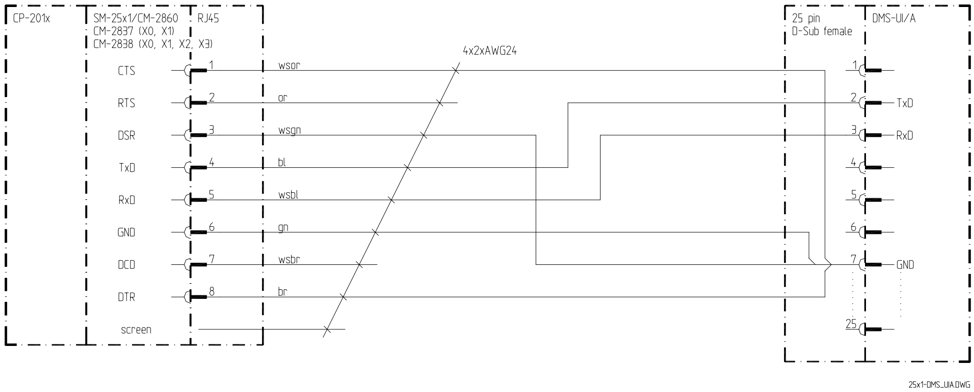
Variant without the deactivation function of the TC35 GSM modem controlled by ACP 1703.



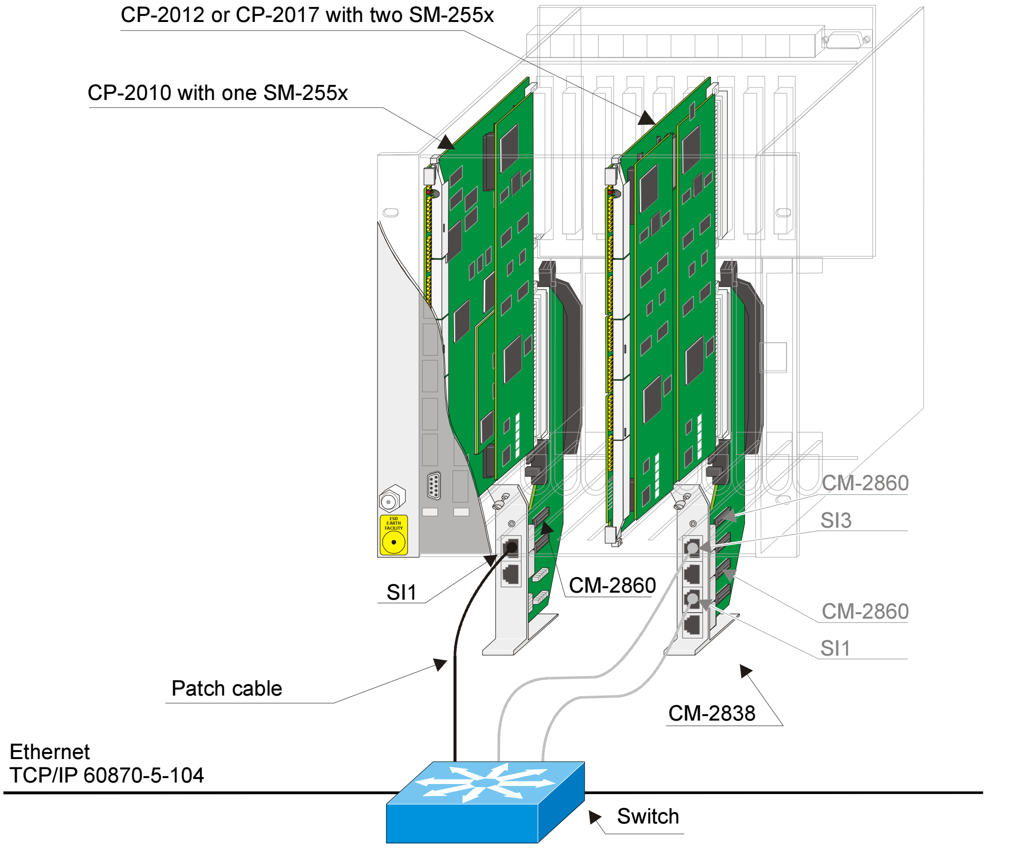
5.3.1.6. Serial communication with DMS (Digitales Multiplexer-System)



Cable Circuitry Connection Board – DMS

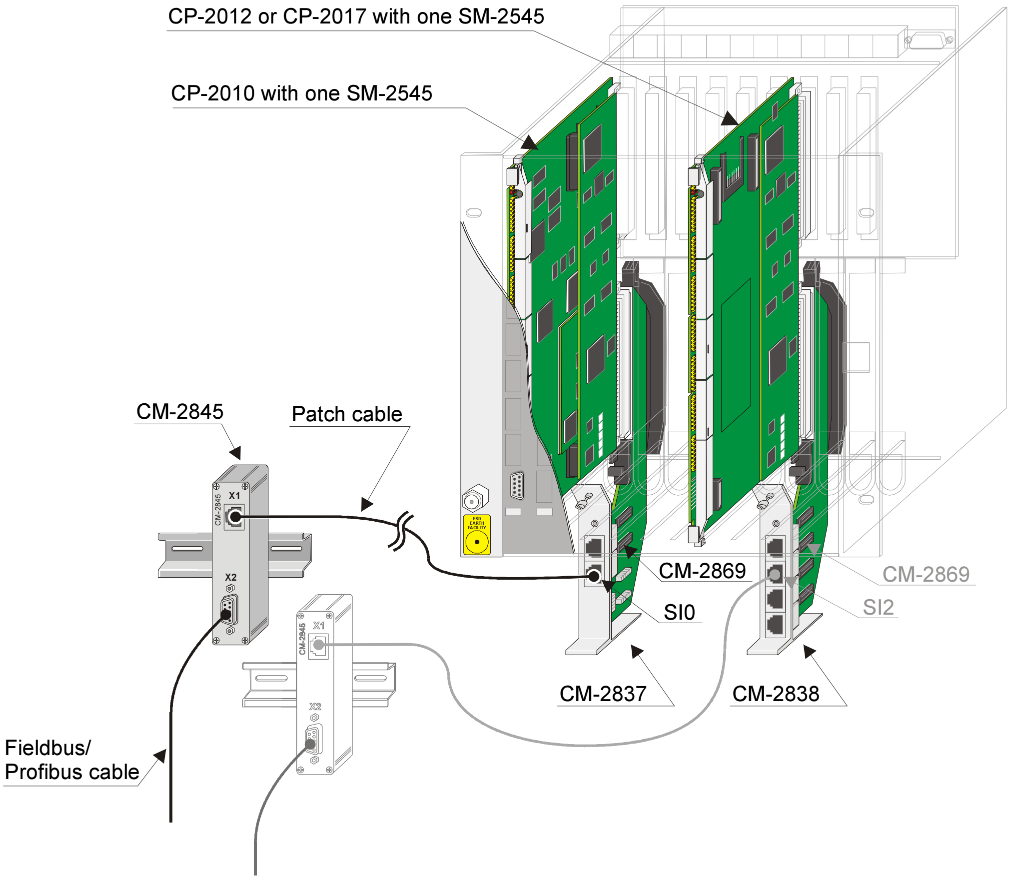


5.3.2. Wiring LAN Communication (Ethernet TCP/IP)



	Designation	Item-Number / MLFB
	Patch cable CAT5 (4x2) AWG26/7 (used for cabinet-<u>internal</u> wirings up to 10m)	T41-255 (1m) 6MF13040BC550AA0 T41-251 (2m) 6MF13040BC510AA0 T41-252 (3m) 6MF13040BC520AA0 T41-253 (5m) 6MF13040BC530AA0 T41-254 (10m) 6MF13040BC540AA0
	Patch cable CAT5 (4x2x0.5) AWG24 (used for cabinet-<u>external</u> wirings up to 100m) Details for connections over 10m can be taken from document: *Configuration Automation Units and Automation Networks* (DC2-021); Appendix A; Chapter: Electrical Connection, Cable longer than 10 m.	TF5-101 6MF13140FB010AA0 TF5-200 (1.5m) 6MF13140FC000AA0 TF5-201 (3m) 6MF13140FC010AA0 TF5-202 (5m) 6MF13140FC020AA0

5.3.3. Wiring Fieldbus/Profibus Communication



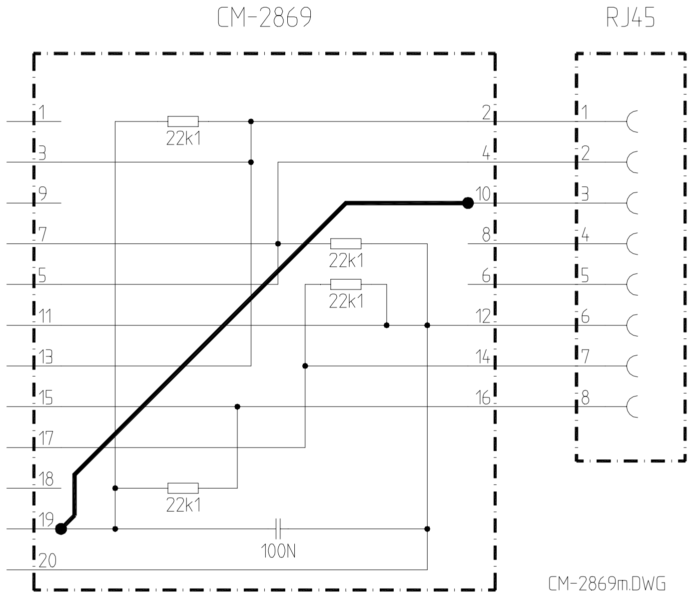
	Designation	Item-Number / MLFB
	Connectionbox for SM-2545 in AK	GA2-845 6MF11110CJ450AA0
	Patch Cable CAT5 (4x2) AWG26/7 (used for cabinet-internal wirings up to 10m)	T41-255 (1m) 6MF13040BC550AA0 T41-251 (2m) 6MF13040BC510AA0 T41-252 (3m) 6MF13040BC520AA0 T41-253 (5m) 6MF13040BC530AA0 T41-254 (10m) 6MF13040BC540AA0

Fieldbus/Profibus Patch Plug

Patch Plug CM-2869 is used for fieldbus/profibus communikation. It is enclosed with SIM SM-2545.

Designation	Item-Number
CM-2869 Patch Plug SMI-RS485	CA2-869-B 6MF12112CJ600AC0

The latest version of this patch plug, item number CA2-869-B, is already modified according to following illustration (thick line). This modification must be added on older versions of the patch plug.

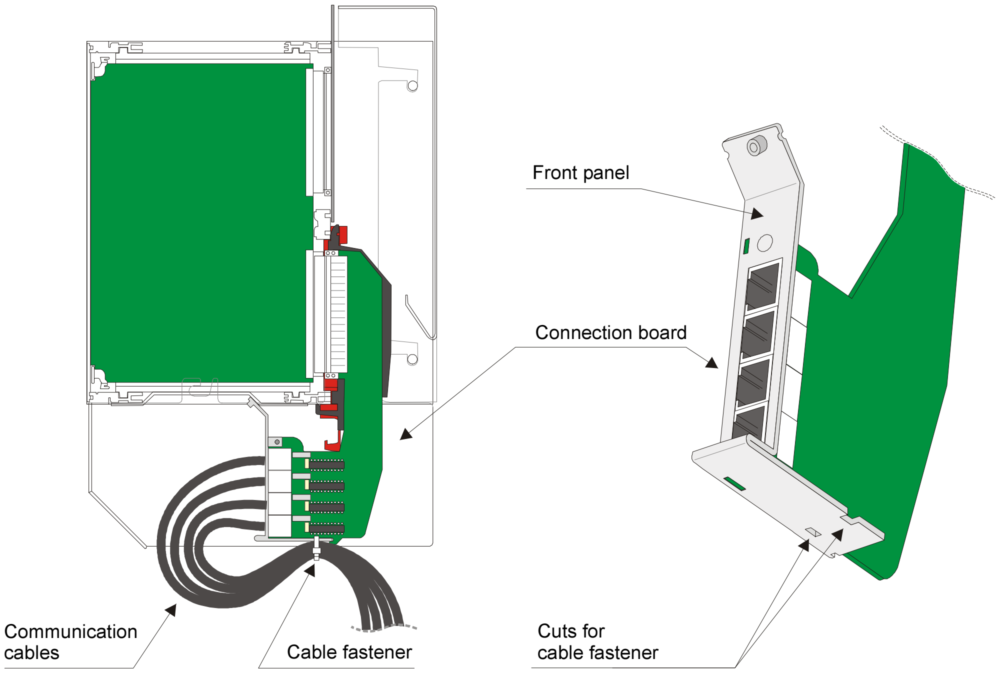


5.4. Cable Routing

It is recommended to fasten the communication cables to the connection board by means of a cable fastener.

The cable fastener must be led through the cuts at the bottom side of the front panel of the connection board.

The cable fastener (Becom NA65340; Panduit PLT 1,5M) is enclosed to the connection board.



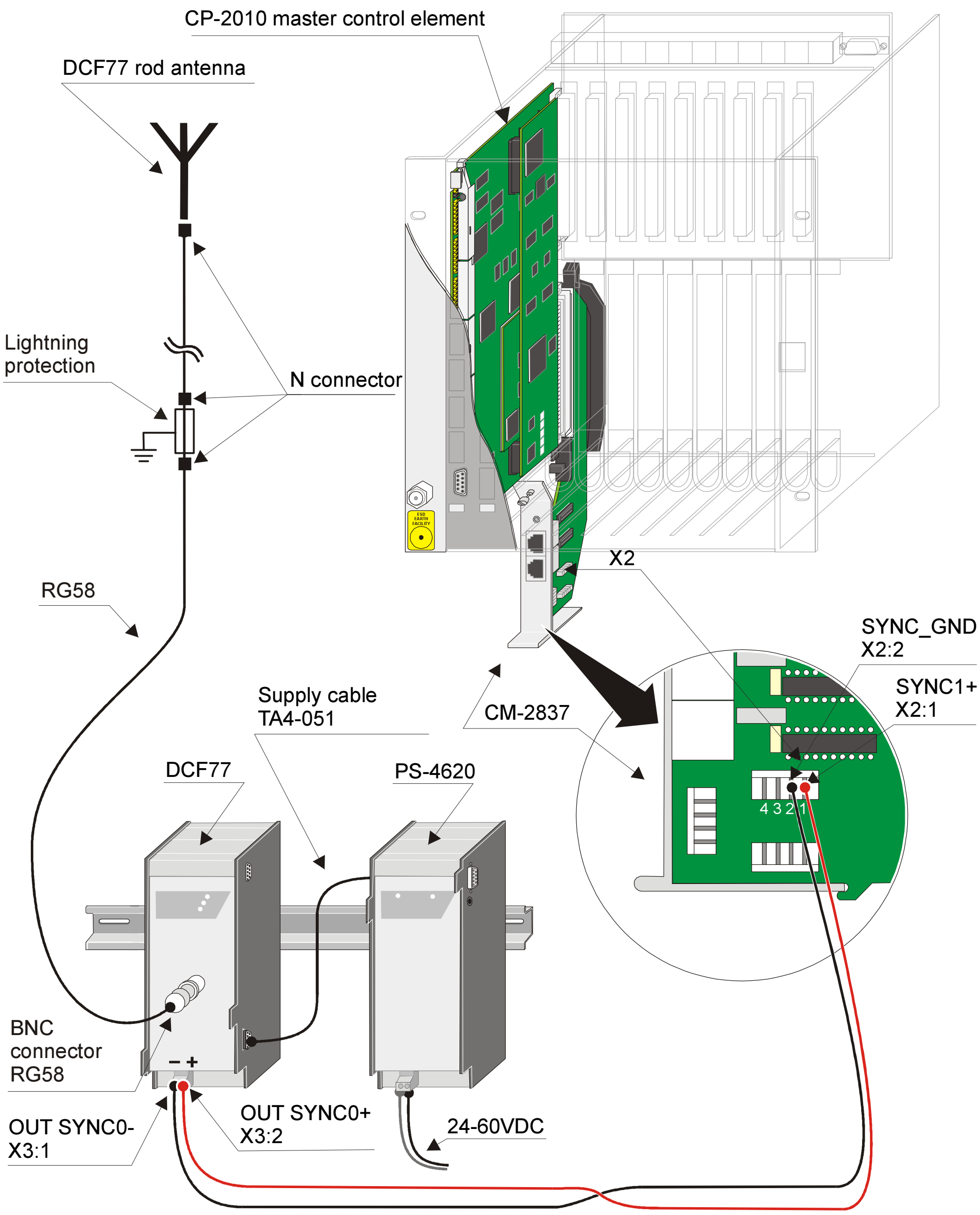
6. Wiring for the Reception of Time Signals

Content

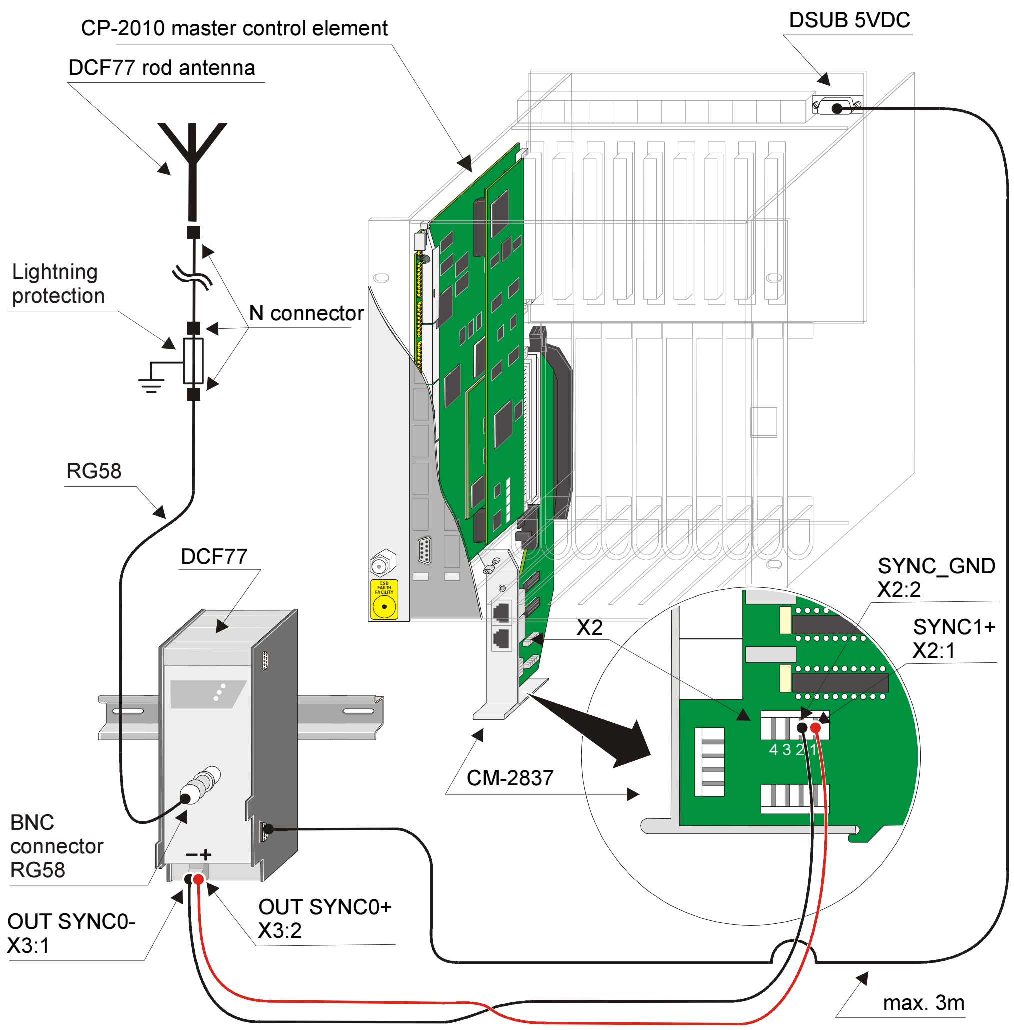
6.1. DCF77 Receiver

On the DCF77 receiver there is a 2-pole screw terminal (X11 SYNC) available over which the minute pulse or a serial time signal, adjustable via jumper, can be transmitted.

Wiring with external power supply of the receiver



Wiring with internal power supply of the receiver



Standard DCF77 Receiver

	Designation	Item-Number / MLFB
	DCF77- Receiver for DIN rail	GA0-806 6MF11110AJ060AA0
	KR-DCF/STAA DCF77 Stand. Outdoor Antenna	G12-107 6MF11010CB070AA0
	KR-DCF/V4 DCF77 Antenna Terminal Block	G12-106 6MF11010CB060AA0
	Antenna cable (Coaxial cable RG58 C/U 50 Ohm)	TF5-050 6MF13140FA500AA0
	BNC-Plug RG58	

Accessoires for external Power Supply of Modems

	PS-4620 Additional Power Supply 24-60VDC	GA4-620 6MF11110EG200AA0
	Supply cable for Modem	TA4-051 6MF13110EA510AA0

Accessoires for internal Power Supply of Modems

Designation	Item-Number / MLFB
Cable, 3m, 2xDSUB-9pol, 1:1 (Attention, modification required)	T41-289 6MF13042BC800AA0
D-SUB Socket Connector , 9-pole, solder version	SAT105104
D-SUB Housing 9-pole	TF0-004 6MF13140AA040AA0

Attention

The cable used for the internal power supply (T41-289) must be modified for this use. The female D-SUB plug must be pinched off and be repaced by a new, which is wired according the following cable circuitry.

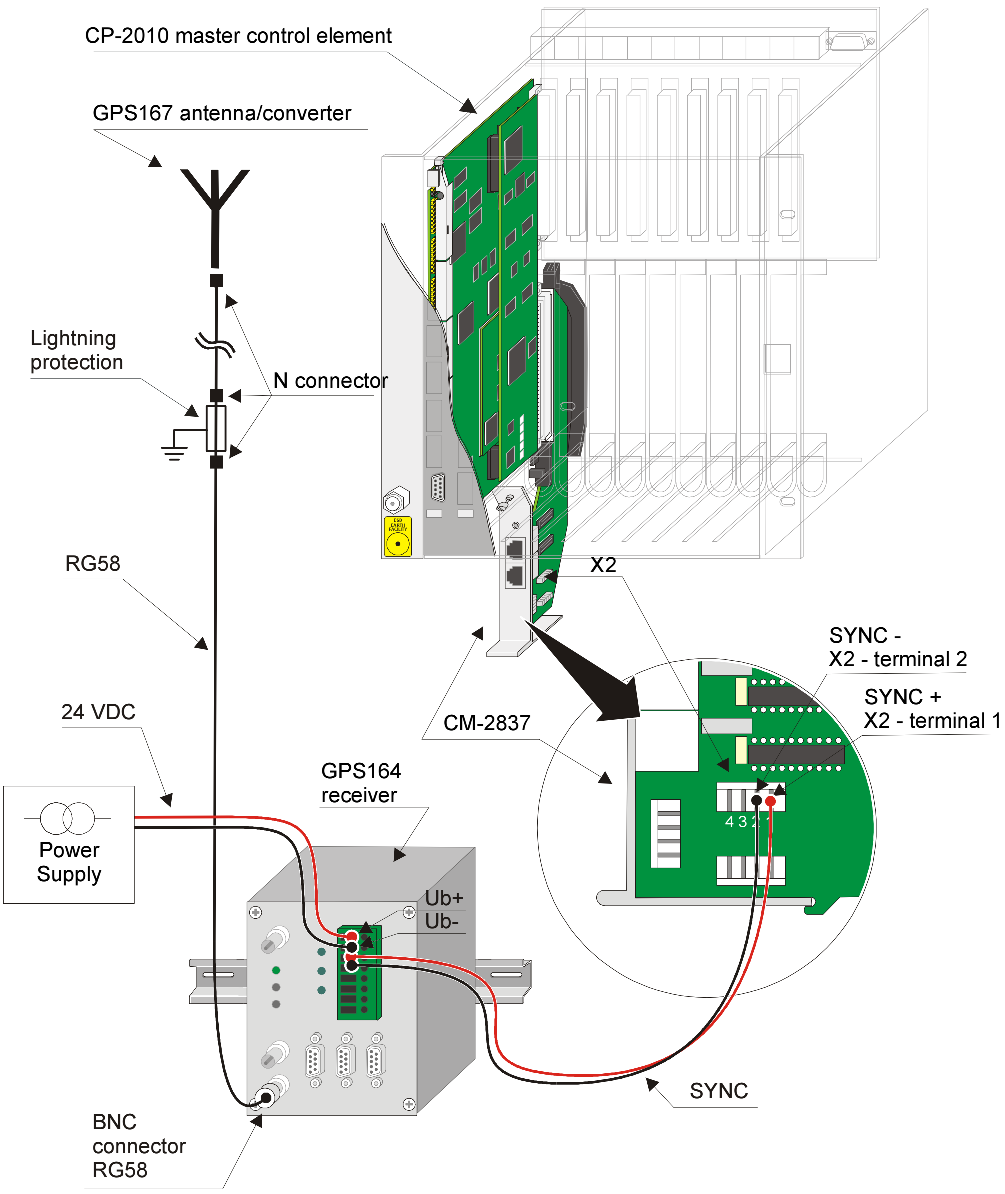
Cable Circuitry of Power Supply Cable: AK 1703 ACP – DCF77 Receiver

6.2. GPS Satellite Receiver

The GPS164 satellite receiver offers a minute pulse and a serial time signal. The settings are made with software GPSMON32. This software is delivered with the GPS164 receiver.

6.2.1. Wiring for the receipt of the minute pulse

The minute pulse is supplied on the screw terminal on the front panel of the GPS163 receiver.



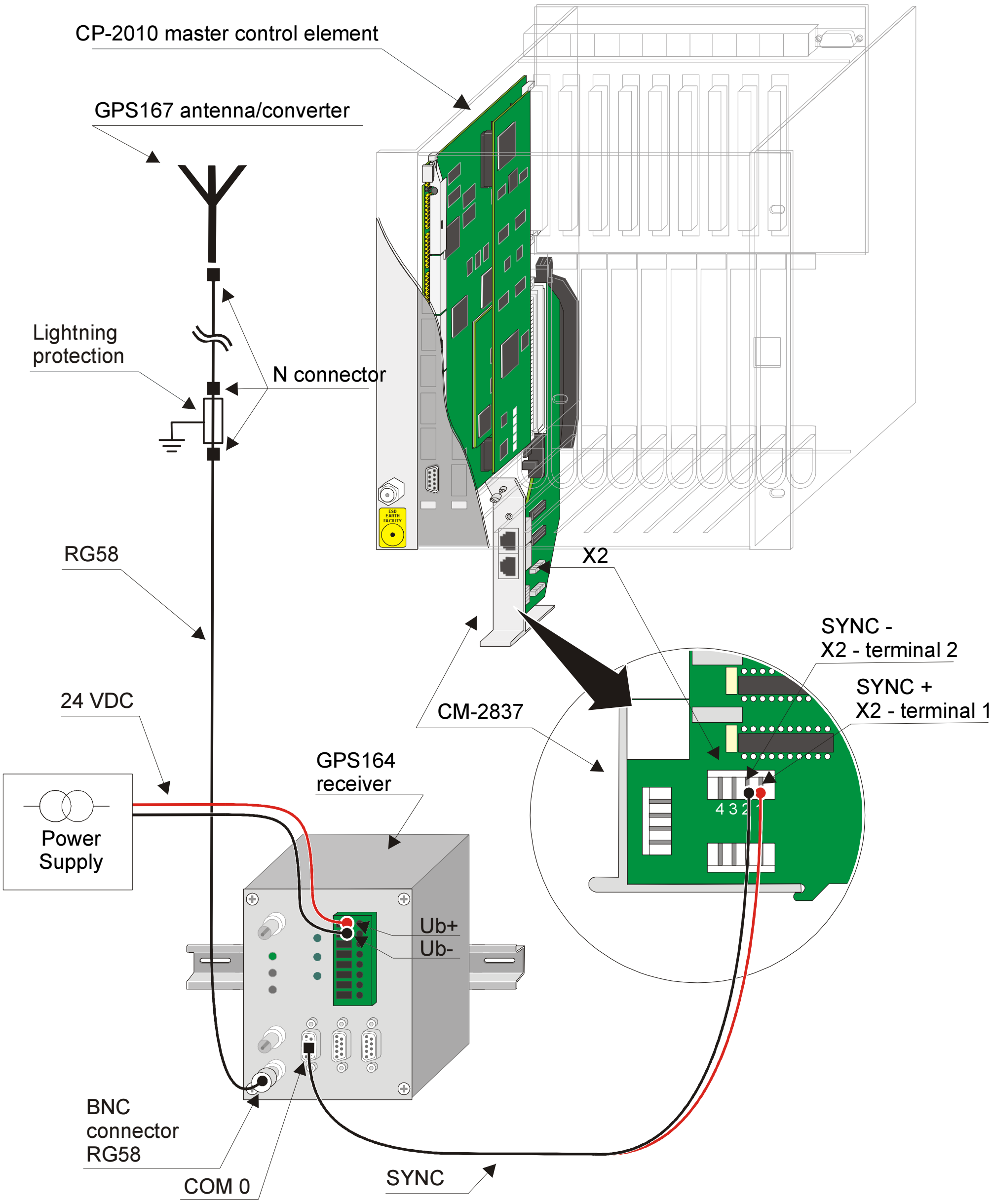
Accessories see Appendix A, Recommended and tested purchased products

6.2.2. Wiring for the receipt of the serial time signal

The serial time signal is supplied on the COM 0 interface on the front panel of the GPS164 receiver.

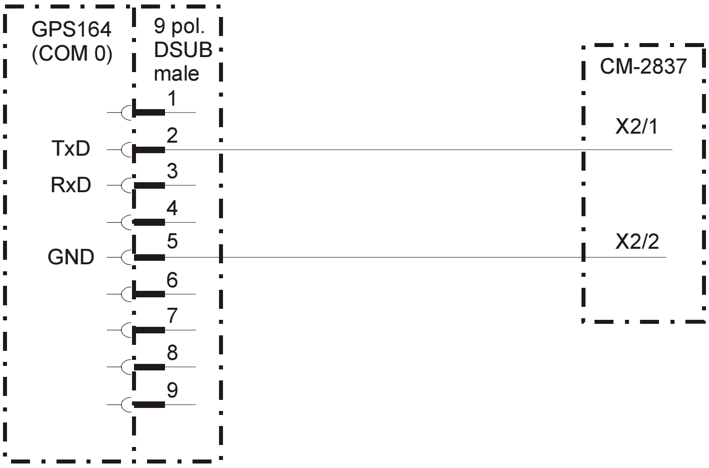
The necessary adjustments for the SAT-STRING must be made with software GPSMON32 as follows:

Format	7E2
Baud rate	2400
Serial time signal	per second



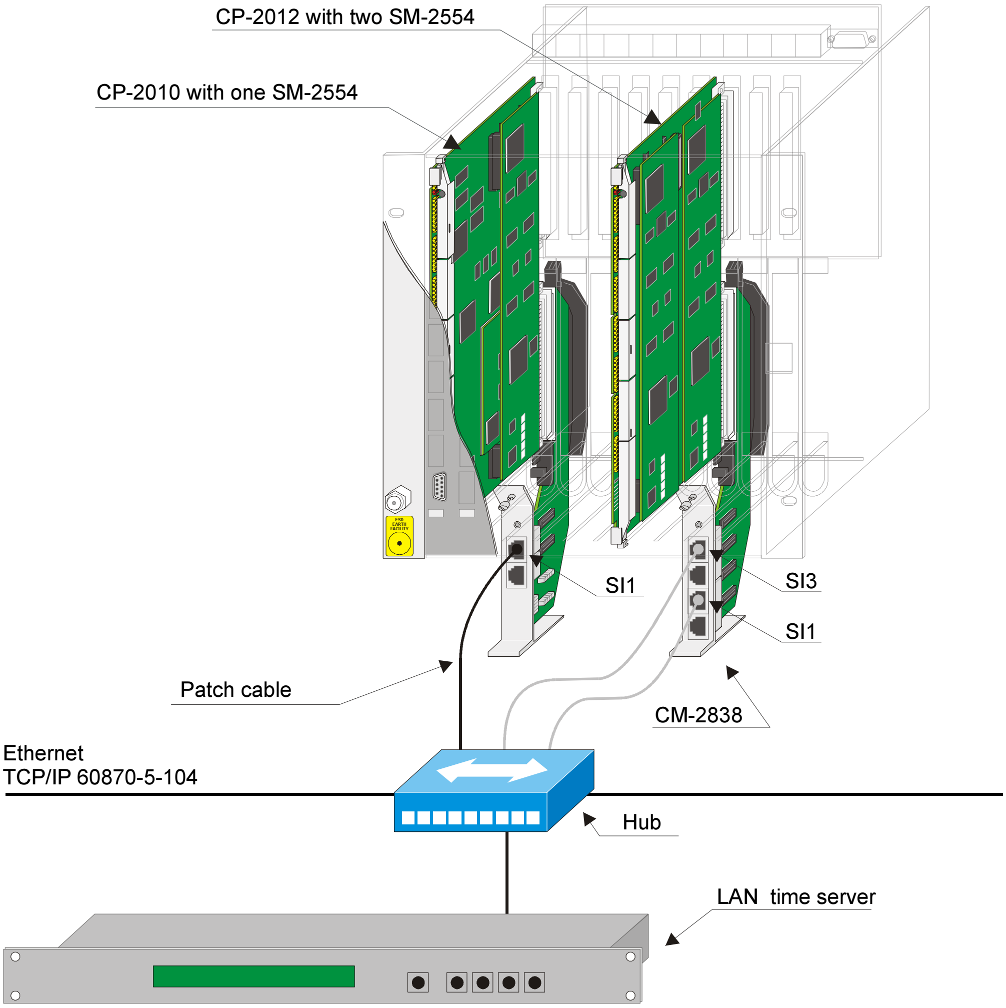
Accessories see Appendix A, Recommended and tested purchased products

Wiring cable GPS164 – CM-2837



6.3. LAN Time Server

NTP Server in 19" case for TCP/IP networks with integrated HTTP-Server. The internal reference time is derived from a built-in GPS receiver of the type GPS167.



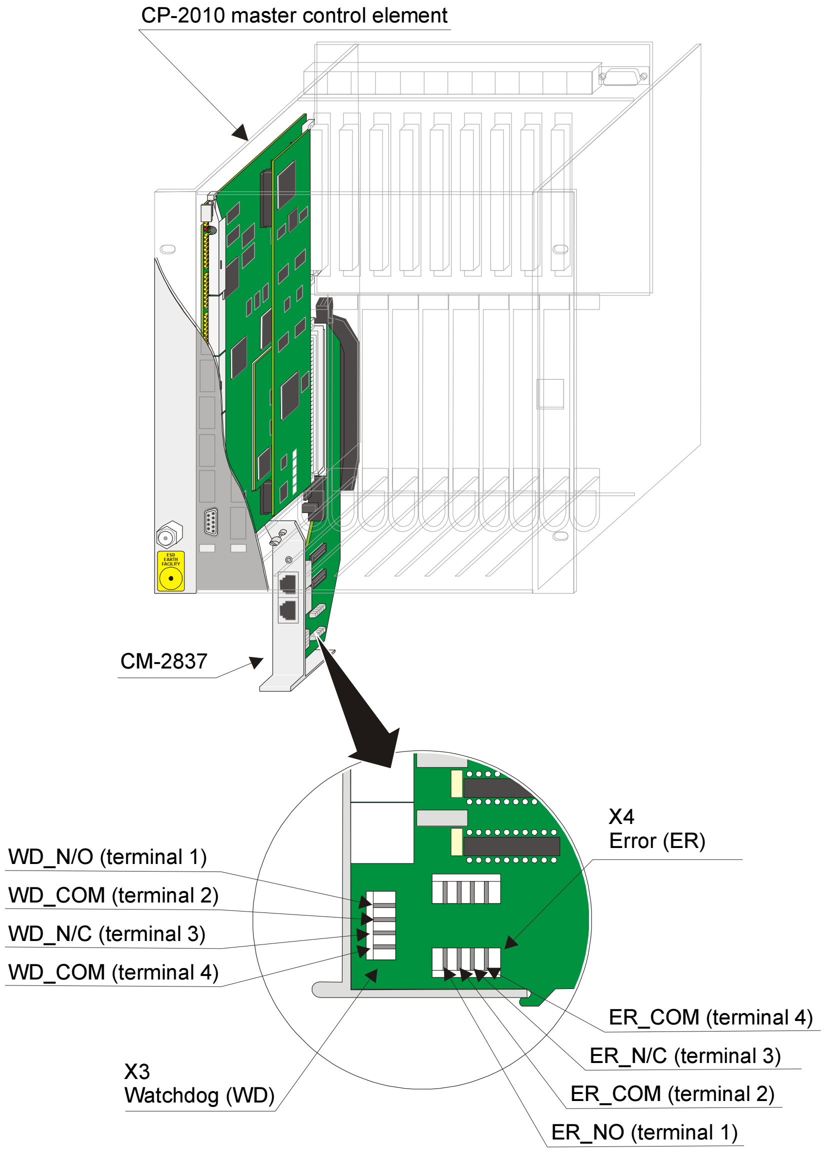
Accessories see Appendix A, Recommended and tested purchased products

7. Wiring Watchdog and Error

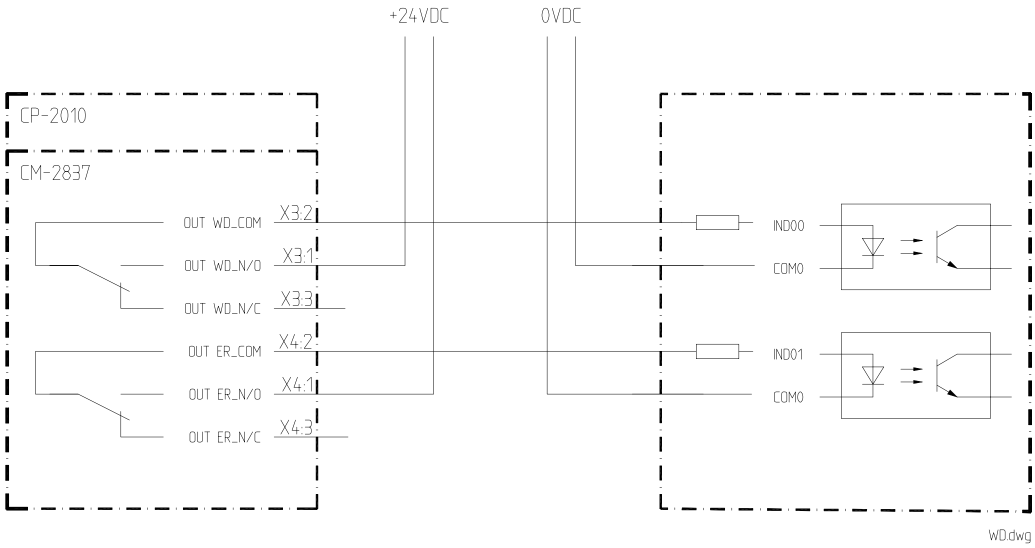
Content

7.1. General

The relay contacts (open- and closed contact) of watchdog and error are available on the communication interface CM-2837 on connectors X3 and X4.



Example (Circuitry for external alarm annunciation system)

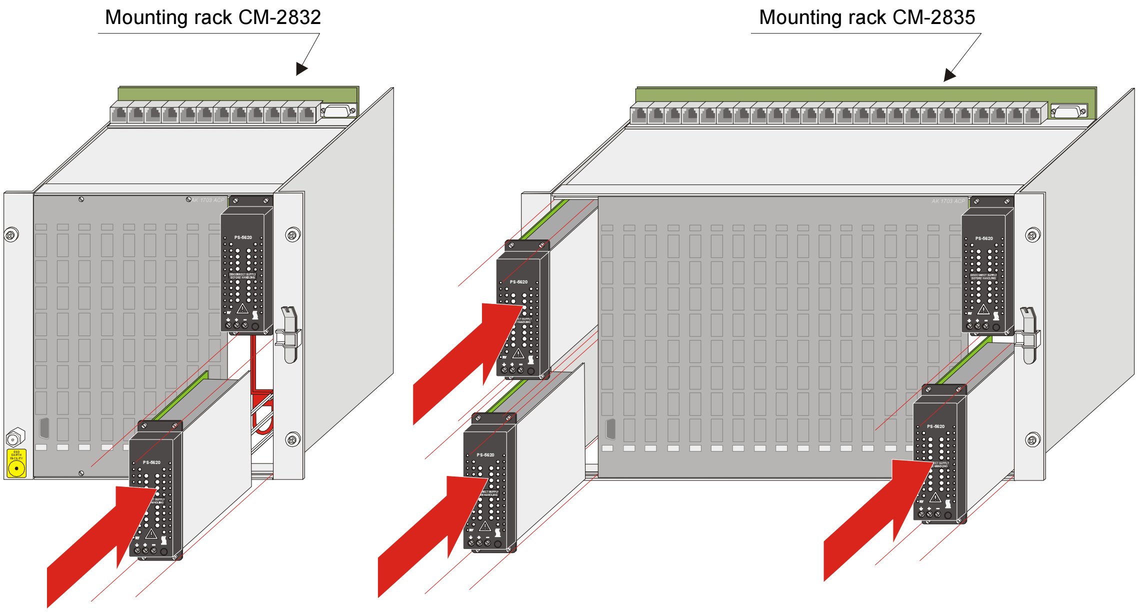


8. Power Supply

Content

8.1. General

AK 1703 ACP can be supplied with the power supplies PS-5620 and PS-5622. Depending on requirements and mounting rack, a differing number are used.



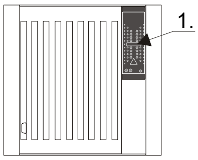
The power supplies are slotted into the compartments provided from the front. The cabling is carried out from the rear of the mounting rack. Systems that are mounted on the wall must therefore be removed beforehand.

The supply can be carried out with single leads of the type H07V-K (2.5) or a cable of the type H05VV-F3G (2.5).

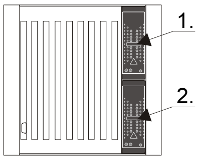
	Designation	Item-Number / MLFB
	Power Supply 24-60VDC/80W	BC5-620 6MF10130FG200AA0
	Power Supply 110-220VDC,230VAC/80W	BC5-622 6MF10130FG220AA0
	Mains cable 3x2,.5 (H05 VV-F 3G 2.5 flexible lead)	TF5-002 6MF13140FA020AA0

8.2. Possible Configurations

Mounting Rack CM-2832

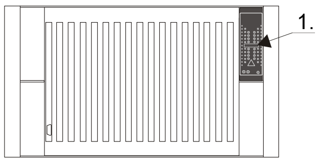


In the basic configuration the mounting rack is fitted with one power supply.
This is installed in the upper slot.

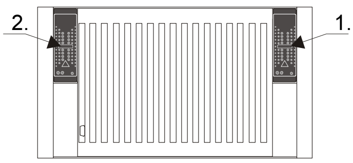


The 2nd power supply can be used either for increasing performance or for redundancy.
This is installed in the lower slot.

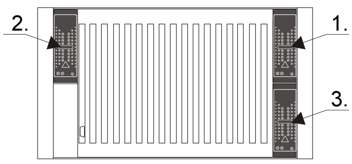
Mounting Rack CM-2835



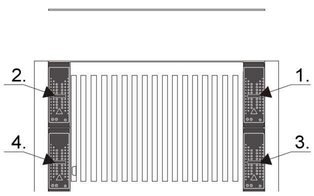
In the basic configuration the mounting rack is fitted with one power supply.
This is installed in the upper right slot.



The 2nd power supply can be used either for increasing performance or for redundancy.
This is installed in the upper left slot.



The 3rd power supply can be used solely as redundancy for the first or second power supply.
It is installed in the lower right slot. This must be expanded with CM-2826 beforehand.

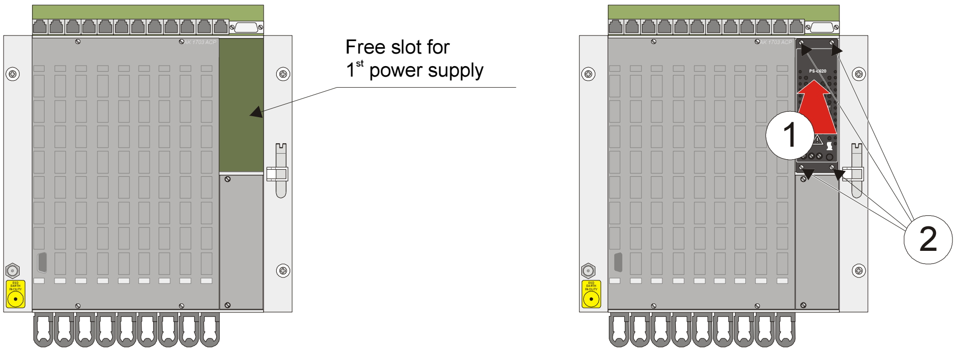


The 4th power supply can be used solely as redundancy for the first or second power supply.
It is installed in the lower left slot. This must be expanded with CM-2826 beforehand.

8.3. Fitting Power Supply in CM-2832

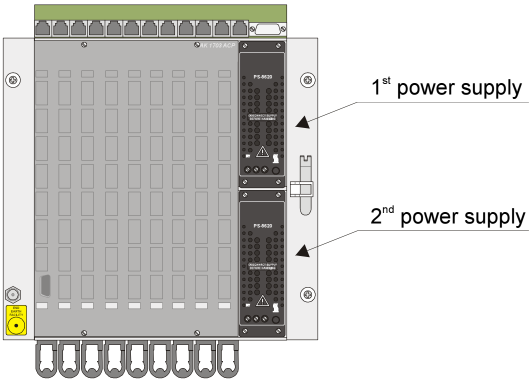
8.3.1. Installing First Power Supply

The first power supply is installed in the upper right slot. It only needs to be pushed into the compartment ① and fastened with screws ②. The associated network cable is already cabled with the reverse side of the backplane.



8.3.2. Installing Second Power Supply

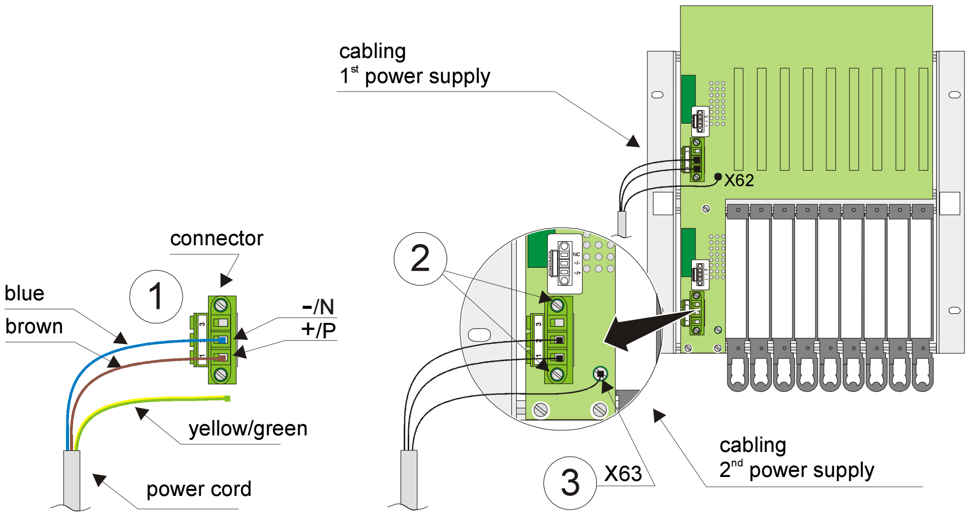
The second power supply is installed in the lower right slot.



Proceed as follows:

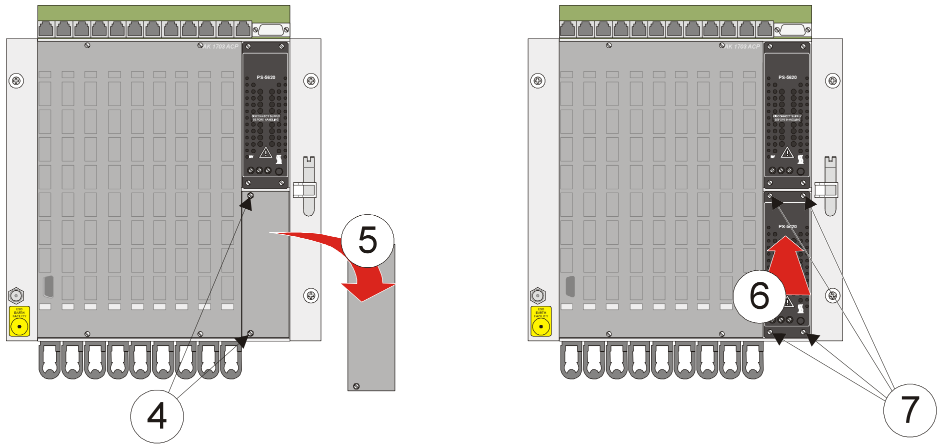
Cabling at the rear of the mounting rack

- Unscrew the connector ① from the backplane and connect it with the network cable.
- Screw the connector back on to the backplane of the mounting rack ②.
- Fasten the ground lead of the network cable to X63 of the backplane ③.



Installing the power supply

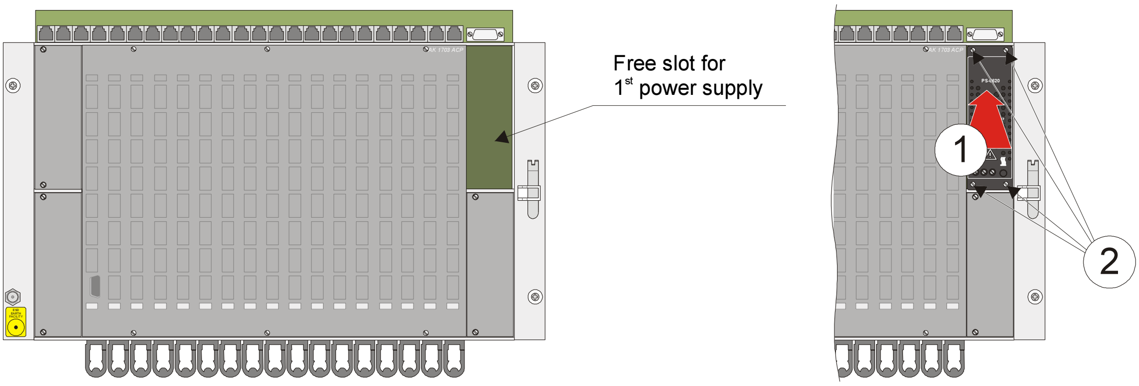
- Open the 2 screws ④ of the plate that covers the slot for the second power supply.
- Remove the plate ⑤.
- Install the power supply ⑥.
- Screw on the power supply ⑦.



8.4. Fitting Power Supply in CM-2835

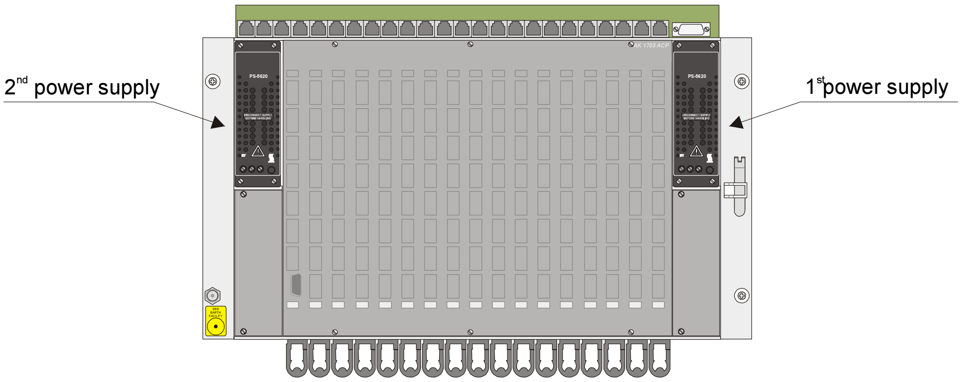
8.4.1. Installing First Power Supply

The first power supply is installed in the upper right slot. It only needs to be pushed into the compartment ① and fastened with screws ②. The associated network cable is already cabled with the reverse side of the backplane.



8.4.2. Installing Second Power Supply

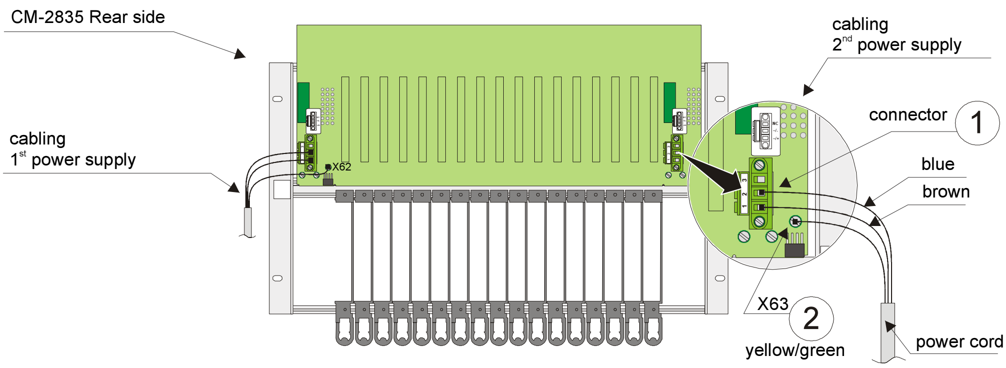
The second power supply is installed in the upper left slot.



Proceed as follows:

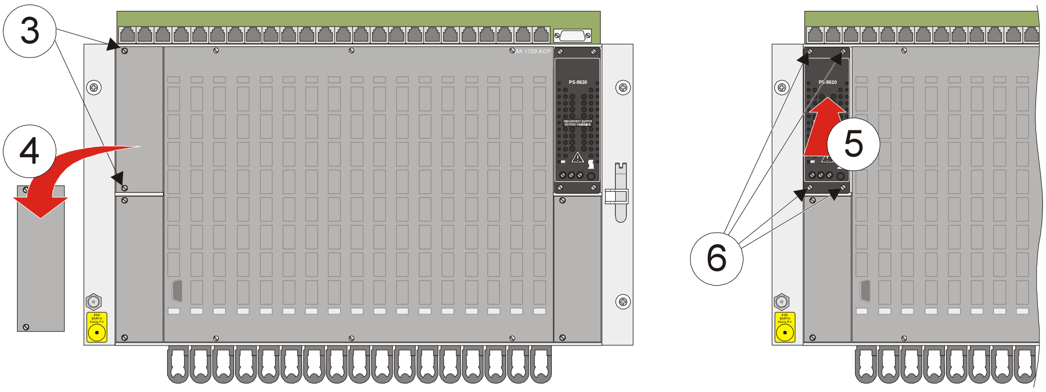
Cabling at the rear of the mounting rack

- Connect the network cable with the connector ①.
- Fasten the ground lead of the network cable to the backplane at X63 ②.



Installing the power supply

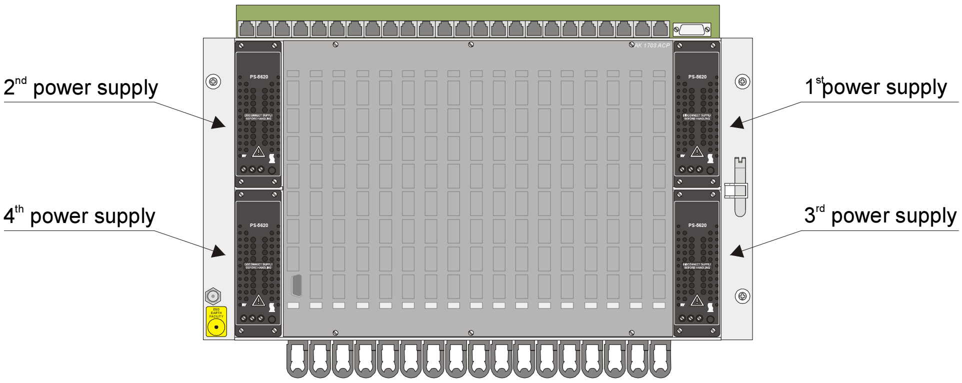
- Open the 2 screws ③ of the plate that covers the slot for the second power supply.
- Remove the plate ④.
- Install the power supply ⑤.
- Screw on the power supply ⑥.



8.4.3. Installing the Third and Fourth Power Supply

The third power supply is installed in the lower right slot and the fourth in the lower left slot.

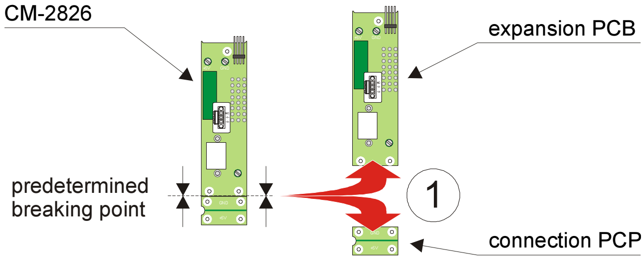
In contrast to the installation of the first and second power supply, for the 3rd and 4th power supply an additional PCB must be fitted beforehand.



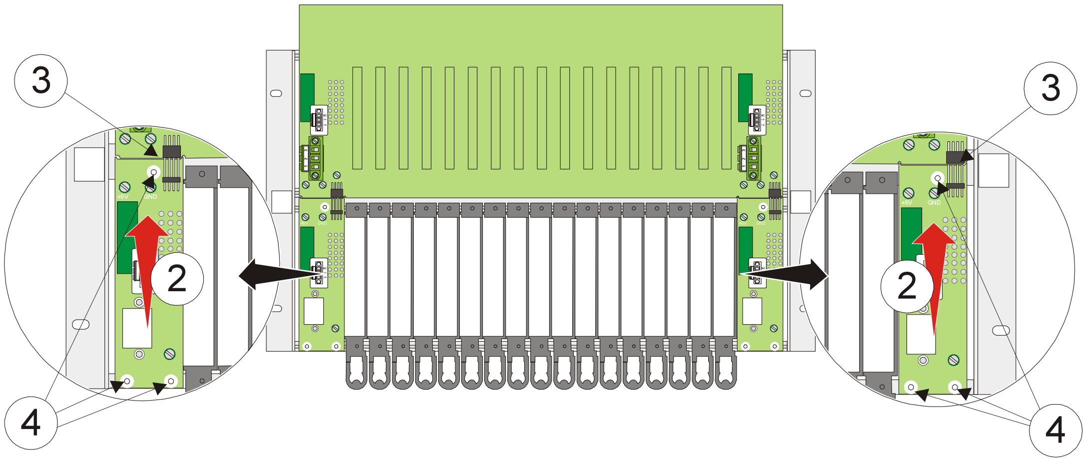
Designation	Item-Number / MLFB
CM-2826 Print for redundant power supply (only for 3rd and 4th power supply in CM-2835)	BC2-826 6MF10130CJ260AA0

Installation of the PCB for redundant supply

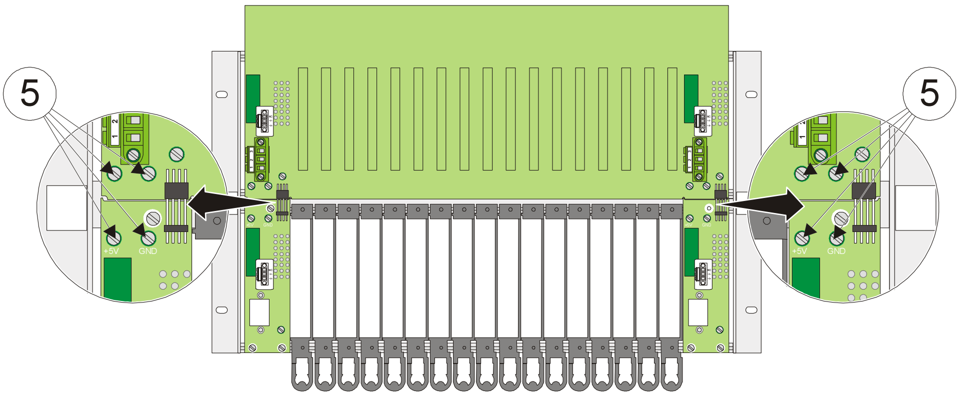
- Break the CM-2826 into 2 parts at the predetermined breaking point ①. You consequently obtain an expansion PCB and a connection PCB.



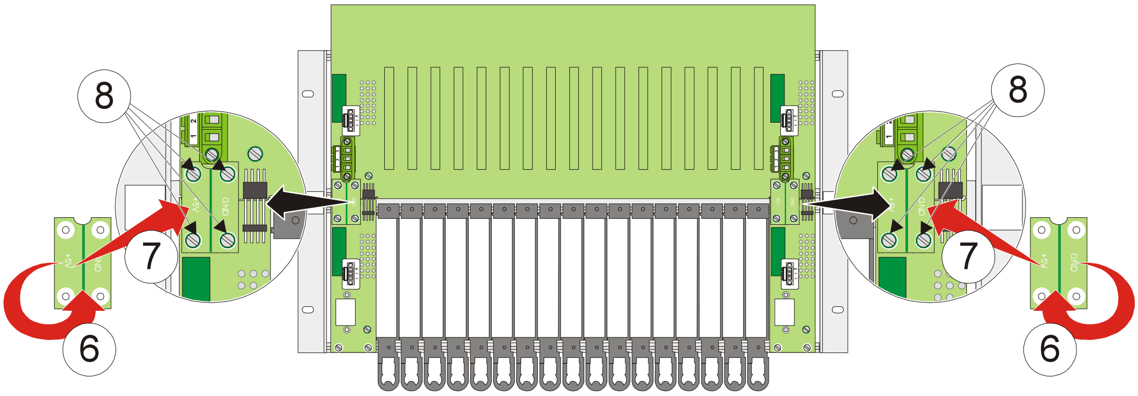
- Take the expansion PCB and guide it towards the backplane from below ②. Ensure correct plug connection between both PCB's ③.
- Screw the expansion PCB firmly to the mounting rack ④.



- Remove the screws for the connection PCB ⑤. There are 2 on both the backplane and expansion PCB..

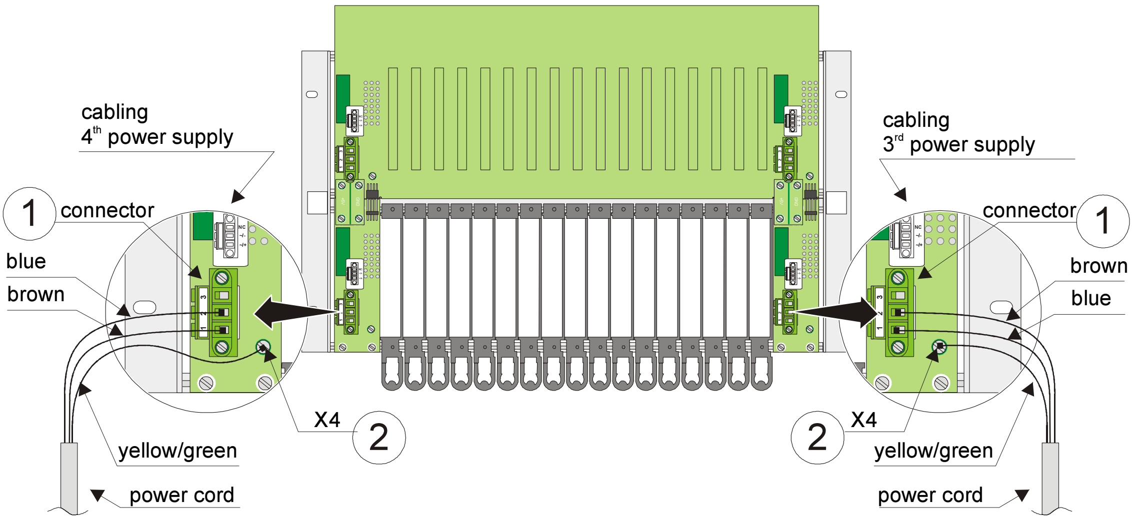


- Rotate the connection PCB to the position shown ⑥ and fasten it with backplane and expansion PCB ⑦+⑧.



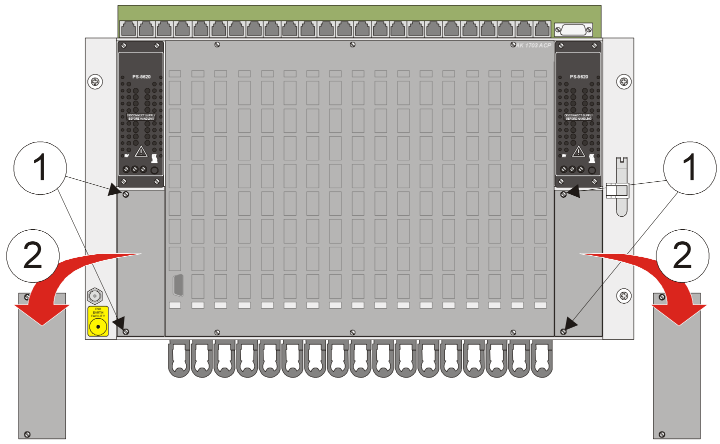
VerkaCabling at the rear of the mounting rack

- Connect the network cable with the connector ① and screw the connector to the expansion PCB.
- Fasten the ground lead of the network cable to the expansion PCB at X4 ②.

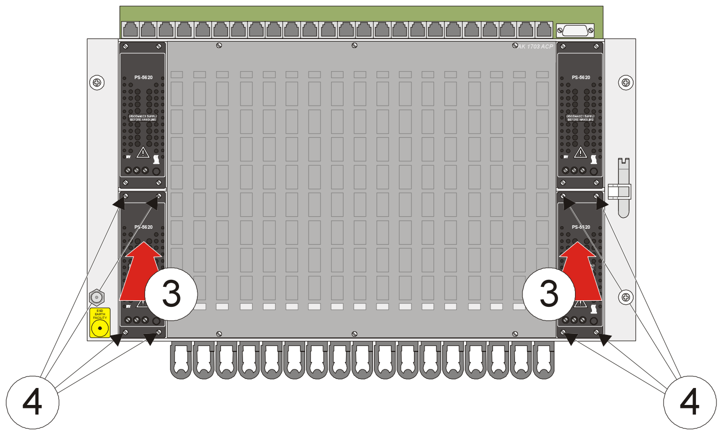


Installing the power supply

- Open the screws of the plates ① that cover the slots for the third and fourth power supply.
- Remove the plates ②.



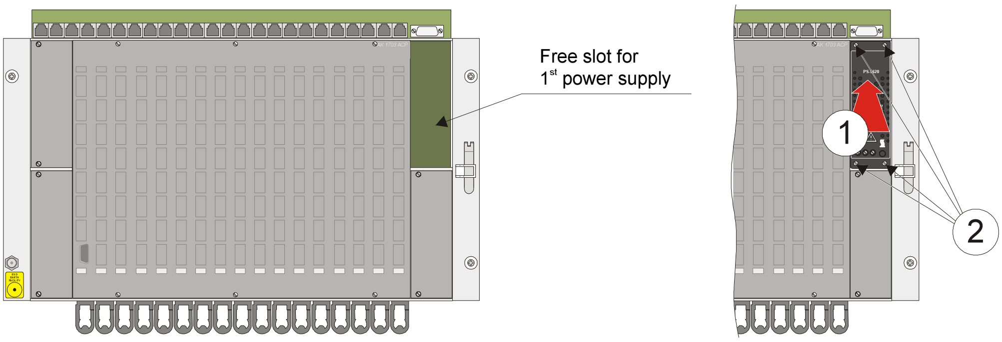
- Install the power supplies ③.
- Screw on the power supplies ④.



8.5. Fitting Power Supply in CM-2833

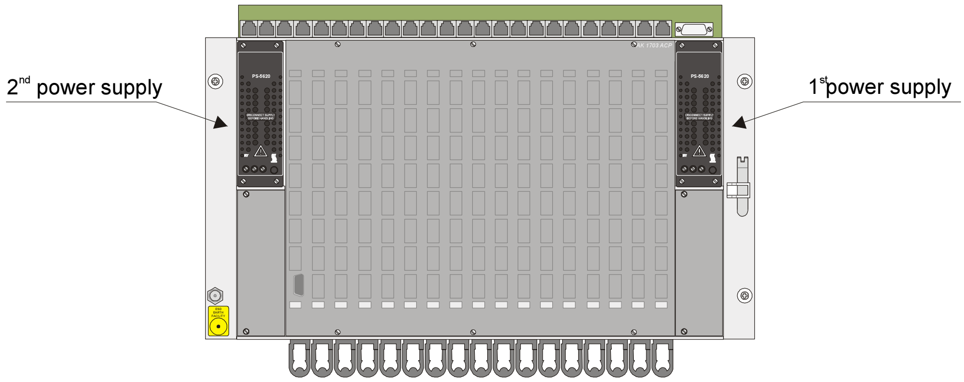
8.5.1. Installing First Power Supply

The first power supply is installed in the upper right slot. It only needs to be pushed into the compartment ① and fastened with screws ②. The associated submodule (SM-5860) for the supply from the reverse side of the mounting rack is already mounted and cabled.



8.5.2. Installing Second Power Supply

The second power supply is installed in the upper left slot.

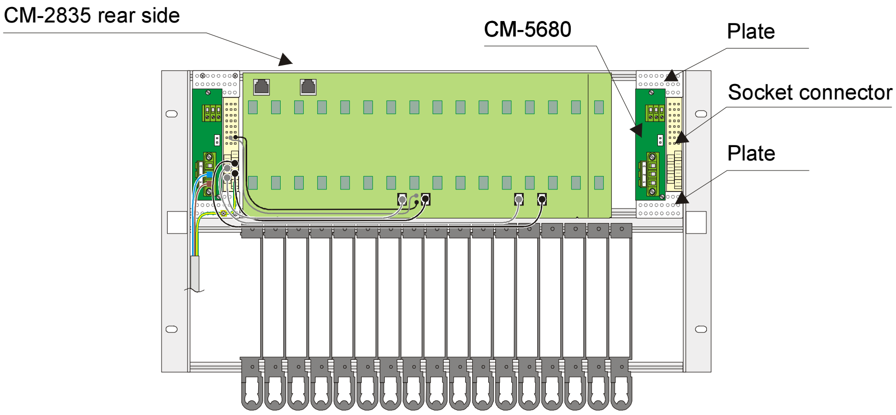


Therefor the CM-5680 connection p.c.b. and socket connector must be installed before. The same parts as used for the first power supply have to be used. For detail see parts list GC2-833.

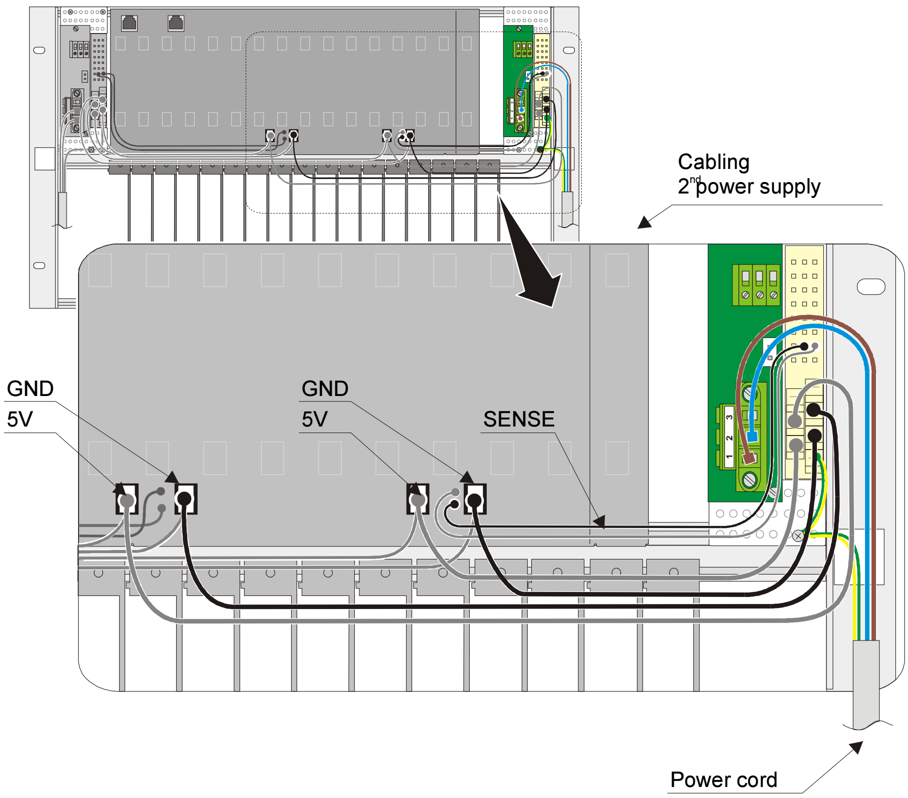
Designation	Item-Number / MLFB
CM-5680 connection p.c.b. Supply/display PS-562x	BC5-680 6MF10130FG800AA0
Socket Connector 24+7	NC78762
Plate 8,5 TE (43mm) (2 parts for fastening connection p.c.b. and socket connector)	409120026 (Intermas)

Installing connection p.c.b. CM-5680

- Open the screws of the plate that covers the slot for the second power supply.
- Install the plates on which connection p.c.b. and socket connector will be mounted.
- Screw the socket connector from inside the mounting rack to the plates.
- Screw the connection p.c.b. from the rear side of the mounting rack to the plates.

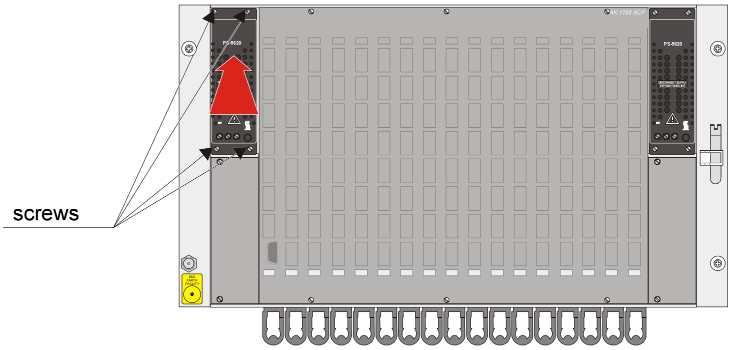


Cabling at the rear of the mounting rack



Installing the power supply

- Install the power supply.
- Screw on the power supplies.



9. Wiring Process Peripherals

Content

9.1. General

The process peripherals are connected with the system by means of prefabricated peripheral cables.

	Designation	Item-Number / MLFB
	CM-2890CM-2890CM-2890CM-2890 Periphery cable Crimp 5m 100pins	TC2-890 6MF13131CJ000AA0

9.2. Fitting Peripheral Cable

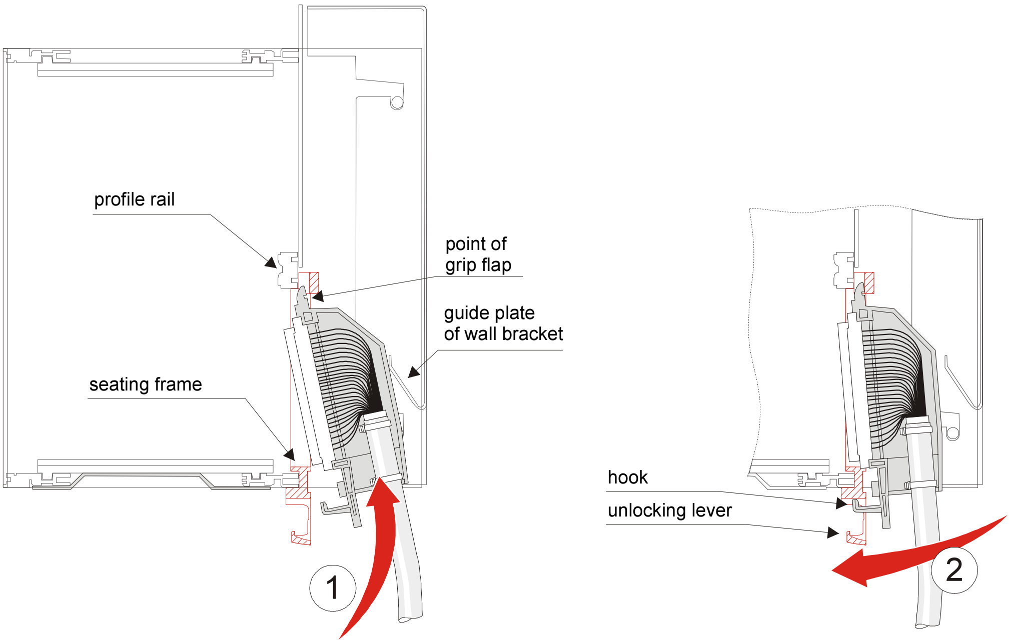
Attention

The peripheral modules must **NOT** be installed in the mounting rack during the fitting of the associated peripheral cable.

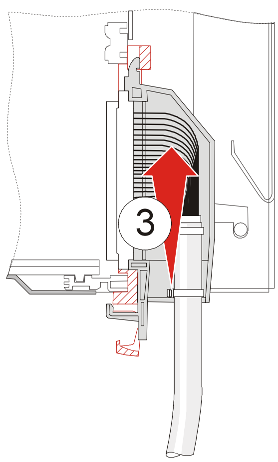
The fitting is carried out by engaging the peripheral cable in the red seating frame.

Proceed as follows:

- Insert the peripheral cable into the mounting rack from below, until the point of the grip flap is fixed between profile rail and seating frame ①. In the case of wall mounting a plate on the wall bracket serves as a guide during installation.
- Then swing the peripheral cable forwards up to the stop ②. Thereby the hook of the grip flap must be guided through the unlocking lever.



- Now push the peripheral cable upwards. Thereby grip flap and seating frame are finally fixed together ③.

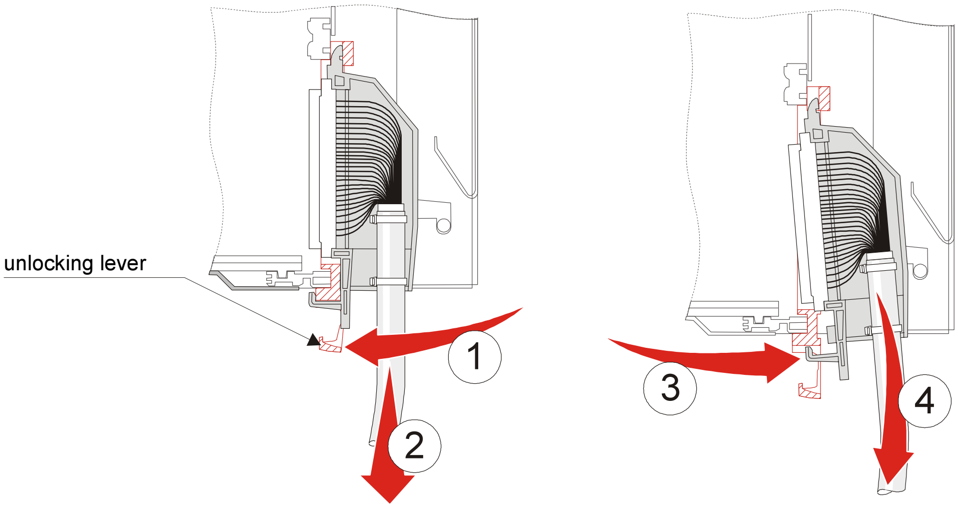


9.3. Removing Peripheral Cable

Âttention

The peripheral modules must **NOT** be installed in the mounting rack during the fitting of the associated peripheral cable.

- Push the unlocking lever of the seating frame slightly forwards and hold it in this position ①.
- Pull the peripheral cable vertically 4-5mm downwards ②.
- Swing the connector of the peripheral cable to the back ③ (the hook of the grip flap is thereby guided through the unlocking lever) and pull it downwards out of the mounting rack ④.

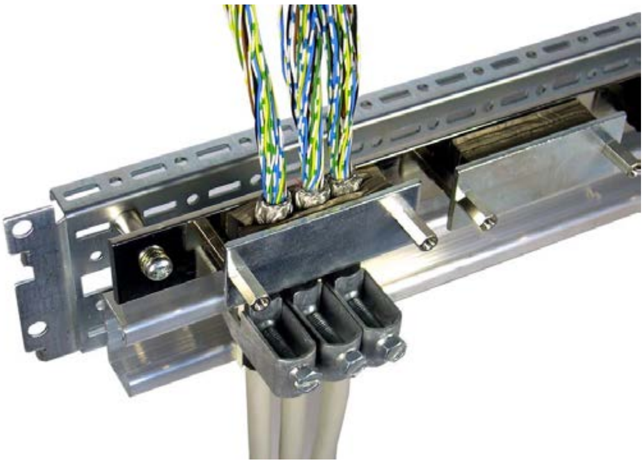


10. Shielding / Protective Earthing

Content

10.1. Shielding

Normally, shielded cables are strain-relieved directly after the cabinet/rack entry and then grounded on a large-surface screening rail installed for this purpose.



10.2. Protective Earthing / Grounding

When installing a AK 1703 ACP system attention is to be paid that the cabinet or rack used features a proper protective earthing / grounding. That means, that all electrical conducting parts must be connected large-surface and as short as possible with the cabinet spar. The cabinet spar itself must be connected with the existing grounding system.

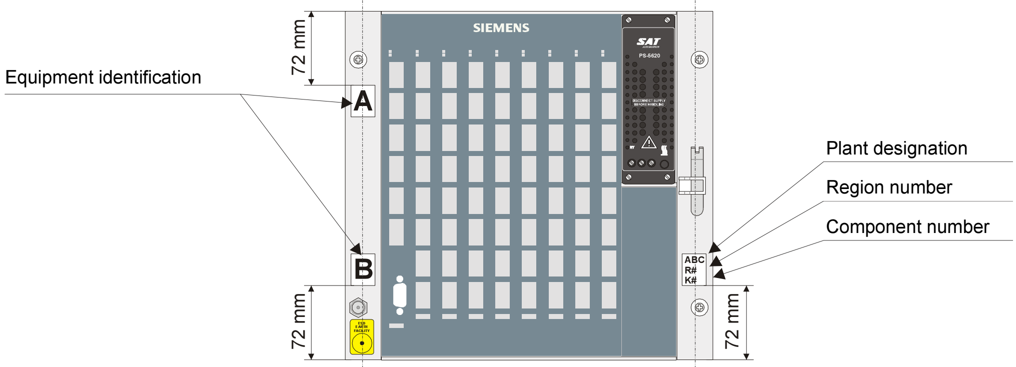
11. Labeling

Content

11.1. General

This chapter shows how an AK 1703 ACP system is to be labeled according to Standard. The following labels are provided:

- Region- and Component Number
- Plant Designation
- Equipment Identification



12. Parameter Setting Preparation

Content

12.1. General

AK 1703 ACP is parameterized with a PC on which the TOOLBOX II software is installed. For this, the Toolbox PC and automation unit must be connected with each other and a Flash Card installed on the master control element (CP-2010).

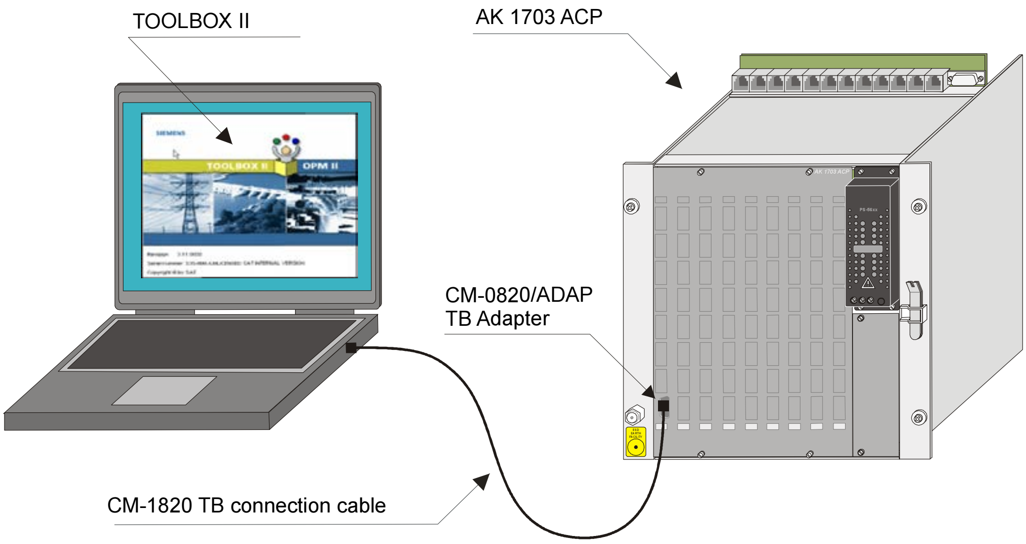
12.2. Physical connection between Toolbox PC and AK 1703 ACP

Following connections are differentiated:

- direct local via toolbox cable
- serial via modems
- Ethernet (TCP/IP, TELNET)
- Ethernet (TCP/IP, TELNET) and Terminal Server (serial)

12.2.1. Direct connection via toolbox cable

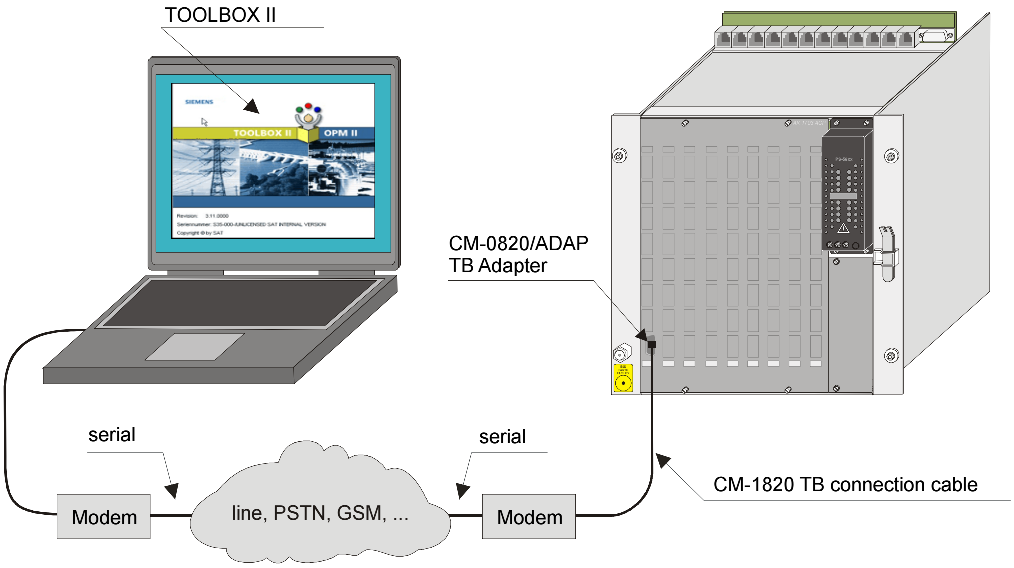
Toolbox PC and AK 1703 ACP are connected direct via the toolbox cable.



	Designation	Item-Number / MLFB
	CM-1820 TB Cable Set	BA1-820 6MF10110BJ200AA0
	CM-0820/ADAP TB Adapter	GA0-820 6MF11110AJ200AA0

12.2.2. Serial via modem

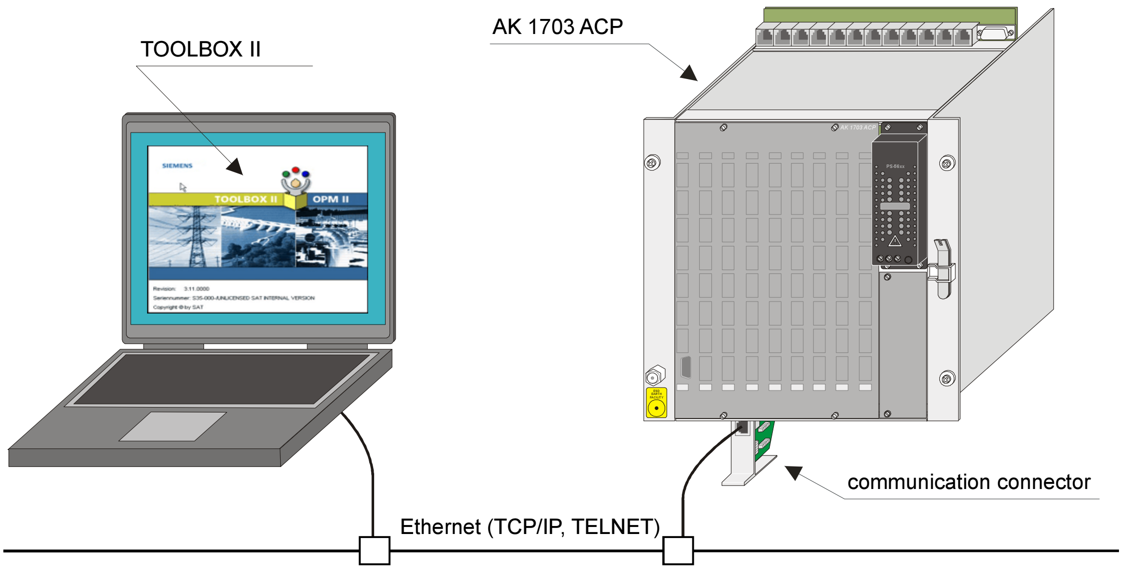
Toolbox PC and AK 1703 ACP are connected via 2 serial modems.



12.2.3. Via Ethernet (TCP/IP, TELNET)

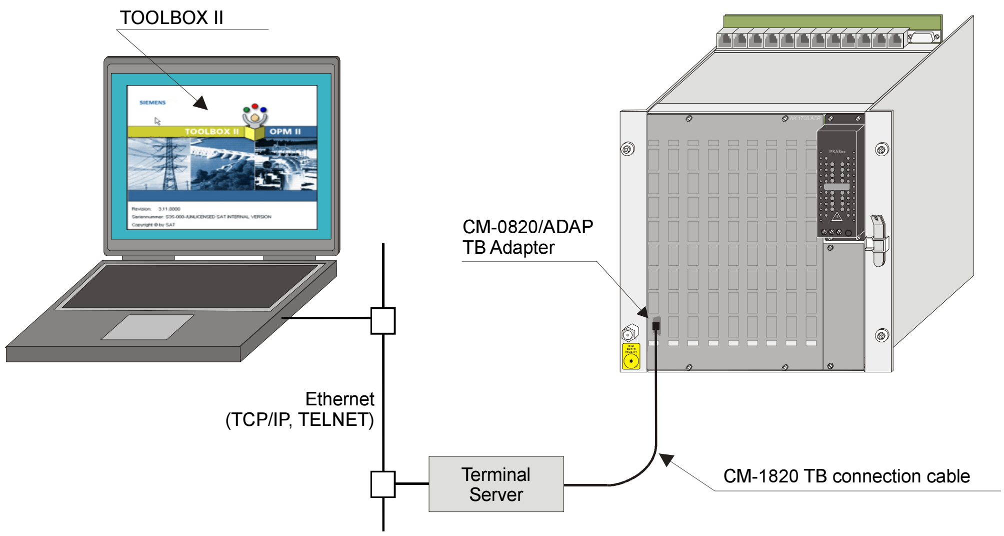
Toolbox PC and AK 1703 ACP are connected via Ethernet (TCP/IP, TELNET).

For this, the master control element or processing- and communication element must be fitted with additional SIM and connection boards. For details refer to section 5 Setup of external Communication Connections.



12.2.4. Via Ethernet (TCP/IP, TELNET) and Terminal Server (serial)

Toolbox PC and AK 1703 ACP are connected via ethernet (TCP/IP, TELNET) and terminal server (serial).



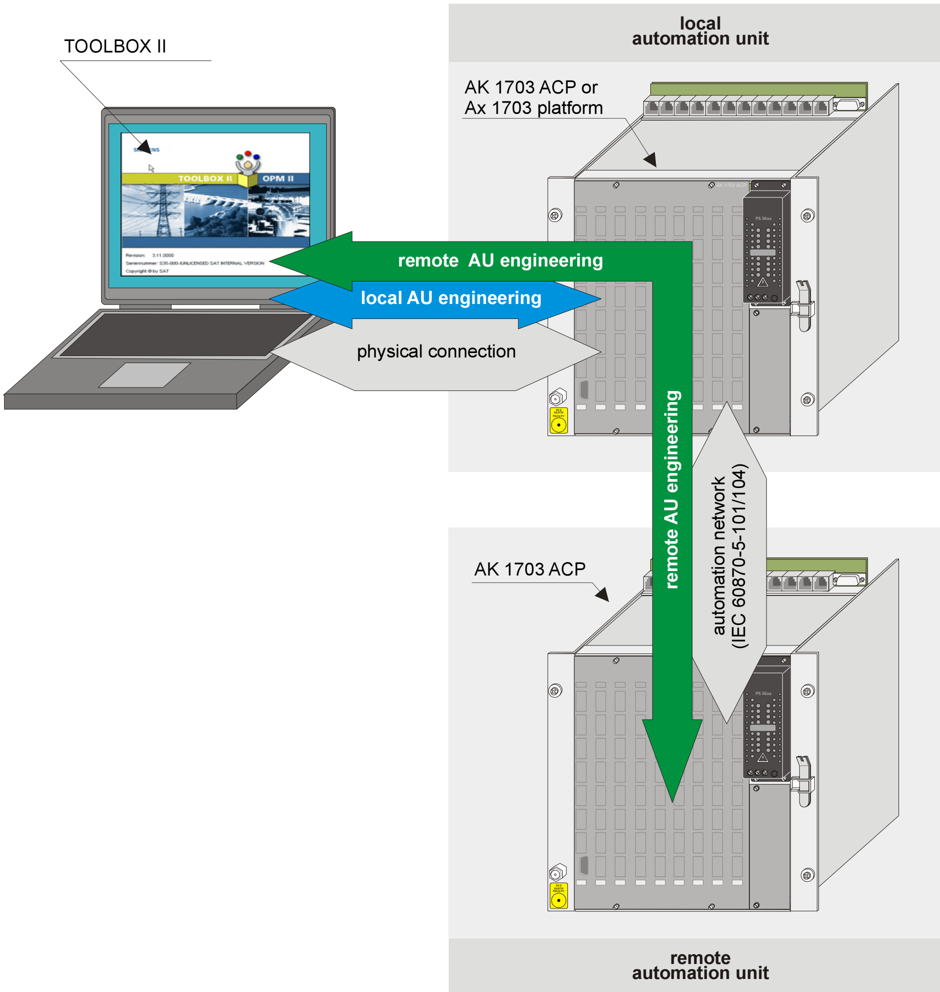
12.3. Logical Connection between Toolbox PC und AK 1703 ACP

For engineering there must be a logical connection to those automation unit, which is subject of engineering. Following two cases are differentiated:

- Engineering of a local automation unit
  (that is that one, to which an physical connection exists)

Attention
A first initialization of the automation unit is only possible in this way.

- Engineering of a remote automation unit
  (automation unit, which can be accessed via a local automation unit; a continuous remote communication according to IEC 60870-5-101 or -104 is required)



12.4. Flash Card

AK 1703 ACP uses a flash card for the storage of parameters, user programs and firmware codes for basic system elements and supplementary system elements (e.g. peripheral elements, protocol elements). This is located on the master control element.

A. Recommended and tested purchased products

Content

A.1. Modem

Standard modem for analog dial-up traffic

	Designation	Item-Number / MLFB
	Eurocom 24	G21-014 6MF11020BA140AA0
	Westermo TD-36 AV (20 – 264 VAC / 14 – 300 VDC) Westermo TD-36 LV (DC) (10 – 30 VAC / 10 – 60 VDC)	www.westermo.com

Standard modem for ISDN dial-up traffic

	Designation	Manufacturer
	Westermo IDW-90	www.westermo.com

Standard modem for GSM dial-up traffic

	Designation	Item-Number
	Siemens TC35 (dual band modem) Including DIN-rail installation kit and connection cable (1.5m) Without antenna!	G21-030-- 6MF11020BA300AA1
	GSM antenna for outside installation	G21-031-- 6MF11020BA310AA1

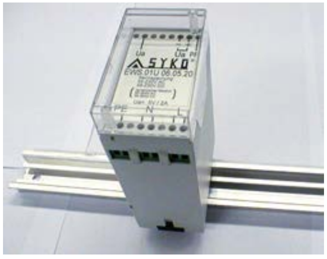
A.2. GPS Satellite Receiver

	Designation	Manufacturer
	Meinberg GPS164DHS Receiver for TS35	www.meinberg.de
	GPS167-ANT/CON GPS-Antenna-/Converter.	www.meinberg.de
	GPS-AV4 Antenna Terminal Block	www.meinberg.de

A.3. LAN Time Server

Designation	Manufacturer
Meinberg NTP Server + Antenna + Cable	www.meinberg.de

A.4. Power Supplies



Designation	Manufacturer
Power Supply 5V/2A SYKO EWS 01 U.06.05.20 82 – 264VAC or 36 – 350VDC	www.syko.de
Power S. 115/230VAC,100-375VDC-24VDC-50W	www.mtm-power.com

Literature

ACP 1703 Common Functions System and Basic System Elements	DC0-015-2
System Element Manual CP-2010/CPC25	DC2-005-2
System Element Manual CP-2012/PCCE25	DC2-007-2
System Element Manual DI-2100/BISI25	DC0-001-2
System Element Manual DI-211x/BISI26	DC0-005-2
System Element Manual DO-2201/BISO25	DC0-017-2
System Element Manual AI-2300/PASI25	DC0-007-2
System Element Manual DO-2210/PCCO2X	DC0-003-2
System Element Manual MX-2400/USIO2X	DC0-019-2
Data Sheet CM-2832	MC2-005-2
Data Sheet CM-2835	MC2-007-2
Data Sheet CM-0842	MC0-021-2
Data Sheet CM-0843	MC0-023-2
Data Sheet SM-2541/PROTOCOL	MC0-001-2
Data Sheet SM-2551/PROTOCOL	MC0-002-2
Data Sheet SM-25x4/ET02	MC0-005-2
Data Sheet SM-2545/DPM00	MC0-007-2
Data Sheet PS-562x	MC2-013-2